

LOCAL OCNEANU RIGIDITY FOR SEPARABLE ALGEBRA OBJECTS

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ABSTRACT. We establish local Ocneanu rigidity for separable algebra objects in Hom-finite monoidal categories $\mathcal{C}$ over an algebraically closed field $\mathbb{k}$. A separability morphism contracts the first two Hochschild cohomology groups without requiring an abelian ambient category. Separable algebra structures and homomorphisms from separable sources have open algebraic-group orbits, and affine Bézout estimates give effective finiteness results. For Frobenius subalgebras, exchange relations replace the source and embedding data by a single self-dual idempotent. We prove that their separable inner-conjugacy classes are open in the exchange locus, yielding a bound of $2^{\dim_{\mathbb{k}} \operatorname{End}_{\mathcal{C}}(X)}$ for nonzero ambient algebras; connectedness gives the same bound on the actual number of subalgebras. As an application, we obtain an effective form of the Etingof–Walton finiteness theorem: every finite-dimensional semisimple Hopf algebra H over $\mathbb{C}$ has at most $2^{\dim_{\mathbb{C}} H}$ left coideal subalgebras. In the unitary setting, we bound E -compatible intermediates of C^* -algebra and von Neumann algebra inclusions up to unitary conjugacy, under finite-index and finite-center hypotheses. For irreducible subfactors, these improve the $9^{[M:N]}$ bound of Bakshi–Das–Liu–Ren to $2^{[M:N]}$.

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Internal detail 1. This expanded version supplements the submission manuscript. The orange passages give additional calculations, explanations of logical steps, classical references, and diagrams to help readers verify the arguments. The main statements and hypotheses agree with the submission version; results imported from the literature are identified by their references.

1. INTRODUCTION

Ocneanu rigidity is a global finiteness principle for fusion categories, tensor functors, and semisimple module categories: once the relevant discrete data are fixed, the corresponding structures admit no nontrivial infinitesimal deformations. Following the historical account in Etingof–Nikshych–Ostrik [16], the unitary argument was suggested by Ocneanu and was subsequently developed algebraically by Blanchard and Wassermann; that paper gives a general published proof for fusion categories. We use the terminology of [15]. Related deformation complexes were introduced by Crane–Yetter [12] and Davydov [13].

This paper studies a local counterpart. The ambient monoidal category and the underlying object are fixed, while an algebra structure, an algebra map, or a subalgebra embedding is allowed to vary. Our purpose is to isolate the algebraic mechanism, determine the exact hypotheses, make the conclusion effective, and connect it with Frobenius subalgebras, rational representations of linearly reductive groups, left coideal subalgebras of semisimple Hopf algebras, and intermediate operator algebras.

The deformation-theoretic core is elementary and categorical. If A is a separable algebra object and M is an A – A -bimodule, a separability morphism defines contractions in Hochschild degrees one and two. No abelian structure is needed, and

$$\mathrm{HH}^1(A, M) = \mathrm{HH}^2(A, M) = 0.$$

The two vanishings serve different purposes. The equality $\mathrm{HH}^2(A, A) = 0$ identifies infinitesimal deformations of the multiplication with infinitesimal changes of coordinates, whereas $\mathrm{HH}^1(A, {}_fX_f) = 0$ identifies infinitesimal deformations of an algebra map $f : A \rightarrow X$ with infinitesimal inner conjugations. This is the algebra-object analogue of Gerstenhaber’s deformation theory [19]; Hochschild cohomology in abelian monoidal categories is developed, for example, in [2]; the contraction used here is proved in Proposition 3.1.

Our first main result is qualitative.

Theorem 1.1. *Let $\mathbb{k}$ be an algebraically closed field and let $\mathcal{C}$ be a $\mathbb{k}$ -linear monoidal category with finite-dimensional morphism spaces.*

- (1) *Each fixed object B in $\mathcal{C}$ supports only finitely many isomorphism classes of separable unital algebra structures (Theorem 5.1).*
- (2) *For separable A and any algebra object X , the unital algebra maps $A \rightarrow X$ form finitely many inner-conjugacy orbits. If X is connected, the map set itself is finite (Theorem 6.1).*
- (3) *If only finitely many isomorphism classes of objects occur as subobjects of X , then the separable algebra subobjects of X form finitely many inner-conjugacy classes and, when X is connected, finitely many equivalence classes over X (Theorem 7.1).*

The third assertion cleanly separates the local deformation argument from the only global input needed to count all subalgebras of a fixed ambient algebra: finiteness of the isomorphism classes of objects occurring as subobjects. This condition is automatic for a finite-length object in a semisimple category in the sense of Definition 2.2, but it can also hold in nonsemisimple situations, as Example 7.3 shows. In particular, the theorem applies to semisimple tensor categories in the sense of [15], including $\mathrm{Rep}(G)$ for a linearly reductive algebraic group G over $\mathbb{k}$ (in particular, for a reductive group when $\mathrm{char} \mathbb{k} = 0$). Thus a finite-dimensional rational G -algebra X with $X^G = \mathbb{k}1$ has only finitely many G -stable separable unital subalgebras; this is Corollary 7.2

specialized to $\text{Rep}(G)$. The corresponding fixed-module assertion follows from Theorem 5.1. The local orbit statements themselves require only a Hom-finite $\mathbb{k}$ -linear monoidal category.

Finite-dimensional parameter spaces also yield explicit estimates. For an object B , set

$$\begin{aligned} a_B &= \dim_{\mathbb{k}} \text{Hom}_{\mathcal{C}}(B \otimes B, B), & u_B &= \dim_{\mathbb{k}} \text{Hom}_{\mathcal{C}}(\mathbf{1}, B), \\ t_B &= \dim_{\mathbb{k}} \text{Hom}_{\mathcal{C}}(B^{\otimes 3}, B), & E_B &= \dim_{\mathbb{k}} \text{End}_{\mathcal{C}}(B). \end{aligned}$$

After bases are chosen, associativity and unitality become polynomial equations of total degree at most two in the coordinates of the multiplication and unit. Moreover, the change-of-coordinates orbit representing each separable isomorphism class is open. An affine Bézout inequality therefore gives the following bound.

Theorem 1.2 (Effective local rigidity). *The number of isomorphism classes of separable unital algebra structures on B is at most*

$$2^{\min\{a_B+u_B, t_B+2E_B\}}.$$

This is proved in Theorem 5.2. If A is separable and X is an algebra object, set

$$\begin{aligned} b_{A,X} &= \dim_{\mathbb{k}} \ker(\text{Hom}_{\mathcal{C}}(A, X) \rightarrow \text{Hom}_{\mathcal{C}}(\mathbf{1}, X), f \mapsto f \circ e_A), \\ c_{A,X} &= \dim_{\mathbb{k}} \text{Hom}_{\mathcal{C}}(A \otimes A, X). \end{aligned}$$

Then the number of inner-conjugacy classes of unital algebra homomorphisms $A \rightarrow X$ is at most $2^{\min\{b_{A,X}, c_{A,X}\}}$. For connected X , this bounds the homomorphism set itself (Theorem 6.2). The explicit resulting sums for algebra subobjects are given in Theorem 7.4, with a bound depending only on the ambient algebra in Corollary 7.5.

Internal detail 2. For a compact ambient-object form of the last estimate, suppose that $\mathcal{C}$ is semisimple with simple tensor unit and that the connected algebra X has $X \cong \mathbf{1} \oplus \bigoplus_i S_i^{\oplus n_i}$. Put $P_X := \prod_i (n_i + 1)$, $E_X := \dim_{\mathbb{k}} \text{End}_{\mathcal{C}}(X)$, $M_X := \dim_{\mathbb{k}} \text{Hom}_{\mathcal{C}}(X \otimes X, X)$, and $T_X := \dim_{\mathbb{k}} \text{Hom}_{\mathcal{C}}(X^{\otimes 3}, X)$. Then

$$N_{\text{sep}}(X) \leq P_X 2^{\min\{M_X-1, T_X\} + \min\{E_X-1, M_X\}},$$

where $N_{\text{sep}}(X)$ counts equivalence classes over X ; see Corollary 7.5.

Frobenius self-duality permits a smaller parameter space: it records both the source algebra and its embedding by an endomorphism of the ambient Frobenius algebra. The exchange-relations theorem of Ghosh–Palcoux [20, Theorem 4.1] encodes a Frobenius subalgebra by a self-dual unital idempotent satisfying two quadratic equations. Our additional observation is that the joint deformation of a separable source algebra and its embedding has an open orbit. Its inverse image in the exchange locus makes each separable inner-conjugacy class open, even though ambient inner conjugation need not preserve the chosen Frobenius counit. Counting these disjoint open sets gives the following bound, with no hypothesis on the isomorphism classes of subobjects of X .

Theorem 1.3 (Quadratic exchange bound). *Let $\mathcal{C}$ be a Hom-finite $\mathbb{k}$ -linear monoidal category and let $X \neq 0$ be a Frobenius algebra in $\mathcal{C}$. Put*

$$r_X := \dim_{\mathbb{k}} \{h \in \text{End}_{\mathcal{C}}(X) : h^* = h, h \circ e_X = 0\},$$

where $*$ denotes the Frobenius dual. The number of inner-conjugacy classes of separable Frobenius subalgebras, with conjugacy referring to the underlying algebra subobjects, is at most

$$2^{r_X} \leq 2^{\dim_{\mathbb{k}} \text{End}_{\mathcal{C}}(X)-1}.$$

If X is connected, this bounds the number of equivalence classes over X . This is Theorem 8.9.

We first apply the exchange bound to left coideal subalgebras. The regular comodule of a finite-dimensional semisimple Hopf algebra is a connected Frobenius algebra, and the Frobenius-subalgebra correspondence of Ghosh–Palcoux identifies its Frobenius subalgebras with left coideal subalgebras [21, Theorem 11.11]. After proving the required categorical separability, we obtain an effective refinement of the theorem of Etingof–Walton [17, Theorem 3.6].

Corollary 1.4 (Semisimple Hopf algebras). *A semisimple Hopf algebra H over $\mathbb{C}$ with $\dim_{\mathbb{C}} H = N$ has at most 2^{N-1} left coideal subalgebras (Corollary 9.4).*

For a finite-index inclusion $N \subseteq M$ of II_1 factors, the exchange bound controls intermediate von Neumann algebras up to conjugation by unitaries in $N' \cap M$. Under irreducibility, these classes are singletons, and the dimension comparison $\dim_{\mathbb{C}}(N' \cap M_1) \leq [M : N]$ in (85) gives the following consequence.

Corollary 1.5 (Intermediate subfactors). *For every finite-index irreducible II_1 subfactor $N \subset M$,*

$$|\mathcal{L}(N \subset M)| \leq 2^{\dim_{\mathbb{C}}(N' \cap M_1) - 1} \leq 2^{[M:N] - 1}.$$

The estimate is proved in Theorem 10.8. This improves the general whole-lattice estimate $9^{[M:N]}$ of Bakshi–Das–Liu–Ren [4].

§10 also treats unital C^* -inclusions with a conditional expectation of finite Watatani index and a finite-dimensional center for the smaller algebra. It bounds E -compatible intermediates up to unitary inner conjugacy; irreducibility gives an actual count and a corresponding index bound. These C^* -algebra bounds also apply to E -compatible intermediate von Neumann algebras under the same hypotheses; the separate subfactor statements express the bounds in terms of the Jones index (see Remark 10.5).

The quantitative estimates imply qualitative finiteness. Their common input is the low-degree separability contraction: degree two gives open orbits for algebra structures, while degree one gives open inner-conjugacy orbits for homomorphisms. Combining them for a varying source and embedding proves openness of the separable inner-conjugacy classes in the exchange locus. In the connected case these classes are single points. Affine degree then bounds their number. We retain the qualitative statements to expose the local mechanism and its precise hypotheses.

The paper is organized as follows. §2 fixes the terminology and graphical conventions, and §3 proves the low-degree Hochschild contraction. §4 records the algebraic-geometric tools. §§5–7 treat algebra structures, homomorphisms, and subobjects. §8 develops the exchange locus and the general Frobenius bound. Hopf algebras and finite groups are treated in §9, followed by unitary tensor categories and operator-algebra inclusions in §10. Examples appear in §11, and the scope, limitations, open questions, and comparison with the formal-angle method are discussed in §12.

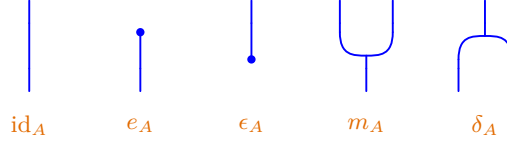
2. SETTING AND TERMINOLOGY

2.1. Categorical conventions. Let $\mathbb{k}$ be an algebraically closed field. We use the terminology of [15] whenever tensor categories are involved. For the local statements, however, $\mathcal{C}$ is only a $\mathbb{k}$ -linear monoidal category whose composition and tensor product are $\mathbb{k}$ -bilinear and whose morphism spaces are finite-dimensional.

We suppress associators and unitors, interpreting all tensor products and composites by monoidal coherence. No braiding is assumed, and the order of the tensor factors is preserved throughout.

Internal detail 3. The coherence convention above is justified by [34, Chapter VII, §2]. All categorical identities below are equations between morphisms. Words such as “point” and “subset” refer only to ordinary finite-dimensional morphism spaces and affine varieties. Generalized elements are used only in explicitly concrete categories.

Diagrams are read from top to bottom, following the conventions of [21]. The basic symbols are displayed below. The counit ϵ_A and comultiplication δ_A are used only when a Frobenius structure has been specified, as in §8. The action symbols are introduced in §2.3.



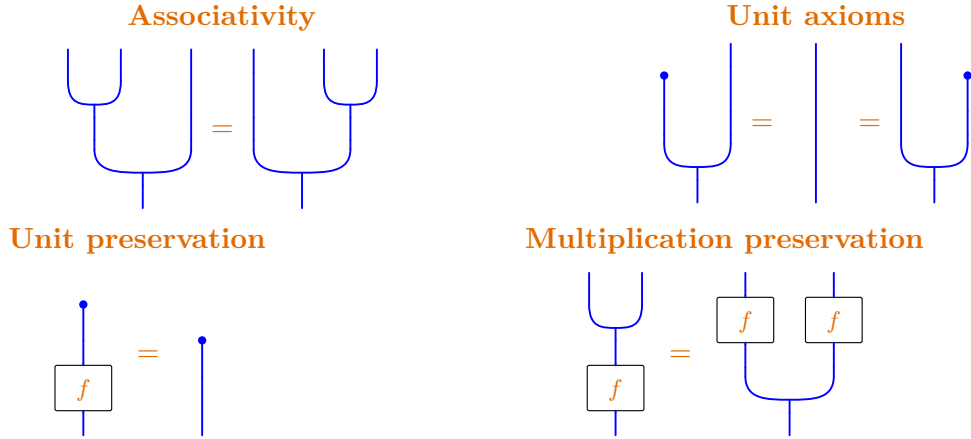
An *algebra object* $A = (A, m_A, e_A)$ consists of morphisms $m_A : A \otimes A \rightarrow A$ and $e_A : \mathbf{1} \rightarrow A$ such that

$$m_A \circ (m_A \otimes \text{id}_A) = m_A \circ (\text{id}_A \otimes m_A), \quad m_A \circ (e_A \otimes \text{id}_A) = \text{id}_A = m_A \circ (\text{id}_A \otimes e_A).$$

A *unital algebra homomorphism* $f : A \rightarrow X$ satisfies

$$f \circ e_A = e_X, \quad f \circ m_A = m_X \circ (f \otimes f).$$

Internal detail 4. The algebra axioms and the unital-homomorphism axioms have the following Ghosh–Palcoux diagrams.

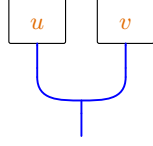
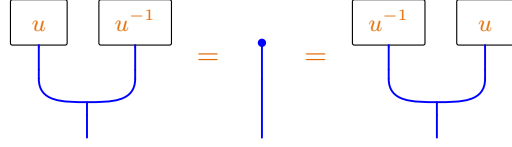


An *algebra monomorphism* is a unital algebra homomorphism whose underlying morphism in $\mathcal{C}$ is a monomorphism, that is, is left-cancellable. An *algebra subobject* of X is a pair (A, i) consisting of an algebra object A and an algebra monomorphism $i : A \rightarrow X$. The pairs (A, i) and (A', i') are *equivalent over X* if there is an algebra isomorphism $h : A \rightarrow A'$ such that $i = i' \circ h$.

The algebra object X is *connected* if $\dim_{\mathbb{k}} \text{Hom}_{\mathcal{C}}(\mathbf{1}, X) = 1$. Its unit morphism is nonzero: if $e_X = 0$, the left-unit identity gives $\text{id}_X = 0$, and then every morphism $\mathbf{1} \rightarrow X$ is zero. Consequently $\text{Hom}_{\mathcal{C}}(\mathbf{1}, X) = \mathbb{k}e_X$.

2.2. The morphism algebra and inner conjugacy. The finite-dimensional vector space $\Gamma(X) := \text{Hom}_{\mathcal{C}}(\mathbf{1}, X)$ is an ordinary unital associative $\mathbb{k}$ -algebra, with product $u \star v := m_X \circ (u \otimes v)$ and unit e_X . Write $\Gamma(X)^\times$ for its group of invertibles $u : \mathbf{1} \rightarrow X$, i.e. those for which there is $u^{-1} : \mathbf{1} \rightarrow X$ satisfying $u \star u^{-1} = e_X = u^{-1} \star u$.

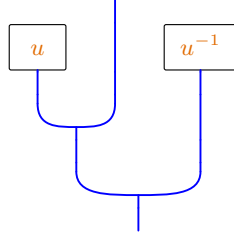
Internal detail 5. The product and inverse equations are pictured as follows; suppressed unitors identify the top boundary with $\mathbf{1}$.

$u \star v$ **The two inverse equations**For such u , define

$$(1) \quad c_u := m_X \circ (m_X \otimes \text{id}_X) \circ (u \otimes \text{id}_X \otimes u^{-1}) : X \longrightarrow X.$$

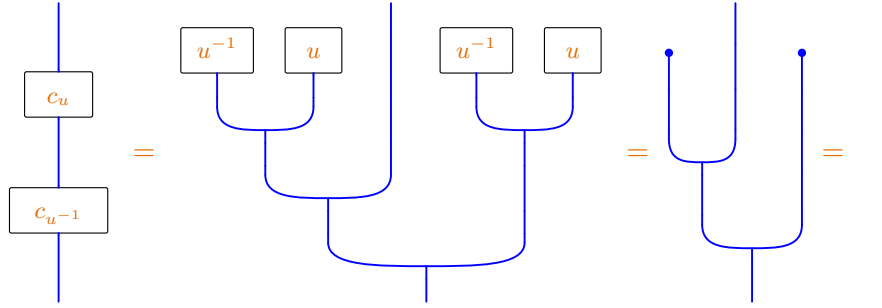
Associativity, unitality, and the inverse identities give

$$c_u \circ e_X = e_X, \quad c_u \circ m_X = m_X \circ (c_u \otimes c_u), \quad c_{u^{-1}} \circ c_u = \text{id}_X = c_u \circ c_{u^{-1}}.$$

Internal detail 6. The conjugation morphism itself is the three-input multiplication tree

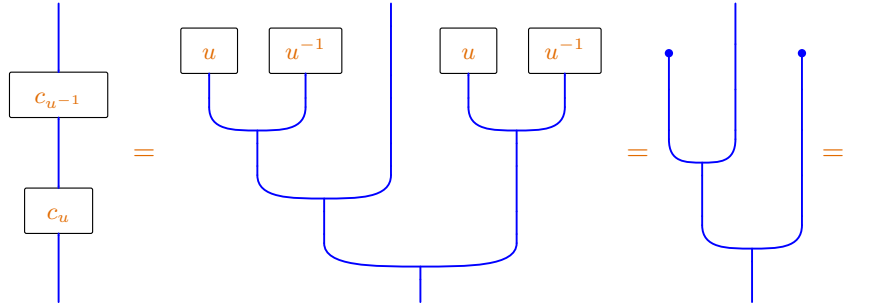
$$c_u = m_X \circ (m_X \otimes \text{id}_X) \circ (u \otimes \text{id}_X \otimes u^{-1}).$$

We now display every reduction separately. For the first inverse identity, the two conjugation coupons are expanded, associativity flattens the five factors, both adjacent inverse pairs are replaced by units, and both units are deleted:



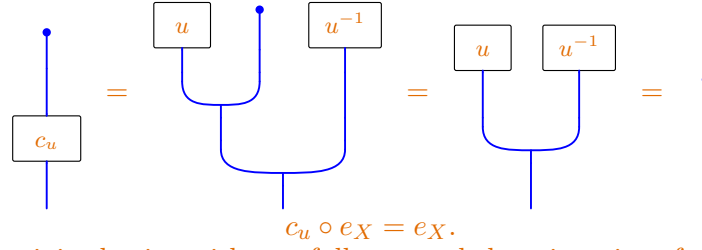
$$c_{u^{-1}} \circ c_u = \text{id}_X.$$

For the second inverse identity the expansion has the other ordered five factors; its two different inverse pairs and its unit deletion are all shown:

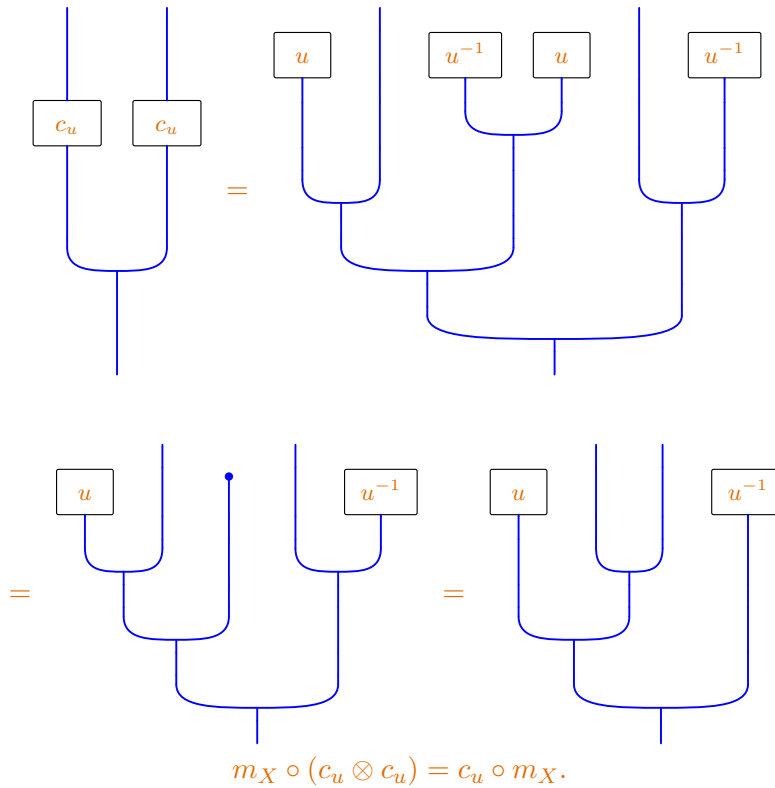


$$c_u \circ c_{u^{-1}} = \text{id}_X.$$

For unit preservation, the input unit is inserted into the conjugation tree; first the middle unit disappears, then the remaining inverse pair becomes the unit:



Finally, multiplicativity begins with two fully expanded conjugations feeding a multiplication. Associativity flattens the six ordered factors; the middle u^{-1}, u pair becomes a unit; deleting it and reassociating the two free inputs produces one conjugation around their product:



Thus unit preservation, multiplication preservation, and both inverse identities have been verified by the displayed planar reductions.

The same identities give $c_{e_X} = \text{id}_X$ and $c_{u \star v} = c_u \circ c_v$. Thus c_u is an algebra automorphism, and $\Gamma(X)^\times$ acts on algebra homomorphisms $f : A \rightarrow X$ by

$$(2) \quad u \cdot f := c_u \circ f.$$

We call the resulting equivalence relation *inner conjugacy*. If X is connected, $\Gamma(X) = \mathbb{k}e_X$, so the action is trivial.

Two algebra subobjects (A, i) and (A', i') are *inner conjugate* if there are $u \in \Gamma(X)^\times$ and an algebra isomorphism $h : A \rightarrow A'$ such that $c_u \circ i = i' \circ h$. For connected X , this is exactly equivalence over X .

2.3. Bimodules and separability. An A – A -bimodule is an object M equipped with morphisms $\ell_M : A \otimes M \rightarrow M$ and $r_M : M \otimes A \rightarrow M$ satisfying

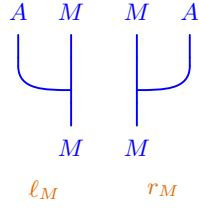
$$\begin{aligned} \ell_M \circ (m_A \otimes \text{id}_M) &= \ell_M \circ (\text{id}_A \otimes \ell_M), & \ell_M \circ (e_A \otimes \text{id}_M) &= \text{id}_M, \\ r_M \circ (r_M \otimes \text{id}_A) &= r_M \circ (\text{id}_M \otimes m_A), & r_M \circ (\text{id}_M \otimes e_A) &= \text{id}_M, \\ r_M \circ (\ell_M \otimes \text{id}_A) &= \ell_M \circ (\text{id}_A \otimes r_M). \end{aligned}$$

A bimodule morphism $q : M \rightarrow N$ satisfies

$$q \circ \ell_M = \ell_N \circ (\text{id}_A \otimes q), \quad q \circ r_M = r_N \circ (q \otimes \text{id}_A).$$

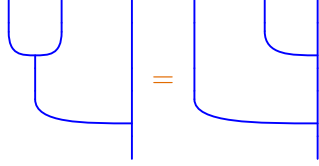
Internal detail 7. The side convention is not cosmetic: it makes the two bimodule-morphism identities literal mirror images.

The left and right action symbols are

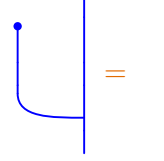


Each bimodule axiom is displayed below, including both handed unit and associativity identities.

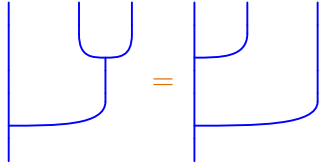
Left associativity



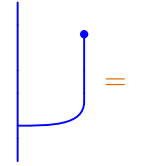
Left unit



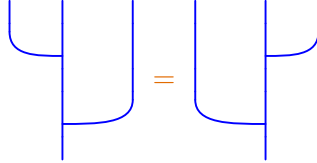
Right associativity



Right unit

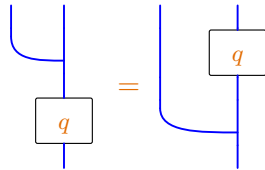


Compatibility of the left and right actions

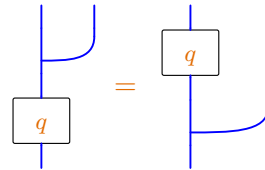


The two equations for a bimodule morphism are also displayed independently.

Left A -linearity of q



Right A -linearity of q



The regular bimodule on A has both actions m_A . On $A \otimes A$ we use the outer regular actions $m_A \otimes \text{id}_A$ and $\text{id}_A \otimes m_A$.

An algebra object A is *separable* if m_A has an A – A -bimodule section $s_A : A \rightarrow A \otimes A$. The following equivalent formulation will be used in the Hochschild contraction.

Lemma 2.1 (Separability morphism). *For an algebra object A , the following are equivalent.*

(i) *There is an A – A -bimodule morphism $s_A : A \rightarrow A \otimes A$ such that $m_A \circ s_A = \text{id}_A$. Its associated separability morphism is*

$$(3) \quad p_A := s_A \circ e_A : \mathbf{1} \longrightarrow A \otimes A.$$

(ii) *There is a morphism $p_A : \mathbf{1} \rightarrow A \otimes A$ such that*

$$(4) \quad m_A \circ p_A = e_A, \quad (m_A \otimes \text{id}_A) \circ (\text{id}_A \otimes p_A) = (\text{id}_A \otimes m_A) \circ (p_A \otimes \text{id}_A).$$

Conversely, the section corresponding to p_A is

$$(5) \quad s_A = (m_A \otimes \text{id}_A) \circ (\text{id}_A \otimes p_A) = (\text{id}_A \otimes m_A) \circ (p_A \otimes \text{id}_A).$$

Proof. Given s_A , put $p_A = s_A \circ e_A$. The section identity gives $m_A \circ p_A = e_A$. Precomposing the left and right A -linearity identities for s_A with $\text{id}_A \otimes e_A$ and $e_A \otimes \text{id}_A$, respectively, shows that both composites in (4) equal s_A .

Conversely, let p_A satisfy (4) and define s_A by either of the equal composites in (5). Associativity proves left and right A -linearity. Moreover,

$$m_A \circ s_A = m_A \circ (\text{id}_A \otimes m_A) \circ (\text{id}_A \otimes p_A) = m_A \circ (\text{id}_A \otimes e_A) = \text{id}_A.$$

Thus s_A is a bimodule section; the unit identities also give $s_A \circ e_A = p_A$. $\square$

Internal detail 8. Here is the detailed algebraic verification of Lemma 2.1. If $p_A = s_A \circ e_A$, then

$$\begin{aligned} (m_A \otimes \text{id}_A) \circ (\text{id}_A \otimes p_A) &= (m_A \otimes \text{id}_A) \circ (\text{id}_A \otimes s_A) \circ (\text{id}_A \otimes e_A) \\ &= s_A \circ m_A \circ (\text{id}_A \otimes e_A) = s_A, \end{aligned}$$

while right linearity gives, independently,

$$\begin{aligned} (\text{id}_A \otimes m_A) \circ (p_A \otimes \text{id}_A) &= (\text{id}_A \otimes m_A) \circ (s_A \otimes \text{id}_A) \circ (e_A \otimes \text{id}_A) \\ &= s_A \circ m_A \circ (e_A \otimes \text{id}_A) = s_A. \end{aligned}$$

Conversely, let $L_p = R_p = s_A$. Using $s_A = L_p$, associativity gives

$$\begin{aligned} s_A \circ m_A &= (m_A \otimes \text{id}_A) \circ (m_A \otimes \text{id}_A \otimes \text{id}_A) \circ (\text{id}_A \otimes \text{id}_A \otimes p_A) \\ &= (m_A \otimes \text{id}_A) \circ (\text{id}_A \otimes m_A \otimes \text{id}_A) \circ (\text{id}_A \otimes \text{id}_A \otimes p_A) \\ &= (m_A \otimes \text{id}_A) \circ (\text{id}_A \otimes s_A). \end{aligned}$$

Using $s_A = R_p$ instead gives

$$\begin{aligned} s_A \circ m_A &= (\text{id}_A \otimes m_A) \circ (\text{id}_A \otimes \text{id}_A \otimes m_A) \circ (p_A \otimes \text{id}_A \otimes \text{id}_A) \\ &= (\text{id}_A \otimes m_A) \circ (\text{id}_A \otimes m_A \otimes \text{id}_A) \circ (p_A \otimes \text{id}_A \otimes \text{id}_A) \\ &= (\text{id}_A \otimes m_A) \circ (s_A \otimes \text{id}_A). \end{aligned}$$

These are exactly the left and right bimodule-morphism equations. Finally, $m_A \circ s_A = m_A \circ (\text{id}_A \otimes m_A) \circ (\text{id}_A \otimes p_A) = m_A \circ (\text{id}_A \otimes e_A) = \text{id}_A$, and $s_A \circ e_A = p_A$ by unitality.

Pictorially, first the definition $p_A = s_A \circ e_A$ and its normalization are the following complete reductions:

Definition of p_A

$$p_A = s_A \circ e_A.$$

Normalization from the section identity

$$m_A \circ p_A = m_A \circ s_A \circ e_A = e_A.$$

The normalization and centrality relations themselves are

$$m_A \circ p_A = e_A.$$

$$L_p = R_p = s_A.$$

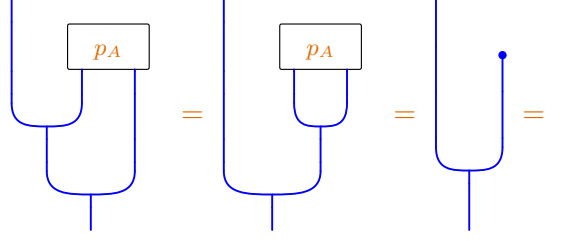
Here are the two centrality reductions from a bimodule section. They are drawn independently, including the unit insertion, the corresponding bimodule-morphism move, and the unit deletion.

Left centrality: $L_p = s_A$
Right centrality: $R_p = s_A$

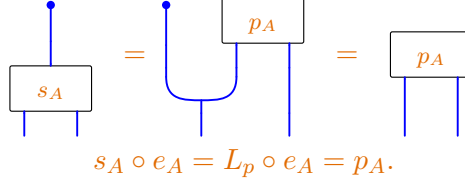
The two bimodule equations for s_A are the following distinct associativity moves:

Left A -linearity
Right A -linearity

Attaching a multiplication at the bottom, reassociating, replacing $m_A \circ p_A$ by e_A , and deleting the unit gives the section identity:



Finally, the equality $s_A \circ e_A = p_A$ in the converse direction is also a three-panel planar reduction:



2.4. Semisimple categories and the subobject hypothesis. Several statements below do not require an abelian category, so we record the precise additive terminology used in the paper.

Definition 2.2. A $\mathbb{k}$ -linear category is *Hom-finite* if every morphism space is finite-dimensional, and *Karoubian* if every idempotent splits. An additive $\mathbb{k}$ -linear category is *Schur* if every indecomposable object S satisfies $\text{End}_{\mathcal{C}}(S) = \mathbb{k}$, and $\text{Hom}_{\mathcal{C}}(S, T) = 0$ for nonisomorphic indecomposable objects S, T . In this paper, a *semisimple $\mathbb{k}$ -linear category* means an additive, Karoubian, Hom-finite Schur category in which every object is a finite direct sum of indecomposable objects; these indecomposable objects are then called simple.

In this setting only finitely many isomorphism classes of underlying objects can occur as subobjects of a fixed object X . Indeed, decompose X into pairwise nonisomorphic simple objects with finite multiplicities. A monomorphism into X induces an injection on each space $\text{Hom}_{\mathcal{C}}(S, -)$ with S simple, so its source contains no additional simple object and no larger multiplicity. This assertion concerns the isomorphism class of the source, not the equivalence class of its embedding over X .

Internal detail 9. Explicitly, if $X \cong \bigoplus_{r=1}^t S_r^{\oplus n_r}$ with pairwise nonisomorphic simple objects S_r , then every subobject is isomorphic to $\bigoplus_{r=1}^t S_r^{\oplus m_r}$ with $0 \leq m_r \leq n_r$. Hence X has at most $\prod_{r=1}^t (n_r + 1)$ isomorphism classes of subobjects. The later subobject theorem uses only this finiteness, not semisimplicity itself.

3. LOW-DEGREE HOCHSCHILD CONTRACTION

Throughout this section, $\mathcal{C}$ is a Hom-finite $\mathbb{k}$ -linear monoidal category; no abelian or rigid structure is used. Let A be an algebra object and M an A - A -bimodule. Put $C^n(A, M) := \text{Hom}_{\mathcal{C}}(A^{\otimes n}, M)$ for $n \geq 0$, where $A^{\otimes 0} = \mathbf{1}$. We write $d^n : C^n(A, M) \rightarrow C^{n+1}(A, M)$ for the Hochschild differential in degree n . These cochain operators are distinct from the geometric differential d of a regular map used later.

Internal detail 10. Thus the four cochain spaces used in the calculation are $C^0(A, M) = \text{Hom}_{\mathcal{C}}(\mathbf{1}, M)$, $C^1(A, M) = \text{Hom}_{\mathcal{C}}(A, M)$, $C^2(A, M) = \text{Hom}_{\mathcal{C}}(A \otimes A, M)$, and $C^3(A, M) = \text{Hom}_{\mathcal{C}}(A^{\otimes 3}, M)$.

The three operators needed here are

$$(6) \quad d^0 z = \ell_M \circ (\text{id}_A \otimes z) - r_M \circ (z \otimes \text{id}_A),$$

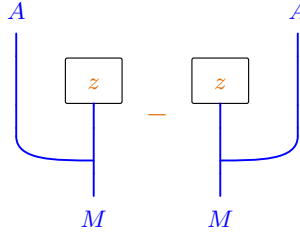
$$(7) \quad d^1 f = \ell_M \circ (\text{id}_A \otimes f) - f \circ m_A + r_M \circ (f \otimes \text{id}_A),$$

$$(8) \quad d^2 \varphi = \ell_M \circ (\text{id}_A \otimes \varphi) - \varphi \circ (m_A \otimes \text{id}_A) + \varphi \circ (\text{id}_A \otimes m_A) - r_M \circ (\varphi \otimes \text{id}_A).$$

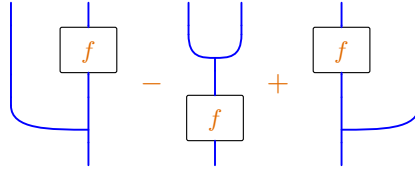
The algebra and bimodule axioms give $d^1 \circ d^0 = 0$ and $d^2 \circ d^1 = 0$. For $n = 1, 2$, write $Z^n(A, M) := \ker d^n$, $B^n(A, M) := \text{im } d^{n-1}$, and $\text{HH}^n(A, M) := Z^n(A, M)/B^n(A, M)$. Thus $Z^1(A, M)$ consists of derivations and $B^1(A, M)$ of inner derivations; elements of $Z^2(A, M)$ are 2-cocycles and elements of $B^2(A, M)$ are 2-coboundaries. These are kernels, images, and quotients of ordinary finite-dimensional $\mathbb{k}$ -vector spaces, so their definition does not require kernels or cokernels in $\mathcal{C}$. In fact, the calculations in this section use neither Hom-finiteness nor algebraic closedness; these assumptions enter the later geometric arguments.

Internal detail 11. The three differentials are the following signed sums of planar pictures. Every summand is drawn separately; the left and right action vertices retain the side convention fixed above.

The two summands of $d^0 z$

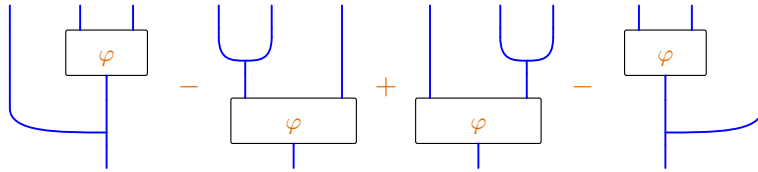


The three summands of $d^1 f$



$$\ell_M \circ (\text{id}_A \otimes f) - f \circ m_A + r_M \circ (f \otimes \text{id}_A).$$

The four summands of $d^2 \varphi$



$$\ell_M \circ (\text{id}_A \otimes \varphi) - \varphi \circ (m_A \otimes \text{id}_A) + \varphi \circ (\text{id}_A \otimes m_A) - r_M \circ (\varphi \otimes \text{id}_A).$$

If one substitutes the first row into the second and then the second into the third, associativity, the two module-associativity axioms, and action compatibility pair every resulting planar tree with the identical tree of opposite sign. This gives the diagrammatic check of $d^1 \circ d^0 = 0$ and $d^2 \circ d^1 = 0$.

The contraction below is the categorical form of the standard separability homotopy; compare [2, Theorem 1.30]. Only degrees one and two are needed here, and the argument in these degrees is an identity of morphisms that does not require an abelian ambient category.

Proposition 3.1 (Low-degree separability vanishing). *If A is separable, then for every A - A -bimodule M , $\text{HH}^1(A, M) = \text{HH}^2(A, M) = 0$.*

Proof. Choose a separability morphism $p_A : \mathbf{1} \rightarrow A \otimes A$ as in Lemma 2.1. For $n \geq 1$ define

$$(9) \quad H_n : C^n(A, M) \longrightarrow C^{n-1}(A, M), \quad H_n(\psi) := \ell_M \circ (\text{id}_A \otimes \psi) \circ (p_A \otimes \text{id}_A^{\otimes(n-1)}).$$

In particular,

$$(10) \quad H_1(f) = \ell_M \circ (\text{id}_A \otimes f) \circ p_A,$$

$$(11) \quad H_2(\varphi) = \ell_M \circ (\text{id}_A \otimes \varphi) \circ (p_A \otimes \text{id}_A).$$

Internal detail 12. The contractions are obtained by inserting p_A at the left and letting its first leg act on the cochain output:

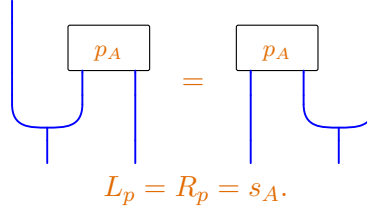


To verify the contraction identities, put

$$L_p := (m_A \otimes \text{id}_A) \circ (\text{id}_A \otimes p_A), \quad R_p := (\text{id}_A \otimes m_A) \circ (p_A \otimes \text{id}_A).$$

By Lemma 2.1, $L_p = R_p = s_A$.

Internal detail 13. The two maps L_p and R_p are the two centrality pictures:



Contraction lemma. For $f \in C^1(A, M)$ and $\varphi \in C^2(A, M)$, one has

$$(12) \quad d^0(H_1(f)) = \ell_M \circ (\text{id}_A \otimes f) \circ L_p - r_M \circ (H_1(f) \otimes \text{id}_A),$$

$$(13) \quad H_2(d^1 f) = f - \ell_M \circ (\text{id}_A \otimes f) \circ R_p + r_M \circ (H_1(f) \otimes \text{id}_A),$$

$$(14) \quad d^1(H_2(\varphi)) = \ell_M \circ (\text{id}_A \otimes \varphi) \circ (L_p \otimes \text{id}_A) - H_2(\varphi) \circ m_A + r_M \circ (H_2(\varphi) \otimes \text{id}_A),$$

$$(15) \quad H_3(d^2 \varphi) = \varphi - \ell_M \circ (\text{id}_A \otimes \varphi) \circ (R_p \otimes \text{id}_A) + H_2(\varphi) \circ m_A - r_M \circ (H_2(\varphi) \otimes \text{id}_A).$$

For the first and third identities, substitute the definitions of d^0 and d^1 and use left-module associativity to expose L_p in the left-action term. In the second and fourth identities, expand $H_2(d^1 f)$ and $H_3(d^2 \varphi)$, respectively. Left-module associativity, $m_A \circ p_A = e_A$, and left unitality turn the first summand into f or φ . The middle summands follow by functoriality of the tensor product and the definition of R_p ; compatibility of the two actions moves the final right action outside H_n . Thus these four expansions use normalization and the bimodule axioms. Centrality, namely $L_p = R_p$, identifies the terms that cancel when the equations are added.

Internal detail 14. For a term-by-term algebraic check, substitute $H_n(\psi) = \ell_M \circ (\text{id}_A \otimes \psi) \circ (p_A \otimes \text{id}_A^{\otimes(n-1)})$. In the first and third formulas, the left-action term is re-associated to expose

L_p , while the right-action term is already the displayed one. In the second formula, the three summands of $d^1 f$ become

$$f, \quad -\ell_M \circ (\text{id}_A \otimes f) \circ R_p, \quad r_M \circ (H_1(f) \otimes \text{id}_A).$$

The first equality uses left-module associativity, $m_A \circ p_A = e_A$, and left unitality; action compatibility gives the third. In the fourth formula the four summands become

$$\varphi, \quad -\ell_M \circ (\text{id}_A \otimes \varphi) \circ (R_p \otimes \text{id}_A), \quad H_2(\varphi) \circ m_A, \quad -r_M \circ (H_2(\varphi) \otimes \text{id}_A).$$

The displayed calculations below give a separate full planar panel for every summand in all four identities.

Pictorial expansion of (12)

$$d^0(H_1(f)) = \underbrace{\begin{array}{c} \text{---} \\ | \\ \boxed{L_p} \\ | \\ \boxed{f} \\ | \\ \text{---} \end{array}}_{\ell_M \circ (\text{id}_A \otimes f) \circ L_p} - \underbrace{\begin{array}{c} \text{---} \\ | \\ \boxed{H_1(f)} \\ | \\ \text{---} \end{array}}_{r_M \circ (H_1(f) \otimes \text{id}_A)}.$$

Pictorial expansion of (13)

$$H_2(d^1 f) = \underbrace{\begin{array}{c} \text{---} \\ | \\ \boxed{f} \\ | \\ \text{---} \end{array}}_f - \underbrace{\begin{array}{c} \text{---} \\ | \\ \boxed{R_p} \\ | \\ \boxed{f} \\ | \\ \text{---} \end{array}}_{\ell_M \circ (\text{id}_A \otimes f) \circ R_p} + \underbrace{\begin{array}{c} \text{---} \\ | \\ \boxed{H_1(f)} \\ | \\ \text{---} \end{array}}_{r_M \circ (H_1(f) \otimes \text{id}_A)}.$$

Pictorial expansion of (14)

$$d^1(H_2(\varphi)) = \underbrace{\begin{array}{c} \text{---} \\ | \\ \boxed{L_p} \\ | \\ \boxed{\varphi} \\ | \\ \text{---} \end{array}}_{\ell_M \circ (\text{id}_A \otimes \varphi) \circ (L_p \otimes \text{id}_A)} - \underbrace{\begin{array}{c} \text{---} \\ | \\ \boxed{H_2(\varphi)} \\ | \\ \text{---} \end{array}}_{H_2(\varphi) \circ m_A} + \underbrace{\begin{array}{c} \text{---} \\ | \\ \boxed{H_2(\varphi)} \\ | \\ \text{---} \end{array}}_{r_M \circ (H_2(\varphi) \otimes \text{id}_A)}.$$

Pictorial expansion of (15)

$$\begin{aligned}
H_3(d^2\varphi) = & \underbrace{\varphi}_{\varphi} - \underbrace{\begin{array}{c} R_p \\ \varphi \end{array}}_{\ell_M \circ (\text{id}_A \otimes \varphi) \circ (R_p \otimes \text{id}_A)} \\
& + \underbrace{\begin{array}{c} \text{cup} \\ H_2(\varphi) \end{array}}_{H_2(\varphi) \circ m_A} - \underbrace{\begin{array}{c} H_2(\varphi) \\ \text{cap} \end{array}}_{r_M \circ (H_2(\varphi) \otimes \text{id}_A)}.
\end{aligned}$$

The panels above record the four identities. For completeness, here are the source-to-target planar moves for every nontrivial summand that was compressed there into an L_p , R_p , or H_n coupon.

The left-action summand in (12)

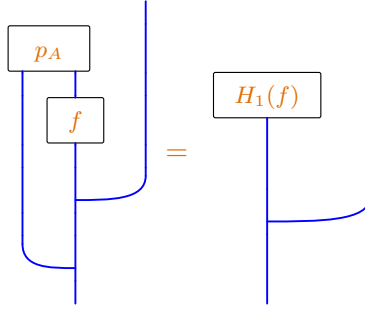
$$\begin{array}{c} \text{cup} \\ p_A \\ f \end{array} = \begin{array}{c} L_p \\ f \end{array}$$

$$\ell_M \circ (\text{id}_A \otimes H_1(f)) = \ell_M \circ (\text{id}_A \otimes f) \circ L_p.$$

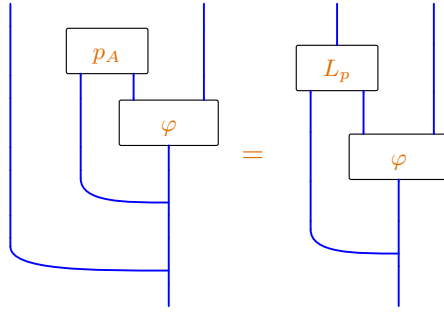
The middle summand in (13)

$$\begin{array}{c} p_A \\ \text{cup} \\ f \end{array} = \begin{array}{c} R_p \\ f \end{array}$$

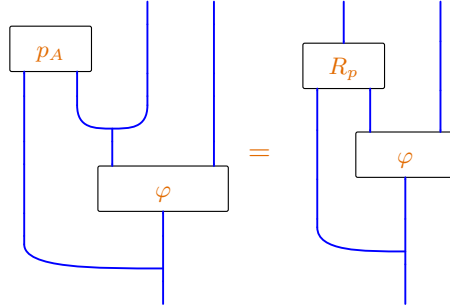
$$H_2(f \circ m_A) = \ell_M \circ (\text{id}_A \otimes f) \circ R_p.$$

The right-action summand in (13)

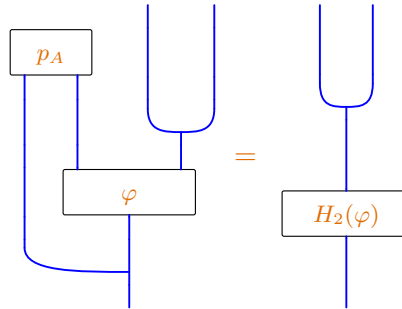
$$H_2(r_M \circ (f \otimes \text{id}_A)) = r_M \circ (H_1(f) \otimes \text{id}_A).$$

The left-action summand in (14)

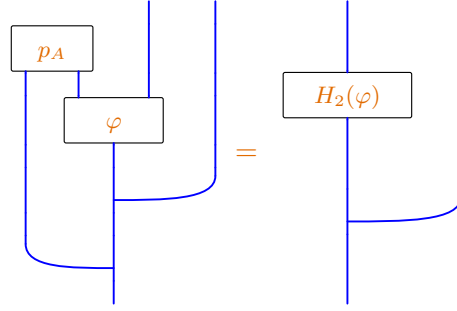
$$\ell_M \circ (\text{id}_A \otimes H_2(\varphi)) = \ell_M \circ (\text{id}_A \otimes \varphi) \circ (L_p \otimes \text{id}_A).$$

The second summand in (15)

$$H_3(\varphi \circ (m_A \otimes \text{id}_A)) = \ell_M \circ (\text{id}_A \otimes \varphi) \circ (R_p \otimes \text{id}_A).$$

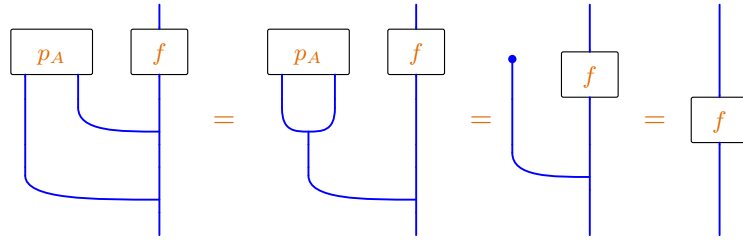
The third summand in (15)

$$H_3(\varphi \circ (\text{id}_A \otimes m_A)) = H_2(\varphi) \circ m_A.$$

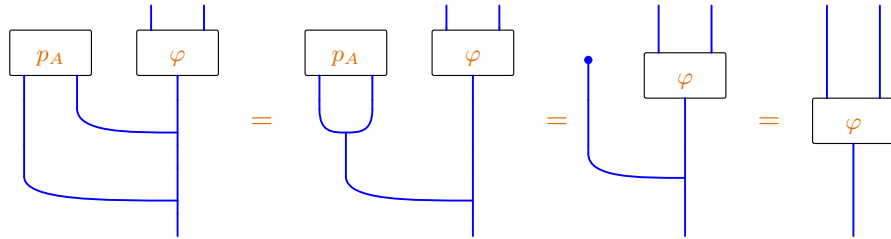
The right-action summand in (15)

$$H_3(r_M \circ (\varphi \otimes \text{id}_A)) = r_M \circ (H_2(\varphi) \otimes \text{id}_A).$$

For completeness, the isolated first term in (13) reduces through all three local moves as follows:



The isolated first term in (15) has its own four-panel reduction, with both inputs of φ drawn throughout:



The L_p and R_p panels are identified by the already displayed centrality slide; the equal right-action panels cancel literally. Thus every summand and every local reduction in (12)–(15) is represented above.

Adding (12) to (13), and (14) to (15), gives

$$(16) \quad d^0 \circ H_1 + H_2 \circ d^1 = \text{id}_{C^1(A,M)},$$

$$(17) \quad d^1 \circ H_2 + H_3 \circ d^2 = \text{id}_{C^2(A,M)}.$$

Consequently $d^1 f = 0$ implies $f = d^0(H_1(f))$, while $d^2 \varphi = 0$ implies $\varphi = d^1(H_2(\varphi))$. Hence both cohomology groups vanish. $\square$

4. AFFINE GEOMETRY AND DEGREE

This section recalls the elementary algebraic geometry used below; see [11, Chapters 1–4 and 9] for the affine notions and [7, Proposition 1.8] for algebraic-group orbits in arbitrary characteristic. An affine variety over $\mathbb{k}$ means here a reduced common zero locus in some affine space $\mathbb{A}_{\mathbb{k}}^n$; it need not be irreducible. Its Zariski-closed subsets are the common zero loci of polynomial families. A nonempty space is *irreducible* if it is not the union of two proper closed subsets, and the

irreducible components of a variety are its maximal irreducible closed subsets; there are finitely many of them. We denote their set by $\text{Irr}(V)$.

A *quasi-affine variety* is an open subvariety of an affine variety, or equivalently a locally closed subvariety of an affine space. The *dimension* of a nonempty (quasi-)affine variety V is the largest integer r for which there is a chain

$$Z_0 \subsetneq Z_1 \subsetneq \cdots \subsetneq Z_r \subseteq V$$

of nonempty irreducible closed subsets of V . Its *local dimension* at $v \in V$ is

$$\dim_v V := \min_{U \ni v} \dim U = \max\{\dim C : C \in \text{Irr}(V), v \in C\},$$

where U ranges over open neighborhoods of v in V ; equivalently, $\dim_v V$ is the Krull dimension of the local ring $\mathcal{O}_{V,v}$. For affine V , $\dim V$ is the Krull dimension of $\mathbb{k}[V]$; see [7, AG, §§1.4, 3.8, and 9.1].

For affine V , if the radical ideal $I(V) \subseteq \mathbb{k}[x_1, \dots, x_n]$ is generated by $q_1, \dots, q_s$ and $v \in V$, its Zariski tangent space is

$$T_v V = \{w \in \mathbb{k}^n : Dq_j(v)w = 0 \text{ for } 1 \leq j \leq s\}, \quad Dq(v)w := \sum_{i=1}^n \frac{\partial q}{\partial x_i}(v)w_i.$$

Here $Dq(v)$ is the differential of the polynomial q at v . Let $R = \mathbb{k}[\varepsilon]/(\varepsilon^2)$, where ε is a formal symbol with $\varepsilon^2 = 0$. Every element of R is uniquely $a + \varepsilon b$, and

$$q(v + \varepsilon w) = q(v) + \varepsilon Dq(v)w.$$

Thus a tangent vector is a first-order lift $v + \varepsilon w$ satisfying all equations in $I(V)$ over R . Terms containing two or more copies of ε vanish; this is the linearization used below. The definition is independent of the chosen generators of $I(V)$. For a quasi-affine variety, the tangent space is computed in any affine open neighborhood of the point. Linearizing equations defining V only as a zero set always gives necessary conditions on tangent vectors, but these need not be sufficient. The local dimension satisfies $\dim_v V \leq \dim_{\mathbb{k}} T_v V$, with equality precisely when v is smooth.

Internal detail 15. If $I(V) \subseteq \mathbb{k}[x_1, \dots, x_n]$ is the radical ideal of V and $\mathfrak{m}_v$ is the maximal ideal of v , then $T_v V \cong \text{Hom}_{\mathbb{k}}(\mathfrak{m}_v/\mathfrak{m}_v^2, \mathbb{k})$. This shows both that the Jacobian definition is independent of a selected generating set and that dual-number lifts give the same tangent space. A point is smooth when its local ring $\mathcal{O}_{V,v}$ is regular, equivalently when its Krull dimension equals $\dim_{\mathbb{k}} \mathfrak{m}_v/\mathfrak{m}_v^2$. The symbol ε records first-order coefficients; no nonzero nilpotent element of $\mathbb{k}$ is chosen. The calculations below take place in morphism spaces tensored with R , with composition and tensor product extended bilinearly, and require no new object of $\mathcal{C}$. For example, the reduced zero set of x^2 is $\{0\}$, with radical ideal (x) and zero tangent space. Linearizing only x^2 gives no restriction. This explains why we first obtain an inclusion in the linearized solution space and then use the orbit differential to obtain equality.

Algebraic groups are understood here as reduced algebraic varieties with regular multiplication and inversion; over $\mathbb{k}$ they are smooth [7, Proposition 1.2]. All groups used below are affine. An orbit in an affine variety is smooth and locally closed [7, Proposition 1.8]; with its reduced induced structure it is therefore a quasi-affine variety.

Lemma 4.1 (Surjective orbit differential). *Let an algebraic group G over $\mathbb{k}$ act algebraically on an affine variety V , let $v \in V$, and let $\theta_v : G \rightarrow V$, $g \mapsto g \cdot v$, be the orbit map. If $(d\theta_v)_{1_G} : T_{1_G} G \rightarrow T_v V$ is surjective, then $G \cdot v$ is open in V . Consequently V has only finitely many orbits satisfying this differential-surjectivity condition.*

Proof. Let G° be the identity component, put $O = G^\circ \cdot v$, and let $Z = \overline{O}$. The subgroup G° is open and irreducible; see [7, Proposition 1.2, pp. 46–47]. Restricting the orbit map to G° therefore leaves its differential at 1_G unchanged. Since the restricted map factors through O ,

$$\mathrm{im}(\mathrm{d}\theta_v)_{1_G} \subseteq T_v O \subseteq T_v V.$$

Surjectivity gives $T_v O = T_v V$. The orbit O is smooth and open in the irreducible variety Z by [7, Proposition 1.8, p. 53]. Hence

$$\dim Z = \dim O = \dim_{\mathbb{k}} T_v O = \dim_{\mathbb{k}} T_v V.$$

Because $Z \subseteq V$ passes through v , we also have

$$\dim Z \leq \dim_v V \leq \dim_{\mathbb{k}} T_v V,$$

where the second inequality is [7, AG, §3.9 and Theorem 17.1]. Equality follows throughout. Thus v is a smooth point of V and lies on a unique irreducible component C ; see [7, AG, Theorem 17.1, p. 40]. Since $Z \subseteq C$ and $\dim Z = \dim C$, we have $Z = C$. Therefore O is open in C .

Removing from V the union of the irreducible components other than C gives an open neighborhood of v contained in C . Intersecting it with O shows that O contains an open neighborhood of v in V ; translating this neighborhood by G° proves that O is open in V . Hence $G \cdot v = \bigcup_{g \in G} gO$ is open as well.

Finally, V has finitely many irreducible components, and an irreducible component meets at most one open orbit because any two nonempty open subsets of an irreducible variety intersect. Thus only finitely many open orbits occur. $\square$

Internal detail 16. The first inclusion in the proof comes from the chain rule for $G^\circ \rightarrow O \hookrightarrow V$; see [7, AG, §16.1, pp. 36–37]. It does not require the orbit map to be separable. Indeed, in characteristic p , the action $t \cdot x = x + t^p$ of $\mathbb{G}_a$ on $\mathbb{A}^1$ has the whole affine line as its orbit, although every orbit differential is zero. Here the surjectivity hypothesis forces the required tangent-space equality.

For completeness, a proper irreducible closed subset of an irreducible finite-dimensional variety has smaller dimension: a chain in the subset can be extended by the whole variety. Thus $Z \subseteq C$ and $\dim Z = \dim C$ imply $Z = C$. Every affine variety of finite type is Noetherian and has finitely many irreducible components; see [7, AG, Proposition 1.2(a) and §3.7]. Finally, if $U \subseteq O$ is open in V and contains v , then gU is open and contains gv for every $g \in G^\circ$. Since $O = \bigcup_{g \in G^\circ} gU$, the orbit is open in V .

Lemma 4.2 (Zero tangent spaces imply finiteness). *If V is an affine variety and $T_v V = 0$ for every $v \in V$, then V is a finite set.*

Proof. The inequality $\dim_v V \leq \dim_{\mathbb{k}} T_v V$ makes V zero-dimensional. Its coordinate ring is therefore an Artinian reduced finitely generated $\mathbb{k}$ -algebra, hence a finite product of copies of $\mathbb{k}$. Thus V has finitely many points. $\square$

Internal detail 17. More explicitly, zero tangent space forces every local ring to have Krull dimension zero. A finitely generated zero-dimensional algebra over a field is Artinian; see [3, Chapter 8]. Reducedness makes its Artinian coordinate ring a product of fields, and the Nullstellensatz identifies each finite field extension of the algebraically closed field $\mathbb{k}$ with $\mathbb{k}$. This is the commutative-algebra proof that a zero-dimensional reduced affine variety over $\mathbb{k}$ is finite.

Bézout inequality. For an irreducible projective variety $Y \subseteq \mathbb{P}_{\mathbb{k}}^n$ of dimension r , its *algebraic degree* $\deg(Y)$ is the length of its intersection with a general linear subspace of codimension r , counting intersection multiplicities. Equivalently, the leading term of its Hilbert polynomial is $\deg(Y)t^r/r!$. The degree of an irreducible affine variety is the degree of its projective closure.

For a reduced affine algebraic set $V \subseteq \mathbb{A}_{\mathbb{k}}^n$, define its *cumulative degree* by

$$(18) \quad \text{cdeg}(V) := \sum_{Z \in \text{Irr}(V)} \deg(\overline{Z}),$$

where $\overline{Z} \subseteq \mathbb{P}_{\mathbb{k}}^n$ is the projective closure of Z . This version of degree records components of every dimension. In particular,

$$(19) \quad |\text{Irr}(V)| \leq \text{cdeg}(V),$$

since every irreducible component has positive integral degree.

The classical projective Bézout equality says that, when pure-dimensional projective varieties $Y, H \subseteq \mathbb{P}^n$ satisfy $\dim Y + \dim H \geq n$ and meet properly,

$$\deg(Y) \deg(H) = \sum_Z i(Z; Y, H) \deg(Z),$$

where Z runs over the irreducible components of $Y \cap H$ and $i(Z; Y, H)$ is the positive intersection multiplicity [23, Theorem 18.4]. In particular, if H is a hypersurface of degree e not containing the irreducible variety Y with $\dim Y \geq 1$, this sum equals $e \deg(Y)$. Discarding multiplicities and components at infinity gives the affine inequality used below.

We use the following standard affine form of Bézout's theorem (cf. [24]).

Theorem 4.3 (Affine Bézout inequality). *Let $V \subseteq \mathbb{A}_{\mathbb{k}}^n$ be the reduced common zero locus of s polynomials of degree at most d , where $d \geq 1$. Then*

$$(20) \quad \text{cdeg}(V) \leq d^{\min\{n, s\}}.$$

Consequently, V has at most $d^{\min\{n, s\}}$ irreducible components.

Proof. Projective hypersurface Bézout, applied to the projective closures of the irreducible components, gives

$$(21) \quad \text{cdeg}(W \cap Z(g)) \leq d \text{cdeg}(W)$$

for every reduced affine algebraic set W and polynomial g of degree at most d . A component contained in $Z(g)$ is retained with its original degree, which also satisfies this estimate because $d \geq 1$. Iteration over the s defining equations yields $\text{cdeg}(V) \leq d^s$.

For the bound in terms of n , we prove by induction on $r = \dim W$ that

$$\text{cdeg}(W \cap V) \leq \deg(W) d^r$$

for every irreducible affine variety W . The claim is immediate when $r = 0$ or $W \subseteq V$. Otherwise, some defining polynomial g of V does not vanish identically on W . Every irreducible component Y of $W \cap Z(g)$ has dimension $r - 1$, and hypersurface Bézout gives $\sum_Y \deg(Y) \leq d \deg(W)$. Since $W \cap V = \bigcup_Y (Y \cap V)$, induction gives

$$\text{cdeg}(W \cap V) \leq \sum_Y \text{cdeg}(Y \cap V) \leq d^{r-1} \sum_Y \deg(Y) \leq d^r \deg(W).$$

Apply this with $W = \mathbb{A}^n$, of degree one, and combine the resulting d^n bound with the d^s bound. $\square$

We record the following result, which will be used later.

Lemma 4.4 (Counting open orbits and isolated points). *Let V be a reduced affine algebraic set.*

- (i) *If an algebraic group acts on V , the number of open orbits is at most $\text{cdeg}(V)$.*
- (ii) *The number of isolated points of V is at most $\text{cdeg}(V)$.*

Proof. An irreducible component meets at most one open orbit: two nonempty open subsets of an irreducible space meet, whereas distinct orbits do not. This proves (i) by (19). An isolated point is a zero-dimensional irreducible component, of degree one, which proves (ii). $\square$

5. ALGEBRA STRUCTURES ON A FIXED OBJECT

Throughout this section, $\mathcal{C}$ is a Hom-finite $\mathbb{k}$ -linear monoidal category, where $\mathbb{k}$ is algebraically closed, and B is a fixed object in $\mathcal{C}$. Put $\mathbf{M}_B := \text{Hom}_{\mathcal{C}}(B \otimes B, B)$ and $\mathbf{U}_B := \text{Hom}_{\mathcal{C}}(\mathbf{1}, B)$. Both are finite-dimensional $\mathbb{k}$ -vector spaces. Let $\text{Alg}(B)$ be the reduced common zero locus in $\mathbf{M}_B \oplus \mathbf{U}_B$ of the polynomial maps encoded by

$$(22) \quad m \circ (m \otimes \text{id}_B) = m \circ (\text{id}_B \otimes m),$$

$$(23) \quad m \circ (e \otimes \text{id}_B) = \text{id}_B = m \circ (\text{id}_B \otimes e).$$

Thus $\text{Alg}(B)$ is an affine variety whose points are precisely the unital algebra structures on the fixed object B . Here “reduced” means that the ideal generated by the coordinate equations is replaced by its radical; this does not change the common zero set over the algebraically closed field $\mathbb{k}$.

Internal detail 18. Choose bases of $\mathbf{M}_B$, $\mathbf{U}_B$, $\text{Hom}_{\mathcal{C}}(B^{\otimes 3}, B)$, and $\text{End}_{\mathcal{C}}(B)$. Because tensor product and composition are bilinear on morphisms, every coordinate of $m \circ (m \otimes \text{id}_B) - m \circ (\text{id}_B \otimes m)$ is a homogeneous quadratic polynomial in the coordinates of m , and every coordinate of the two unit equations is bilinear in (m, e) plus the constant coordinate of $-\text{id}_B$. Taking the reduced common zero locus replaces the generated ideal by its radical but does not change its $\mathbb{k}$ -points. Thus its $\mathbb{k}$ -points are exactly the unital algebra structures on B ; no closure of a smaller parameter set is being taken. This statement about $\mathbb{k}$ -points does not identify the reduced variety with the functor of algebra structures over nonreduced coefficient rings.

The group

$$G_B := \text{Aut}_{\mathcal{C}}(B) = \text{End}_{\mathcal{C}}(B)^{\times}$$

is an algebraic group over $\mathbb{k}$: it is the unit group of a finite-dimensional $\mathbb{k}$ -algebra, hence a smooth Zariski-open subset of $\text{End}_{\mathcal{C}}(B)$, with $T_{\text{id}_B} G_B = \text{End}_{\mathcal{C}}(B)$. It acts by transport of structure,

$$(24) \quad g \cdot (m, e) = (g \circ m \circ (g^{-1} \otimes g^{-1}), g \circ e).$$

Its orbits are exactly the isomorphism classes of unital algebra structures on the object B .

Internal detail 19. For completeness, let $E = \text{End}_{\mathcal{C}}(B)$. Left multiplication $g \mapsto L_g \in \text{End}_{\mathbb{k}}(E)$ is linear. An element g is a unit exactly when $\det L_g \neq 0$: one direction is immediate; in the other, choose $h = L_g^{-1}(\text{id}_B)$, so $g \circ h = \text{id}_B$, and injectivity of L_g applied to $g \circ (h \circ g - \text{id}_B) = 0$ gives $h \circ g = \text{id}_B$. Thus $G_B = \{g : \det L_g \neq 0\}$ is principal open. The formula $g^{-1} = L_g^{-1}(\text{id}_B)$ shows that inversion is regular there, so (24) is a regular action. Finally, $g \cdot (m, e) = (m', e')$ is equivalent to $g \circ e = e'$ and $g \circ m = m' \circ (g \otimes g)$, which is precisely the assertion that $g : (B, m, e) \rightarrow (B, m', e')$ is an algebra isomorphism.

5.1. The tangent calculation with a varying unit. Fix $(m, e) \in \text{Alg}(B)$ and write $A = (B, m, e)$. Every Zariski tangent vector is represented by a pair $(\varphi, v) \in \mathbf{M}_B \oplus \mathbf{U}_B$. Because $\text{Alg}(B)$ is the *reduced* zero locus of the structure equations, such a tangent vector necessarily satisfies their linearizations; we do not need the converse. Linearized associativity is

$$(25) \quad \varphi \circ (m \otimes \text{id}_B) + m \circ (\varphi \otimes \text{id}_B) = \varphi \circ (\text{id}_B \otimes m) + m \circ (\text{id}_B \otimes \varphi),$$

which is precisely $d^2\varphi = 0$ for the regular A -bimodule. The two linearized unit identities are

$$(26) \quad \varphi \circ (e \otimes \text{id}_B) + m \circ (v \otimes \text{id}_B) = 0,$$

$$(27) \quad \varphi \circ (\text{id}_B \otimes e) + m \circ (\text{id}_B \otimes v) = 0.$$

Internal detail 20. To obtain these equations, work over $\mathbb{k}[\varepsilon]/(\varepsilon^2)$ and substitute $m_\varepsilon = m + \varepsilon\varphi$ and $e_\varepsilon = e + \varepsilon v$. The coefficient of ε in $m_\varepsilon \circ (m_\varepsilon \otimes \text{id})$ is $\varphi \circ (m \otimes \text{id}) + m \circ (\varphi \otimes \text{id})$, while that in $m_\varepsilon \circ (\text{id} \otimes m_\varepsilon)$ is $\varphi \circ (\text{id} \otimes m) + m \circ (\text{id} \otimes \varphi)$, yielding (25). Similarly, the ε -coefficient in $m_\varepsilon \circ (e_\varepsilon \otimes \text{id}) = \text{id}$ is (26), and the right unit gives (27). For the regular bimodule, (8) reads

$$d^2\varphi = m \circ (\text{id} \otimes \varphi) - \varphi \circ (m \otimes \text{id}) + \varphi \circ (\text{id} \otimes m) - m \circ (\varphi \otimes \text{id}),$$

so (25) is exactly $d^2\varphi = 0$ after moving two terms across the equality.

Assume now that A is separable. By Proposition 3.1, there is a morphism $h : B \rightarrow B$ with $\varphi = d^1h$. Precomposing (26) with $e : \mathbf{1} \rightarrow B$ gives

$$(28) \quad \varphi \circ (e \otimes e) + m \circ (v \otimes e) = 0.$$

The right-unit identity gives $m \circ (v \otimes e) = v$. Moreover, (7) for the regular bimodule, precomposed with $e \otimes e$, gives

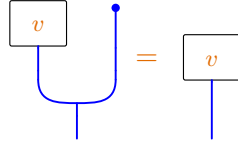
$$(d^1h) \circ (e \otimes e) = h \circ e.$$

Hence

$$(29) \quad h \circ e = -v.$$

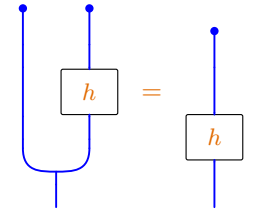
Internal detail 21. The two reductions just used can also be seen without suppressing the unit strings. It is clearest to separate the right-unit reduction from the three signed terms of d^1h .

Right-unit reduction

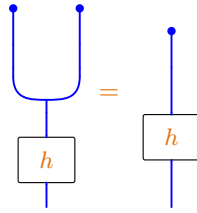


$$m \circ (v \otimes e) = v.$$

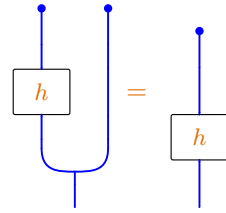
The three terms of d^1h after inserting $e \otimes e$



$$m \circ (e \otimes h \circ e) = h \circ e.$$



$$h \circ m \circ (e \otimes e) = h \circ e.$$



$$m \circ (h \circ e \otimes e) = h \circ e.$$

$$(d^1h) \circ (e \otimes e) = h \circ e - h \circ e + h \circ e = h \circ e.$$

Let

$$\theta_{(m,e)} : G_B \longrightarrow \mathbf{M}_B \oplus \mathbf{U}_B, \quad g \longmapsto g \cdot (m, e)$$

be the orbit map. Its ordinary differential at id_B is

$$(30) \quad (d\theta_{(m,e)})_{\text{id}_B}(f) = (f \circ m - m \circ (f \otimes \text{id}_B) - m \circ (\text{id}_B \otimes f), f \circ e) = (-d^1f, f \circ e).$$

This follows by differentiating multiplication and inversion in the finite-dimensional algebra $\text{End}_{\mathcal{C}}(B)$; it does not require a scalar extension of the object B .

Internal detail 22. Explicitly, the tangent vector $f \in \text{End}_{\mathcal{C}}(B)$ is represented over $\mathbb{k}[\varepsilon]/(\varepsilon^2)$ by $g_{\varepsilon} = \text{id}_B + \varepsilon f$, whose inverse is $\text{id}_B - \varepsilon f$. Expanding

$$g_{\varepsilon} \circ m \circ (g_{\varepsilon}^{-1} \otimes g_{\varepsilon}^{-1})$$

gives

$$m + \varepsilon(f \circ m - m \circ (f \otimes \text{id}_B) - m \circ (\text{id}_B \otimes f)),$$

while $g_{\varepsilon} \circ e = e + \varepsilon(f \circ e)$. This proves (30) and also shows directly why the first component is $-d^1 f$ for the regular bimodule.

Taking $f = -h$, equations (29) and $\varphi = d^1 h$ give

$$(\text{d}\theta_{(m,e)}|_{\text{id}_B})(-h) = (\varphi, v).$$

Thus every tangent vector to $\text{Alg}(B)$ lies in the image of the orbit-map differential. The reverse inclusion is automatic because the orbit map takes values in $\text{Alg}(B)$. We have proved

$$(31) \quad \text{im}(\text{d}\theta_{(m,e)}|_{\text{id}_B}) = T_{(m,e)} \text{Alg}(B)$$

at every separable point.

Theorem 5.1 (Fixed-object rigidity). *Let B be any object of a Hom-finite $\mathbb{k}$ -linear monoidal category. The object B supports only finitely many isomorphism classes of separable unital algebra structures.*

Proof. Equation (31) and Lemma 4.1 show that the G_B -orbit of every separable point of $\text{Alg}(B)$ is open. The same lemma shows that $\text{Alg}(B)$ has only finitely many open G_B -orbits. $\square$

5.2. Effective bounds. For later use, set

$$(32) \quad \begin{aligned} a_B &:= \dim_{\mathbb{k}} \text{Hom}_{\mathcal{C}}(B \otimes B, B), & u_B &:= \dim_{\mathbb{k}} \text{Hom}_{\mathcal{C}}(\mathbf{1}, B), \\ t_B &:= \dim_{\mathbb{k}} \text{Hom}_{\mathcal{C}}(B^{\otimes 3}, B), & E_B &:= \dim_{\mathbb{k}} \text{End}_{\mathcal{C}}(B). \end{aligned}$$

Theorem 5.2 (Fixed-object bound). *The number of isomorphism classes of separable unital algebra structures on B is at most*

$$(33) \quad 2^{\min\{a_B + u_B, t_B + 2E_B\}}.$$

Proof. The variety $\text{Alg}(B)$ lies in an affine space of dimension $a_B + u_B$. Associativity contributes at most t_B scalar quadratic equations, while the two unit identities contribute at most $2E_B$ scalar equations of degree at most two.

Internal detail 23. The phrase “at most” allows linear dependence among coordinates. After a basis of $\text{Hom}_{\mathcal{C}}(B^{\otimes 3}, B)$ is chosen, the associator defect has exactly t_B coordinate functions, each of degree two. Each unit defect takes values in $\text{End}_{\mathcal{C}}(B)$ and therefore has E_B coordinates; these coordinates have total degree at most two in (m, e) . Thus the same reduced zero locus is defined by no more than $t_B + 2E_B$ polynomials of degree at most two in $a_B + u_B$ variables, exactly the two numbers entering the minimum in Theorem 4.3.

Hence Theorem 4.3 gives

$$\text{cdeg}(\text{Alg}(B)) \leq 2^{\min\{a_B + u_B, t_B + 2E_B\}}.$$

For every separable point, equation (31) says that the orbit-map differential is surjective; Lemma 4.1 therefore makes that orbit open. Finally, Lemma 4.4(i) proves the claim. $\square$

Fix now a nonzero morphism $e : \mathbf{1} \rightarrow B$. Let $\text{Alg}_e(B)$ be the fixed-unit slice and let

$$G_{B,e} := \{g \in \text{Aut}_{\mathcal{C}}(B) : g \circ e = e\}$$

be its stabilizer.

The group $G_{B,e}$ is the intersection of the open set G_B with the affine linear space

$$\text{id}_B + \{h \in \text{End}_{\mathcal{C}}(B) : h \circ e = 0\}.$$

It is therefore smooth, with $T_{\text{id}_B} G_{B,e} = \{h \in \text{End}_{\mathcal{C}}(B) : h \circ e = 0\}$.

Internal detail 24. The fixed-unit equations are affine linear, so their ideal is radical. Intersecting their affine solution space with the principal open unit group therefore introduces no additional tangent equations. Substituting $g = \text{id}_B + \varepsilon f$ over $\mathbb{k}[\varepsilon]/(\varepsilon^2)$ gives its Lie algebra exactly as

$$\text{Lie}(G_{B,e}) = \{f \in \text{End}_{\mathcal{C}}(B) : f \circ e = 0\}.$$

This is the identification used in the proof of Proposition 5.3.

Proposition 5.3 (Fixed-unit rigidity). *Every separable point of $\text{Alg}_e(B)$ has an open $G_{B,e}$ -orbit.*

Proof. A tangent vector in the fixed-unit slice has the form $(\varphi, 0)$. The calculation preceding (29) gives $\varphi = d^1 h$ and $h \circ e = 0$. Thus $-h$ belongs to the Lie algebra of $G_{B,e}$, and (30) shows that $(\varphi, 0)$ lies in the image of the stabilizer orbit-map differential. That differential is therefore surjective, so Lemma 4.1 applies. $\square$

Consider the linear map

$$(34) \quad \Lambda_{B,e} : \text{Hom}_{\mathcal{C}}(B \otimes B, B) \longrightarrow \text{End}_{\mathcal{C}}(B)^2, \quad m \longmapsto (m \circ (e \otimes \text{id}_B), m \circ (\text{id}_B \otimes e)),$$

and put

$$(35) \quad q_{B,e} := \dim_{\mathbb{k}} \ker \Lambda_{B,e}.$$

When nonempty, the unit equations cut out an affine space of this dimension.

Theorem 5.4 (Fixed-unit bound). *The number of separable algebra structures on B having unit exactly e , modulo algebra isomorphisms fixing e , is at most*

$$(36) \quad 2^{\min\{q_{B,e}, t_B\}}.$$

If e has a retraction $\pi : B \rightarrow \mathbf{1}$, then

$$(37) \quad q_{B,e} = a_B - 2E_B + u_B.$$

Proof. On the affine unit slice, associativity is a system of at most t_B quadratic equations. Proposition 5.3, Theorem 4.3, and Lemma 4.4(i) give (36).

Assume $\pi \circ e = \text{id}_{\mathbf{1}}$. The image of $\Lambda_{B,e}$ is exactly

$$(38) \quad \{(L, R) \in \text{End}_{\mathcal{C}}(B)^2 : L \circ e = R \circ e\}.$$

The inclusion from left to right follows by evaluation on $e \otimes e$. Conversely, if $L \circ e = R \circ e =: v$, then

$$m = L \circ (\pi \otimes \text{id}_B) + R \circ (\text{id}_B \otimes \pi) - v \circ (\pi \otimes \pi)$$

has $\Lambda_{B,e}(m) = (L, R)$.

Internal detail 25. The evaluation is worth recording. Since $\pi \circ e = \text{id}_{\mathbf{1}}$ and $L \circ e = R \circ e = v$,

$$m \circ (e \otimes \text{id}_B) = L + R \circ e \circ \pi - v \circ \pi = L,$$

$$m \circ (\text{id}_B \otimes e) = L \circ e \circ \pi + R - v \circ \pi = R.$$

Thus the displayed formula really is a preimage of (L, R) and proves the reverse inclusion in (38); no compatibility condition other than $L \circ e = R \circ e$ is hidden.

The map $(L, R) \mapsto L \circ e - R \circ e$ from $\text{End}_{\mathcal{C}}(B)^2$ onto $\text{Hom}_{\mathcal{C}}(\mathbf{1}, B)$ is surjective, since w is the image of $(w \circ \pi, 0)$. Thus the space in (38) has dimension $2E_B - u_B$, and rank-nullity gives (37). $\square$

Corollary 5.5 (Connected fixed object). *Suppose $u_B = 1$ and a nonzero morphism $e : \mathbf{1} \rightarrow B$ has a retraction; this holds, for example, when $\mathcal{C}$ is semisimple with simple tensor unit and B is connected. Then the total number of separable algebra-isomorphism classes on B is at most*

$$(39) \quad 2^{\min\{a_B - 2E_B + 1, t_B\}}.$$

Proof. All nonzero maps $\mathbf{1} \rightarrow B$ are scalar multiples of one another, and a scalar automorphism of B transports any unit to the chosen one. Apply Theorem 5.4.

Internal detail 26. If $e' = \lambda e$ with $\lambda \neq 0$, then the automorphism $g = \lambda^{-1} \text{id}_B$ sends e' to e ; transporting the multiplication at the same time gives an isomorphic algebra in the fixed-unit slice. In a semisimple category with simple tensor unit, write $B \cong \mathbf{1}^{\oplus m} \oplus B'$ with no copy of $\mathbf{1}$ in B' . Since $\text{Hom}_{\mathcal{C}}(\mathbf{1}, B) = \mathbb{k}e$, connectedness forces $m = 1$, and e is a nonzero scalar multiple of the inclusion of that unique $\mathbf{1}$ -summand. It therefore has a retraction. Hence the retraction required in Theorem 5.4 exists in the stated example. $\square$

6. HOMOMORPHISMS AND INNER CONJUGACY

Throughout this section, $\mathcal{C}$ is a Hom-finite $\mathbb{k}$ -linear monoidal category over an algebraically closed field $\mathbb{k}$. Fix a separable algebra $A = (A, m_A, e_A)$ and an arbitrary algebra $X = (X, m_X, e_X)$. Inside the finite-dimensional affine space $\text{Hom}_{\mathcal{C}}(A, X)$, let $\text{Hom}_{\text{alg}}(A, X)$ be the reduced common zero locus of the polynomial maps

$$f \mapsto f \circ e_A - e_X, \quad f \mapsto f \circ m_A - m_X \circ (f \otimes f).$$

Thus its points are precisely the unital algebra morphisms $A \rightarrow X$.

Internal detail 27. After bases are chosen, the unit defect is affine-linear in f , while the multiplicativity defect has coordinates of degree at most two because $f \mapsto f \otimes f$ is quadratic and composition with m_X is linear. For a deformation $f_\varepsilon = f + \varepsilon \xi$ over $\mathbb{k}[\varepsilon]/(\varepsilon^2)$, the ε -coefficient of the unit equation is $\xi \circ e_A$, and

$$m_X \circ (f_\varepsilon \otimes f_\varepsilon) = m_X \circ (f \otimes f) + \varepsilon m_X \circ (\xi \otimes f + f \otimes \xi).$$

Comparison with $f_\varepsilon \circ m_A = f \circ m_A + \varepsilon(\xi \circ m_A)$ gives precisely (40)–(41).

At $f \in \text{Hom}_{\text{alg}}(A, X)$, every Zariski tangent vector ξ necessarily satisfies the linearized equations

$$(40) \quad \xi \circ e_A = 0,$$

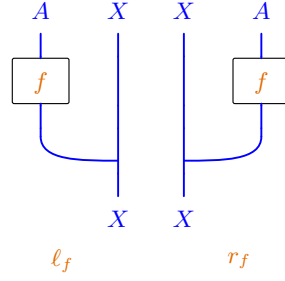
$$(41) \quad \xi \circ m_A = m_X \circ (\xi \otimes f) + m_X \circ (f \otimes \xi).$$

Use f to define actions

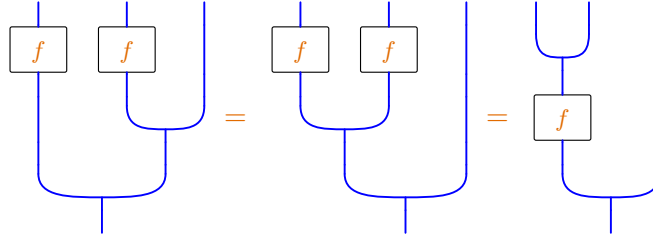
$$(42) \quad \ell_f := m_X \circ (f \otimes \text{id}_X) : A \otimes X \longrightarrow X, \quad r_f := m_X \circ (\text{id}_X \otimes f) : X \otimes A \longrightarrow X.$$

Since f is a unital algebra homomorphism, associativity and unitality of X show that these morphisms make X an A – A -bimodule, denoted ${}_f X_f$.

Internal detail 28. The two actions are

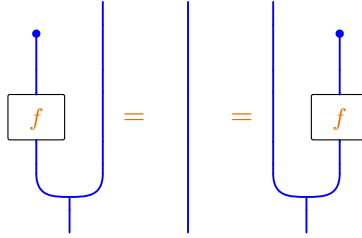


First apply associativity to the two multiplications in X , then use multiplicativity of f . The right module axiom is the exact horizontal reflection, while the compatibility of the two actions is one associativity move in X .



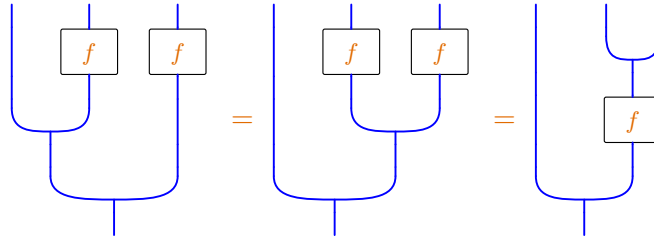
$$\ell_f \circ (\text{id}_A \otimes \ell_f) = \ell_f \circ (m_A \otimes \text{id}_X).$$

Left and right unit axioms

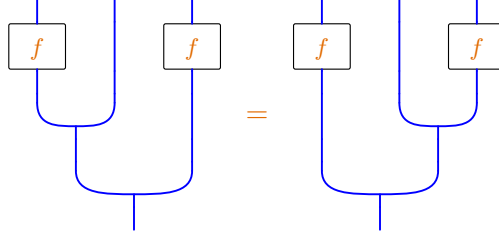


$$\ell_f \circ (e_A \otimes \text{id}_X) = \text{id}_X, \quad r_f \circ (\text{id}_X \otimes e_A) = \text{id}_X.$$

Right associativity



$$r_f \circ (r_f \otimes \text{id}_A) = r_f \circ (\text{id}_X \otimes m_A).$$

Compatibility of the two actions

$$r_f \circ (\ell_f \otimes \text{id}_A) = \ell_f \circ (\text{id}_A \otimes r_f).$$

For comparison with the pictures, all five bimodule identities can also be checked algebraically:

$$\begin{aligned} \ell_f \circ (m_A \otimes \text{id}_X) &= m_X \circ (f \circ m_A \otimes \text{id}_X) = m_X \circ (m_X \otimes \text{id}_X) \circ (f \otimes f \otimes \text{id}_X) \\ &= m_X \circ (\text{id}_X \otimes m_X) \circ (f \otimes f \otimes \text{id}_X) = \ell_f \circ (\text{id}_A \otimes \ell_f), \\ \ell_f \circ (e_A \otimes \text{id}_X) &= m_X \circ (e_X \otimes \text{id}_X) = \text{id}_X, \\ r_f \circ (r_f \otimes \text{id}_A) &= m_X \circ (m_X \otimes \text{id}_X) \circ (\text{id}_X \otimes f \otimes f) = r_f \circ (\text{id}_X \otimes m_A), \\ r_f \circ (\text{id}_X \otimes e_A) &= m_X \circ (\text{id}_X \otimes e_X) = \text{id}_X, \\ r_f \circ (\ell_f \otimes \text{id}_A) &= m_X \circ (m_X \otimes \text{id}_X) \circ (f \otimes \text{id}_X \otimes f) \\ &= m_X \circ (\text{id}_X \otimes m_X) \circ (f \otimes \text{id}_X \otimes f) = \ell_f \circ (\text{id}_A \otimes r_f). \end{aligned}$$

The first and third lines use multiplicativity of f and associativity of m_X ; the second and fourth use unitality of f and X ; the last uses only associativity of m_X .

With these actions, equation (41) is precisely the derivation identity. The unit equation (40) is part of the linearized definition of the homomorphism variety. Conversely, every derivation satisfies it automatically: evaluating the derivation identity on $e_A \otimes e_A$ gives $\xi \circ e_A = 2\xi \circ e_A$, hence $\xi \circ e_A = 0$. Therefore

Internal detail 29. In the pulled-back bimodule, $d^1\xi = 0$ means

$$m_X \circ (f \otimes \xi) - \xi \circ m_A + m_X \circ (\xi \otimes f) = 0,$$

which is exactly (41). Evaluating at $e_A \otimes e_A$ uses $m_A \circ (e_A \otimes e_A) = e_A$ and $f \circ e_A = e_X$ to give

$$0 = \xi \circ e_A - \xi \circ e_A + \xi \circ e_A = \xi \circ e_A$$

when the terms are placed on one side; equivalently the displayed tangent identity gives $\xi \circ e_A = 2\xi \circ e_A$. Thus the unit condition is indeed automatic for every derivation.

$$(43) \quad T_f \text{Hom}_{\text{alg}}(A, X) \subseteq Z^1(A, {}_f X_f).$$

The unit group $\Gamma(X)^\times$ is a smooth affine algebraic group over $\mathbb{k}$, with $T_{e_X} \Gamma(X)^\times = \Gamma(X)$. Indeed, for a polynomial q on a vector space V , write $D(q) := \{v \in V : q(v) \neq 0\}$. Left multiplication in the finite-dimensional $\mathbb{k}$ -algebra $\Gamma(X)$ gives a linear map

$$L : \Gamma(X) \longrightarrow \text{End}_{\mathbb{k}}(\Gamma(X)), \quad u \longmapsto L_u,$$

Put $q := \det \circ L$. If u is invertible, then $q(u) \neq 0$. Conversely, if $q(u) \neq 0$, set $v = L_u^{-1}(e_X)$, so $u \star v = e_X$. Associativity gives $L_u(v \star u - e_X) = 0$, and the injectivity of L_u gives $v \star u = e_X$. Thus $\Gamma(X)^\times = D(q)$. Multiplication is polynomial, while

$$u^{-1} = L_u^{-1}(e_X)$$

is regular on $D(q)$, which is open in the vector space $\Gamma(X)$. Hence $\Gamma(X)^\times$ is a smooth affine algebraic group, and its conjugation action (2) is algebraic.

Internal detail 30. If u is a unit, $L_{u^{-1}}$ is the inverse of L_u . Conversely, if L_u is invertible, put $v = L_u^{-1}(e_X)$; then $u \star v = e_X$. Moreover $u \star (v \star u - e_X) = (u \star v) \star u - u \star e_X = 0$, and injectivity of L_u gives $v \star u = e_X$. Hence $v = u^{-1}$. This proves the claimed equivalence without appealing to any general ring-theoretic result.

For $f \in \text{Hom}_{\text{alg}}(A, X)$, let

$$\omega_f : \Gamma(X)^\times \longrightarrow \text{Hom}_{\text{alg}}(A, X), \quad u \longmapsto u \cdot f$$

be the orbit map. Its Zariski differential at the identity e_X is

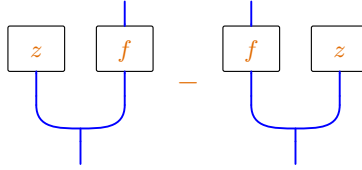
$$(44) \quad (d\omega_f)_{e_X}(z) = m_X \circ (z \otimes f) - m_X \circ (f \otimes z) : A \longrightarrow X.$$

Internal detail 31. Represent $z \in T_{e_X} \Gamma(X)^\times = \Gamma(X)$ over $\mathbb{k}[\varepsilon]/(\varepsilon^2)$ by $u_\varepsilon = e_X + \varepsilon z$, whose inverse is $e_X - \varepsilon z$. Expanding $c_{u_\varepsilon} \circ f = u_\varepsilon \star f \star u_\varepsilon^{-1}$ gives

$$f + \varepsilon(m_X \circ (z \otimes f) - m_X \circ (f \otimes z)),$$

which proves (44). Formula (6) gives $d^0 z = m_X \circ (f \otimes z) - m_X \circ (z \otimes f)$, explaining the minus sign.

Internal detail 32. The two terms of the differential are obtained by placing the 0-cochain z on the two sides of the f -strand:



$$m_X \circ (z \otimes f) - m_X \circ (f \otimes z) = -d^0 z.$$

Formula (6) for the actions (42) shows that (44) is $-d^0 z$. Thus the image of the orbit-map differential is $B^1(A, {}_f X_f)$, and it is contained in the tangent space of the homomorphism variety. Separability and Proposition 3.1 give the following chain of inclusions and equalities:

$$(45) \quad \text{im}(d\omega_f)_{e_X} = B^1(A, {}_f X_f) \subseteq T_f \text{Hom}_{\text{alg}}(A, X) \subseteq Z^1(A, {}_f X_f) = B^1(A, {}_f X_f).$$

Theorem 6.1 (Homomorphism rigidity). *Let A be a separable algebra object and X any algebra object in a Hom-finite $\mathbb{k}$ -linear monoidal category.*

- (i) *There are only finitely many inner-conjugacy classes of unital algebra homomorphisms $A \rightarrow X$.*
- (ii) *If X is connected, then $\text{Hom}_{\text{alg}}(A, X)$ is a finite set.*

Proof. Equation (45) says that every orbit-map differential is surjective. Lemma 4.1 therefore shows that every $\Gamma(X)^\times$ -orbit is open in $\text{Hom}_{\text{alg}}(A, X)$, and there are only finitely many such orbits.

If X is connected, then $\Gamma(X) = \mathbb{k}e_X$ and inner conjugation is trivial. Consequently every orbit is a singleton, so the homomorphism set is finite. Equivalently, every inner derivation is zero; (45) then gives zero tangent space at every point, and Lemma 4.2 applies. $\square$

Internal detail 33. Part (i) is the natural statement for a nonconnected ambient algebra. It should not be weakened to a mere assertion that individual homomorphisms are locally rigid: their only possible infinitesimal motion is precisely the visible motion produced by conjugation with invertible morphisms $1 \rightarrow X$.

Effective bound. The unit condition cuts out either the empty set or an affine translate of

$$(46) \quad K_{A,X} := \ker(\mathrm{Hom}_{\mathcal{C}}(A, X) \longrightarrow \mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, X), f \longmapsto f \circ e_A).$$

Internal detail 34. Suppose there is a unital map $f_0 : A \rightarrow X$. Then all unit-preserving underlying morphisms from A to X are given by $f_0 + K_{A,X}$.

Set

$$(47) \quad b_{A,X} := \dim_{\mathbb{k}} K_{A,X}, \quad c_{A,X} := \dim_{\mathbb{k}} \mathrm{Hom}_{\mathcal{C}}(A \otimes A, X).$$

Theorem 6.2 (Effective homomorphism rigidity). *The number of inner-conjugacy classes of unital algebra homomorphisms $A \rightarrow X$ is at most*

$$(48) \quad 2^{\min\{b_{A,X}, c_{A,X}\}}.$$

If X is connected, the same expression bounds the cardinality of $\mathrm{Hom}_{\mathrm{alg}}(A, X)$. If, in addition, this set is nonempty, then

$$(49) \quad b_{A,X} = \dim_{\mathbb{k}} \mathrm{Hom}_{\mathcal{C}}(A, X) - 1.$$

Proof. If the affine unit slice is empty, the assertion is immediate. Otherwise, inside this slice of dimension $b_{A,X}$, multiplicativity is a system of at most $c_{A,X}$ quadratic equations. Its inner-conjugacy orbits are open by (45) and Lemma 4.1. The bound follows from Theorem 4.3 and Lemma 4.4(i). If X is connected, inner conjugation is trivial. Finally, when a unital map exists, the linear map in (46) has nonzero image in the one-dimensional space $\mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, X)$, hence rank one. Thus rank–nullity proves (49). $\square$

7. SEPARABLE ALGEBRA SUBOBJECTS

Let X be an algebra object in a Hom-finite $\mathbb{k}$ -linear monoidal category over an algebraically closed field $\mathbb{k}$. We now combine the two local results; the only additional global input is that only finitely many isomorphism classes of objects occur as subobjects of X .

Theorem 7.1 (Finiteness of separable algebra subobjects). *Let X be an algebra object in a Hom-finite $\mathbb{k}$ -linear monoidal category. Assume that only finitely many isomorphism classes of objects occur as subobjects of X .*

- (i) *The separable algebra subobjects of X form only finitely many inner-conjugacy classes.*
- (ii) *If X is connected, they form only finitely many equivalence classes over X .*

The algebra X itself need not be separable.

Proof. Choose representatives $B_1, \dots, B_N$ for the isomorphism classes of objects occurring as subobjects of X . By Theorem 5.1, each B_r supports only finitely many separable algebra structures up to algebra isomorphism.

Let (A, i) be a separable algebra subobject whose underlying object is isomorphic to B_r , and choose an object isomorphism $\phi : B_r \xrightarrow{\sim} A$. Transport the algebra structure and the embedding to B_r by

$$m_{B_r} := \phi^{-1} \circ m_A \circ (\phi \otimes \phi), \quad e_{B_r} := \phi^{-1} \circ e_A, \quad i_{B_r} := i \circ \phi.$$

Then $\phi : (B_r, m_{B_r}, e_{B_r}) \rightarrow (A, m_A, e_A)$ is an algebra isomorphism, i_{B_r} is an algebra monomorphism, and separability is preserved. Running over all r therefore leaves a finite list of representative separable source algebras $A_1, \dots, A_s$.

For each fixed A_j , Theorem 6.1(i) gives only finitely many inner-conjugacy classes of algebra maps $A_j \rightarrow X$. Restricting to monomorphisms and taking the finite union over j proves (i). If X is connected, inner conjugacy is trivial, and (ii) follows.

Internal detail 35. No quotient by source automorphisms has been overlooked. Every subobject (A, i) is first transported by an object isomorphism to one of the chosen source objects and

then by an algebra isomorphism to one of the finite list A_j . It is therefore represented by an algebra monomorphism $A_j \rightarrow X$. Two such representatives related by conjugation in X are inner conjugate in the sense of §2; allowing the source algebra isomorphism in that definition also accounts for different choices of transport. We count all algebra maps and then discard the nonmonomorphisms, so no openness of the monomorphism locus is needed. When X is connected, $c_u = \text{id}_X$ for every u , and inner conjugacy reduces exactly to equivalence over X . $\square$

Corollary 7.2 (Semisimple ambient setting). *Let $\mathcal{C}$ be a semisimple $\mathbb{k}$ -linear monoidal category, and let X be a connected algebra object of finite length. Then X has only finitely many separable algebra subobjects up to equivalence over X .*

Proof. By Definition 2.2, a finite-length object has only finitely many isomorphism classes of subobjects. The conclusion is therefore Theorem 7.1(ii). $\square$

Example 7.3 (The subobject hypothesis is strictly weaker). Let $G = \mathbb{G}_a = (\mathbb{k}, +)$ be the additive algebraic group, and consider the tensor category of finite-dimensional rational G -representations over $\mathbb{k}$; see [26, §I.2]. This category is not semisimple. On $V = \mathbb{k}^2$, let

$$\rho(t) = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} \quad (t \in \mathbb{k}).$$

This is a nonsplit extension of the trivial representation by itself. Its only invariant subspaces are 0, the line spanned by the first basis vector, and V . Thus the hypothesis of Theorem 7.1 can hold for a nonsemisimple object in a nonsemisimple tensor category.

Internal detail 36. Choose a basis v_1, v_2 and define $Nv_1 = 0$, $Nv_2 = v_1$; then $\rho(t) = \text{id}_V + tN$. A one-dimensional subspace $L = \mathbb{k}(av_1 + bv_2)$ is G -stable exactly when it is N -stable. If $b \neq 0$, then $N(av_1 + bv_2) = bv_1 \in L$ would force $v_1 \in L$, hence $L = \mathbb{k}v_1$, contradicting $b \neq 0$. Thus the unique invariant line is $\mathbb{k}v_1$, and the only subrepresentations are 0, $\mathbb{k}v_1$, and the whole module.

Internal detail 37. The proof uses: (1) finitely many possible underlying objects; (2) $\text{HH}^2 = 0$ to obtain finitely many separable algebra structures on each; and (3) $\text{HH}^1 = 0$ to obtain finitely many embeddings modulo inner conjugacy. Connectedness is used only to make that conjugacy action trivial.

Effective bounds. Choose representatives $B_1, \dots, B_N$ for the isomorphism classes of objects occurring as subobjects of X . For each r , define a_r, u_r, t_r, E_r from (32), with $B = B_r$, and put

$$(50) \quad h_r := \dim_{\mathbb{k}} \text{Hom}_{\mathcal{C}}(B_r, X), \quad c_r := \dim_{\mathbb{k}} \text{Hom}_{\mathcal{C}}(B_r \otimes B_r, X).$$

Discard any B_r that admits no unital algebra structure with an algebra monomorphism into X ; such an object contributes no algebra subobject.

Theorem 7.4 (Effective subobject bound). *The number of inner-conjugacy classes of separable algebra subobjects of X is at most*

$$(51) \quad \sum_{r=1}^N 2^{\min\{a_r + u_r, t_r + 2E_r\} + \min\{\max\{h_r - 1, 0\}, c_r\}}.$$

If X is connected, the same expression bounds equivalence classes over X .

Proof. For a fixed underlying object B_r , Theorem 5.2 bounds the possible separable source algebras. For each such source algebra A , Theorem 6.2 gives at most $2^{\min\{b_{A,X}, c_r\}}$ inner-conjugacy classes of algebra maps into X . If $e_X \neq 0$, each source that admits a unital map to X witnesses a nonzero value of the unit-evaluation map in (46), so $b_{A,X} \leq h_r - 1$; a source admitting no such

map contributes nothing. If $e_X = 0$, unitality forces $\text{id}_X = 0$, whence $h_r = b_{A,X} = 0$. Thus in all cases $b_{A,X} \leq \max\{h_r - 1, 0\}$, and the homomorphism factor is at most $2^{\min\{\max\{h_r-1,0\}, c_r\}}$. The monomorphisms form a subset. Summing over r proves the result, and connectedness makes inner conjugation trivial.

Internal detail 38. The maximum with zero is needed only for the zero target. If $e_X = 0$, the unit identity gives $\text{id}_X = 0$, hence every morphism to or from X is zero. Thus X is a zero object. A monomorphism $B \rightarrow X$ then forces $\text{id}_B = 0$ as well, and there is exactly one algebra-subobject class. All dimensions in the summand are zero and the corrected bound is $2^0 = 1$. If $e_X \neq 0$, an algebra embedding $f : A \rightarrow X$ has $f e_A = e_X \neq 0$, proving the rank-one lower bound for unit evaluation. Connected targets always lie in this latter case.

□

Suppose now that $\mathcal{C}$ is semisimple with simple tensor unit, and that X is connected, of finite length, and decomposes as

$$(52) \quad X \cong \mathbf{1} \oplus \bigoplus_{i=1}^t S_i^{\oplus n_i},$$

where the $S_i \not\cong \mathbf{1}$ are pairwise nonisomorphic simple objects. Set

$$(53) \quad \begin{aligned} P_X &:= \prod_{i=1}^t (n_i + 1), & E_X &:= \dim_{\mathbb{k}} \text{End}_{\mathcal{C}}(X), \\ M_X &:= \dim_{\mathbb{k}} \text{Hom}_{\mathcal{C}}(X \otimes X, X), & T_X &:= \dim_{\mathbb{k}} \text{Hom}_{\mathcal{C}}(X^{\otimes 3}, X). \end{aligned}$$

Corollary 7.5 (Ambient-object bound). *Let $N_{\text{sep}}(X)$ denote the number of equivalence classes over X of separable algebra subobjects. Then*

$$(54) \quad N_{\text{sep}}(X) \leq P_X 2^{\min\{M_X-1, T_X\} + \min\{E_X-1, M_X\}}.$$

If X is a Frobenius algebra, the same estimate applies to separable Frobenius subalgebras in the sense of [20, Definition 2.6].

Proof. Every algebra subobject of connected X is connected. Its underlying object therefore has the form

Internal detail 39. If $i : B \rightarrow X$ is an algebra monomorphism, composition with i defines an injective linear map $\text{Hom}_{\mathcal{C}}(\mathbf{1}, B) \rightarrow \text{Hom}_{\mathcal{C}}(\mathbf{1}, X)$: if $i \circ f = i \circ g$, left cancellability of i gives $f = g$. Since $i \circ e_B = e_X \neq 0$, the source Hom space is nonzero. Since the codomain has dimension one, the source does too. Thus B is connected.

$$\mathbf{1} \oplus \bigoplus_{i=1}^t S_i^{\oplus m_i}, \quad 0 \leq m_i \leq n_i,$$

so at most P_X subobject isomorphism classes occur. For such a source B , the unit splits and

$$q_{B,e} = a_B - 2E_B + 1 \leq a_B - 1 \leq M_X - 1, \quad t_B \leq T_X.$$

Moreover,

$$\dim_{\mathbb{k}} \text{Hom}_{\mathcal{C}}(B, X) - 1 \leq E_X - 1, \quad \dim_{\mathbb{k}} \text{Hom}_{\mathcal{C}}(B \otimes B, X) \leq M_X.$$

Internal detail 40. All inequalities follow from explicit split injections. Choose a splitting $j : B \rightarrow X$, $p : X \rightarrow B$ with $p \circ j = \text{id}_B$. Then

$$\text{Hom}_{\mathcal{C}}(B^{\otimes k}, B) \longrightarrow \text{Hom}_{\mathcal{C}}(X^{\otimes k}, X), \quad f \longmapsto j \circ f \circ p^{\otimes k}$$

is injective, with left inverse $F \mapsto p \circ F \circ j^{\otimes k}$; this gives $a_B \leq M_X$ and $t_B \leq T_X$. The map $\text{Hom}_{\mathcal{C}}(B, X) \rightarrow \text{End}_{\mathcal{C}}(X)$, $f \mapsto f \circ p$, is injective with left inverse $F \mapsto F \circ j$, and $\text{Hom}_{\mathcal{C}}(B \otimes B, X) \rightarrow \text{Hom}_{\mathcal{C}}(X \otimes X, X)$, $f \mapsto f \circ (p \otimes p)$, is injective with left inverse obtained by precomposing with $j \otimes j$. Finally $q_{B,e} = a_B - 2E_B + 1 \leq a_B - 1$ because $E_B \geq 1$. These are exactly the four estimates used in the proof.

Apply Corollary 5.5 and Theorem 6.2, then sum over the P_X possible isomorphism classes.

Once an algebra embedding is fixed, the compatible Frobenius structure is unique by Lemma 8.1. Hence the same estimate applies. $\square$

Remark 7.6 (Fusion-rule form). If $N_{ij}^k := \dim_{\mathbb{k}} \text{Hom}_{\mathcal{C}}(S_i \otimes S_j, S_k)$ and $n_0 = 1$ denotes the multiplicity of $\mathbf{1} = S_0$ in X , then

$$E_X = 1 + \sum_{i=1}^t n_i^2, \quad M_X = \sum_{i,j,k=0}^t n_i n_j n_k N_{ij}^k.$$

A similar contraction of two fusion matrices computes T_X . Thus

Internal detail 41. Explicitly, with $S_0 = \mathbf{1}$ and the same multiplicities n_i ,

$$T_X = \sum_{i,j,k,\ell=0}^t n_i n_j n_k n_{\ell} \sum_p N_{ij}^p N_{pk}^{\ell},$$

where p ranges over the finitely many simple objects occurring in products $S_i \otimes S_j$. Indeed, $\sum_p N_{ij}^p N_{pk}^{\ell}$ is the multiplicity of S_{ℓ} in $S_i \otimes S_j \otimes S_k$, and multiplying by the four multiplicities in X and summing computes $\dim_{\mathbb{k}} \text{Hom}_{\mathcal{C}}(X^{\otimes 3}, X)$. Thus no associator matrix is needed for this dimension: fusion multiplicities alone suffice.

(54) can be read directly from the fusion rules and the multiplicity vector of X .

8. FROBENIUS SUBALGEBRAS AND EXCHANGE RELATIONS

Throughout this section, $\mathcal{C}$ is a Hom-finite $\mathbb{k}$ -linear monoidal category over the algebraically closed field $\mathbb{k}$. We temporarily assume Karoubianity for the exchange correspondence and remove it by idempotent completion in the effective theorem.

A *Frobenius algebra object* is an algebra (X, m_X, e_X) together with a coalgebra structure

$$\delta_X : X \longrightarrow X \otimes X, \quad \epsilon_X : X \longrightarrow \mathbf{1}$$

satisfying coassociativity, the counit identities, and the Frobenius condition

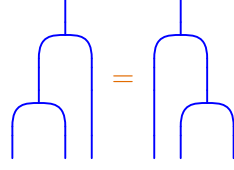
$$(55) \quad \delta_X \circ m_X = (m_X \otimes \text{id}_X) \circ (\text{id}_X \otimes \delta_X) = (\text{id}_X \otimes m_X) \circ (\delta_X \otimes \text{id}_X).$$

The two equalities express that δ_X is a morphism of left and right regular X -modules.

Internal detail 42. The coalgebra structure and all its axioms admit the following string diagrams, read from top to bottom. First, comultiplication and counit are depicted by

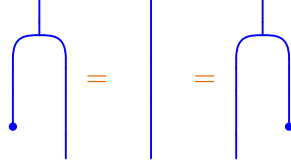
$$\begin{array}{ccc} \begin{array}{c} | \\ \text{---} \text{---} \\ | \end{array} & & \begin{array}{c} | \\ \bullet \end{array} \\ \delta_X & & \epsilon_X \end{array}$$

Coassociativity is the completely displayed equality



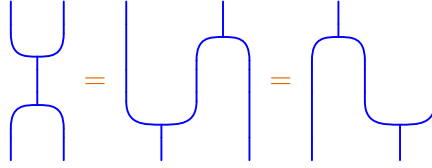
$$(\delta_X \otimes \text{id}_X) \circ \delta_X = (\text{id}_X \otimes \delta_X) \circ \delta_X.$$

The two counit reductions are



$$(\epsilon_X \otimes \text{id}_X) \circ \delta_X = \text{id}_X = (\text{id}_X \otimes \epsilon_X) \circ \delta_X.$$

Finally, the three equal Frobenius diagrams are



and express exactly equation (55).

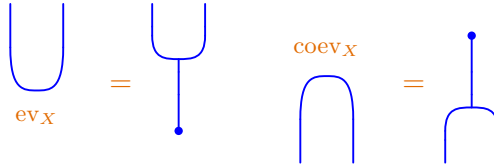
The Frobenius evaluation and coevaluation morphisms are

$$(56) \quad \text{ev}_X := \epsilon_X \circ m_X : X \otimes X \longrightarrow \mathbf{1}, \quad \text{coev}_X := \delta_X \circ e_X : \mathbf{1} \longrightarrow X \otimes X.$$

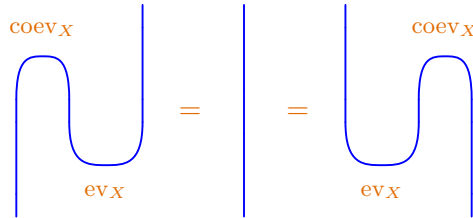
The Frobenius, unit, and counit relations give the two zig-zag identities

$$(57) \quad (\text{id}_X \otimes \text{ev}_X) \circ (\text{coev}_X \otimes \text{id}_X) = \text{id}_X, \quad (\text{ev}_X \otimes \text{id}_X) \circ (\text{id}_X \otimes \text{coev}_X) = \text{id}_X.$$

Internal detail 43. The evaluation is multiplication followed by the counit, while the coevaluation is the unit followed by comultiplication:



The two zig-zag reductions are fully pictorially



where each cap joins the two incoming wires and has no outgoing wire.

For the first zig-zag identity, expand the definitions and use the second Frobenius equality:

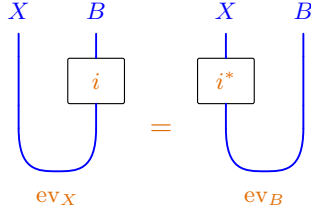
$$\begin{aligned} (\text{id}_X \otimes \text{ev}_X) \circ (\text{coev}_X \otimes \text{id}_X) &= (\text{id}_X \otimes \epsilon_X) \circ (\text{id}_X \otimes m_X) \circ (\delta_X \otimes \text{id}_X) \circ (e_X \otimes \text{id}_X) \\ &= (\text{id}_X \otimes \epsilon_X) \circ \delta_X \circ m_X \circ (e_X \otimes \text{id}_X) = (\text{id}_X \otimes \epsilon_X) \circ \delta_X = \text{id}_X. \end{aligned}$$

The second identity is the reflected calculation, using $(m_X \otimes \text{id}_X) \circ (\text{id}_X \otimes \delta_X) = \delta_X \circ m_X$, right unitality, and the other counit identity. Hence the evaluation morphism is nondegenerate and the two maps in (56) really make X self-dual.

If $i : B \rightarrow X$ is a morphism between Frobenius algebras, its dual $i^* : X \rightarrow B$ is characterized by

$$(58) \quad \text{ev}_X \circ (\text{id}_X \otimes i) = \text{ev}_B \circ (i^* \otimes \text{id}_B).$$

Internal detail 44. The characterizing identity (58) is the following equality of string diagrams:

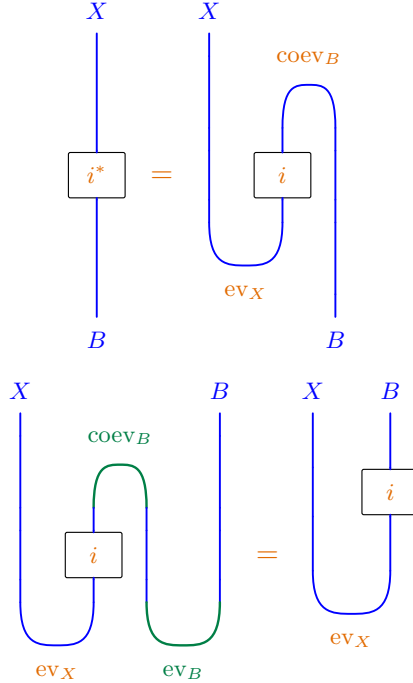


Existence and uniqueness of the dual are explicit. With the conventions above,

$$i^* = (\text{ev}_X \otimes \text{id}_B) \circ (\text{id}_X \otimes i \otimes \text{id}_B) \circ (\text{id}_X \otimes \text{coev}_B) : X \longrightarrow B.$$

Substituting this formula into the right side of (58) and applying the zig-zag identity for B gives the left side. Conversely, if two morphisms satisfy (58), tensor that equation on the right by id_B and precompose with $\text{id}_X \otimes \text{coev}_B$. The same zig-zag identity recovers the displayed formula; the two morphisms are therefore equal. The formula also shows that dualization is $\mathbb{k}$ -linear and contravariant: $(g \circ f)^* = f^* \circ g^*$.

The mate formula and its verification are purely pictorial as well. The first equality below is the displayed formula for i^* ; after substituting it into the right-hand diagram of (58), the highlighted B -cup-cap contracts by the zig-zag identity, giving the second equality.



Following [21], a Frobenius subalgebra is represented by a unital algebra monomorphism $i : B \rightarrow X$ satisfying

$$(59) \quad i^* \circ i = \text{id}_B, \quad i^{**} = i.$$

We call it *separable* when its underlying algebra object is separable.

Lemma 8.1 (Uniqueness of a Frobenius subalgebra structure). *Fix a unital algebra monomorphism $i : (B, m_B, e_B) \rightarrow X$. There is at most one pair (δ_B, ϵ_B) making B a Frobenius algebra for which i is a Frobenius-subalgebra embedding.*

Proof. Assume that two Frobenius structures

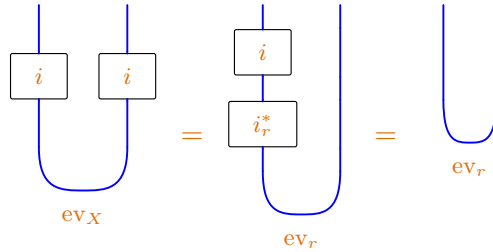
$$F_r = (B, m_B, e_B, \delta_r, \epsilon_r), \quad r = 1, 2,$$

have this property. Let $\text{ev}_r = \epsilon_r \circ m_B$ and let $i_r^* : X \rightarrow B$ be the dual of i with respect to F_r . The defining duality identity and $i_r^* \circ i = \text{id}_B$ give

$$(60) \quad \text{ev}_r = \text{ev}_X \circ (i \otimes i), \quad r = 1, 2.$$

Hence $\text{ev}_1 = \text{ev}_2 =: \text{ev}_B$. Since the algebra structure is fixed,

Internal detail 45. To obtain (60), precompose the defining identity $\text{ev}_X \circ (\text{id}_X \otimes i) = \text{ev}_r \circ (i_r^* \otimes \text{id}_B)$ with $i \otimes \text{id}_B$. The right side becomes $\text{ev}_r \circ (i_r^* \circ i \otimes \text{id}_B) = \text{ev}_r$, because $i_r^* \circ i = \text{id}_B$; the left side is $\text{ev}_X \circ (i \otimes i)$. Thus the evaluation morphism on the source is literally the restriction of the fixed evaluation morphism on X , independent of the proposed Frobenius structure. Pictorially, cancellation of $i_r^* \circ i$ gives



$$\epsilon_r = \text{ev}_B \circ (\text{id}_B \otimes e_B),$$

so $\epsilon_1 = \epsilon_2$.

Put $\text{coev}_r = \delta_r \circ e_B$. Each coev_r is the coevaluation morphism for the same nondegenerate evaluation morphism ev_B . Such a coevaluation morphism is unique: contracting $\text{coev}_1 \otimes \text{coev}_2$ in the middle and applying the two zig-zag identities gives

$$(61) \quad \text{coev}_1 = (\text{id}_B \otimes \text{ev}_B \otimes \text{id}_B) \circ (\text{coev}_1 \otimes \text{coev}_2) = \text{coev}_2.$$

Finally, the Frobenius identity and the unit recover the comultiplication:

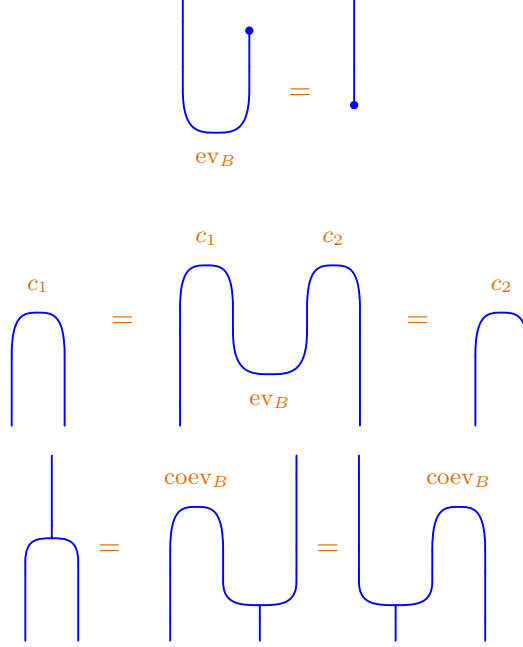
$$(62) \quad \delta_r = (\text{id}_B \otimes m_B) \circ (\text{coev}_r \otimes \text{id}_B) = (m_B \otimes \text{id}_B) \circ (\text{id}_B \otimes \text{coev}_r).$$

Thus $\delta_1 = \delta_2$ and the two Frobenius structures coincide. $\square$

Internal detail 46. The remaining reconstruction formulas also follow directly from the axioms. First, $\text{ev}_B \circ (\text{id}_B \otimes e_B) = \epsilon_B \circ m_B \circ (\text{id}_B \otimes e_B) = \epsilon_B$. Second, if c_1, c_2 are coevaluation morphisms for the same evaluation morphism, then $(\text{id} \otimes \text{ev}_B \otimes \text{id}) \circ (c_1 \otimes c_2)$ reduces to c_1 by the second zig-zag identity for (ev_B, c_2) and to c_2 by the first zig-zag identity for (ev_B, c_1) ; hence $c_1 = c_2$. Finally,

$$(\text{id}_B \otimes m_B) \circ (\text{coev}_B \otimes \text{id}_B) = (\text{id}_B \otimes m_B) \circ (\delta_B \otimes \text{id}_B) \circ (e_B \otimes \text{id}_B) = \delta_B \circ m_B \circ (e_B \otimes \text{id}_B) = \delta_B,$$

by Frobenius and left unitality, and the reflected formula is identical. These three reconstruction steps are depicted, respectively, as follows:



Thus the evaluation morphism determines the counit and its unique coevaluation morphism, while the latter determines the comultiplication.

When the subobject hypothesis of Theorem 7.1 holds, that theorem and Lemma 8.1 already give qualitative finiteness for separable Frobenius subalgebras. The exchange argument below removes that hypothesis and gives a bound directly in the smaller parameter space $\text{End}_{\mathcal{C}}(X)$.

Corollary 8.2 (Frobenius subalgebras up to inner conjugacy). *Let X be a Frobenius algebra object in a Hom-finite $\mathbb{k}$ -linear monoidal category. Its separable Frobenius subalgebras form finitely many classes under inner conjugacy of their underlying algebra subobjects, in the sense of §2. The ambient inner automorphism is not required to preserve the chosen Frobenius counit.*

Proof. This follows from the quadratic bound in Theorem 8.9. If $X = 0$, its only algebra subobject is 0. $\square$

Corollary 8.3 (Connected Frobenius-subalgebra finiteness). *If X is connected, its separable Frobenius subalgebras form finitely many equivalence classes over X .*

Proof. Connectedness makes inner conjugation trivial, and Lemma 8.1 identifies equivalence of underlying algebra embeddings with equivalence of Frobenius subalgebras over X . The conclusion follows from Corollary 8.2. $\square$

8.1. The quadratic exchange locus. For the construction of the exchange correspondence, assume first that $\mathcal{C}$ is Karoubian; no semisimplicity, simple-unit, or finite-length hypothesis is needed. Idempotent completion will remove Karoubianity from the numerical bound. Let $X = (X, m_X, e_X, \delta_X, \epsilon_X)$ be a Frobenius algebra. For $f \in \text{End}_{\mathcal{C}}(X)$, write f^* for its dual with respect to the Frobenius self-duality. The equations

$$b^* = b, \quad b \circ e_X = e_X$$

cut out the affine space

$$(63) \quad \mathcal{A}_X = \text{id}_X + \{h \in \text{End}_{\mathcal{C}}(X) : h^* = h, h \circ e_X = 0\}.$$

Write

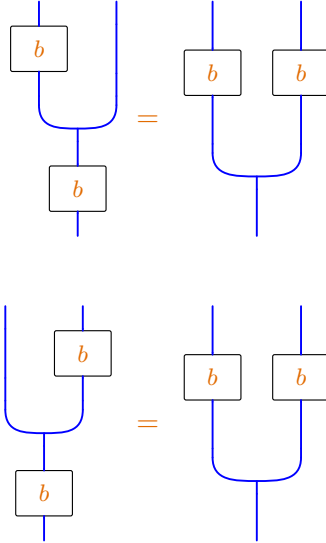
$$(64) \quad r_X := \dim_{\mathbb{k}} \{h \in \text{End}_{\mathcal{C}}(X) : h^* = h, h \circ e_X = 0\}.$$

Inside $\mathcal{A}_X$, consider the equations

$$(65) \quad \begin{aligned} b \circ b &= b, \\ b \circ m_X \circ (b \otimes \text{id}_X) &= m_X \circ (b \otimes b), \\ b \circ m_X \circ (\text{id}_X \otimes b) &= m_X \circ (b \otimes b). \end{aligned}$$

The last two are the *exchange relations*. We call their reduced common zero locus, together with idempotence, the *exchange locus* $\mathfrak{E}(X)$. Its defining equations inside $\mathcal{A}_X$ have degree at most two.

Internal detail 47. The operation $f \mapsto f^*$ is linear, fixes id_X , and reverses composition. It need not be an involution and need not agree with a unitary adjoint. Linearity alone makes $b^* = b$ a linear equation; writing $b = \text{id}_X + h$ gives (63). Each exchange defect contains at most two occurrences of b , so choosing bases gives coordinate polynomials of degree at most two. The exchange relations are depicted by



Unitality and either exchange relation already imply idempotence: compose the first exchange relation with $\text{id}_X \otimes e_X$ to obtain $b^2 = b$. We retain the idempotence equation to make the splitting argument explicit.

Theorem 8.4 (Ghosh–Palcoux). *The points of $\mathfrak{E}(X)$ are naturally in bijection with equivalence classes over X of Frobenius subalgebras of X . A Frobenius-subalgebra embedding $i : B \rightarrow X$ corresponds to $b = i \circ i^*$.*

Proof. Existence in both directions is [20, Theorem 4.1]. For a splitting $b = i \circ p$, $p \circ i = \text{id}_B$, the reconstructed multiplication and unit are

$$m_B = p \circ m_X \circ (i \otimes i), \quad e_B = p \circ e_X.$$

Two splittings of the same idempotent are canonically isomorphic over X ; these formulas make that isomorphism an algebra isomorphism. Lemma 8.1 makes the compatible Frobenius structure unique, proving the asserted bijection. $\square$

Internal detail 48. On the chosen splitting, the remaining reconstruction formulas are

$$\delta_B = (p \otimes p) \circ \delta_X \circ i, \quad \epsilon_B = \epsilon_X \circ i.$$

The proof of the cited theorem verifies $p = i^*$ and $i^{**} = i$ for these structures. Thus the correspondence applies to any chosen splitting, not merely to a preferred choice. Unitality is retained explicitly in our locus, so $b = 0$ is excluded whenever $X \neq 0$.

8.2. Local rigidity with a varying source. An ambient inner automorphism need not preserve the chosen Frobenius counit. Consequently we count the restriction of inner conjugacy of underlying algebra subobjects, as in Corollary 8.2. The following parameter space lets us establish local rigidity without requiring an action on $\mathfrak{E}(X)$.

For an object B , let $\mathbf{P}_B(X)$ be the reduced affine variety of triples (m, e, j) such that (B, m, e) is an algebra object and $j : (B, m, e) \rightarrow X$ is a unital algebra homomorphism. The algebraic group

$$G_B \times \Gamma(X)^\times, \quad G_B := \text{Aut}_{\mathcal{C}}(B),$$

acts by

$$(66) \quad (g, u) \cdot (m, e, j) = (g \circ m \circ (g^{-1} \otimes g^{-1}), g \circ e, c_u \circ j \circ g^{-1}).$$

Lemma 8.5 (Open orbits for a varying source). *If $A = (B, m, e)$ is separable, the orbit of (m, e, j) is open in $\mathbf{P}_B(X)$.*

Proof. Let $(\mu, \eta, v) \in T_{(m, e, j)}\mathbf{P}_B(X)$. Projection to $\text{Alg}(B)$ gives a tangent vector (μ, η) . By (31), there is $z \in \text{End}_{\mathcal{C}}(B)$ whose infinitesimal change of source coordinates gives (μ, η) . Subtracting the tangent to (66) in the direction $(z, 0)$ leaves $(0, 0, w)$, with $w = v + j \circ z$. Linearizing the homomorphism identities now gives

$$w \circ e = 0, \quad w \circ m = m_X \circ (w \otimes j + j \otimes w).$$

Thus w is a derivation from A to the A -bimodule X defined by j . The degree-one contraction makes it inner: for some $a \in \Gamma(X)$,

$$w = m_X \circ (a \otimes j) - m_X \circ (j \otimes a).$$

This is the infinitesimal conjugation direction $(0, a)$ in (66). Hence the orbit-map differential is surjective onto the tangent space of $\mathbf{P}_B(X)$. Both factors of the group are unit groups of finite-dimensional algebras and are smooth and connected. Lemma 4.1 proves the assertion. $\square$

Internal detail 49. The equations for $\mathbf{P}_B(X)$ are associativity, the two unit identities, $j \circ e = e_X$, and $j \circ m = m_X \circ (j \otimes j)$; hence it is an affine variety. At the identity of the acting group, the orbit-map differential sends (z, a) to

$$(z \circ m - m \circ (z \otimes \text{id}_B) - m \circ (\text{id}_B \otimes z), z \circ e, m_X \circ (a \otimes j) - m_X \circ (j \otimes a) - j \circ z).$$

Every tangent vector to the reduced variety satisfies the linearized equations. After removing the first two components by source rigidity, the last two linearized equations are exactly the displayed derivation identities. No identification of the reduced variety with the scheme defined by these equations is needed. Nor is a regular choice of source algebra isomorphisms needed: openness follows from the single joint group action.

Lemma 8.6 (A regular splitting chart). *Let $b \in \text{End}_{\mathcal{C}}(X)$ be an idempotent with a splitting $b = i \circ p$, $p \circ i = \text{id}_B$. For an idempotent c , put*

$$t_c := c \circ b + (\text{id}_X - c) \circ (\text{id}_X - b).$$

On the Zariski-open neighborhood of b where t_c is invertible,

$$i_c := t_c \circ i, \quad p_c := p \circ t_c^{-1}$$

depend regularly on c and satisfy $p_c \circ i_c = \text{id}_B$ and $i_c \circ p_c = c$.

Proof. We have $t_b = \text{id}_X$ and

$$c \circ t_c = c \circ b = t_c \circ b.$$

Invertibility is an open condition in the finite-dimensional algebra $\text{End}_{\mathcal{C}}(X)$, and inversion is regular on its unit group. Thus, on this open neighborhood, $c = t_c \circ b \circ t_c^{-1}$, which gives both splitting identities. $\square$

Internal detail 50. For an arbitrary finite-dimensional unital algebra E , an element t is invertible exactly when left multiplication $L_t : E \rightarrow E$ has nonzero determinant. Hence the open condition above is an ordinary polynomial nonvanishing condition, even when $\text{End}_{\mathcal{C}}(X)$ is not semisimple. This chart uses neither blockwise ranks nor a decomposition of X .

Lemma 8.7 (Open inner-conjugacy classes in the exchange locus). *Every point of $\mathfrak{E}(X)$ corresponding to a separable Frobenius subalgebra has an open neighborhood consisting of points in the same inner-conjugacy class of underlying algebra subobjects.*

Proof. Let $b = i \circ i^*$ correspond to a separable Frobenius subalgebra on B . Apply Lemma 8.6 with $p = i^*$. For $c \in \mathfrak{E}(X)$ in this chart, put

$$(67) \quad e_c := p_c \circ e_X, \quad m_c := p_c \circ m_X \circ (i_c \otimes i_c).$$

The exchange correspondence makes (m_c, e_c, i_c) a point of $\mathbf{P}_B(X)$, and these formulas give a regular map from the chart to that variety. By Lemma 8.5, the orbit of (m_b, e_b, i_b) is open. Its inverse image is therefore an open neighborhood of b . If c lies in this inverse image, (66) supplies a source algebra isomorphism and an ambient inner automorphism identifying the underlying algebra subobjects at b and c . In particular, the source at c is again separable. $\square$

Internal detail 51. Only the map from the chart to $\mathbf{P}_B(X)$ is used. We do not conjugate an exchange idempotent by an arbitrary c_u and assert that it remains in $\mathfrak{E}(X)$: that assertion can fail when c_u does not preserve the ambient Frobenius form. Instead, both endpoints already lie in the exchange locus, and membership in the joint orbit says exactly that their underlying algebra subobjects are inner conjugate. Their source Frobenius structures are the structures induced separately from the fixed ambient Frobenius algebra.

Corollary 8.8 (Separable exchange idempotents are isolated). *If X is connected, every point of $\mathfrak{E}(X)$ corresponding to a separable Frobenius subalgebra is isolated.*

Proof. Connectedness makes inner conjugation trivial. By the exchange correspondence and Lemma 8.1, each equivalence class over X corresponds to a single point. The neighborhood in Lemma 8.7 is therefore that single point. $\square$

8.3. The effective bound.

Theorem 8.9 (Quadratic exchange bound). *Let $\mathcal{C}$ be a Hom-finite $\mathbb{k}$ -linear monoidal category, and let X be a Frobenius algebra in $\mathcal{C}$. Let $N_F^{\text{inn}}(X)$ be the number of classes of separable Frobenius subalgebras under inner conjugacy of their underlying algebra subobjects. Then*

$$(68) \quad N_F^{\text{inn}}(X) \leq 2^{r_X},$$

$$(69) \quad 2^{r_X} \leq 2^{\dim_{\mathbb{k}} \text{End}_{\mathcal{C}}(X) - 1} \quad (X \neq 0).$$

For $X = 0$, the first bound reads $N_F^{\text{inn}}(0) = 1 = 2^{r_0}$. If X is connected, the same bound applies to the number $N_F(X)$ of equivalence classes over X . If, in addition, $\mathcal{C}$ is semisimple with simple tensor unit and X has the decomposition (52), then

$$(70) \quad N_F(X) \leq 2^{\sum_{i=1}^t n_i^2}.$$

Proof. Passing to the Karoubi envelope preserves all spaces and operations entering r_X , the algebra $\Gamma(X)$, and separability. Its fully faithful monoidal embedding also preserves the equivalence relation: an isomorphism between original source objects in the envelope is already a morphism of the original category. Thus the original inner-conjugacy classes inject into those in the envelope, and it suffices to prove the bound when $\mathcal{C}$ is Karoubian. If $X = 0$, its only algebra subobject is 0 and $r_X = 0$; hence assume $X \neq 0$.

The locus $\mathfrak{E}(X)$ is defined by quadratics in the affine space $\mathcal{A}_X$ of dimension r_X . Theorem 4.3 gives

$$\text{cdeg}(\mathfrak{E}(X)) \leq 2^{r_X}.$$

By Lemma 8.7, every separable inner-conjugacy class is a nonempty open subset of $\mathfrak{E}(X)$: take the union of the neighborhoods supplied at its points. Distinct such classes are disjoint. Two nonempty open subsets meeting the same irreducible component must intersect. Consequently the number of these classes is at most the number of irreducible components, and hence at most $\text{cdeg}(\mathfrak{E}(X))$. This proves (68).

Let $E_X^+ := \{h \in \text{End}_{\mathcal{C}}(X) : h^* = h\}$. The linear map

$$E_X^+ \longrightarrow \text{Hom}_{\mathcal{C}}(\mathbf{1}, X), \quad h \longmapsto h \circ e_X$$

sends id_X to $e_X \neq 0$ and therefore has rank at least one. Its kernel has dimension r_X , so $r_X \leq \dim_{\mathbb{k}} E_X^+ - 1 \leq \dim_{\mathbb{k}} \text{End}_{\mathcal{C}}(X) - 1$. This proves (69) without connectedness or a simple-unit hypothesis. Connectedness identifies inner conjugacy with equivalence over X . Finally, $\dim_{\mathbb{k}} \text{End}_{\mathcal{C}}(X) = 1 + \sum_i n_i^2$ under (52), proving (70). $\square$

Internal detail 52. If Z is an irreducible component and two open classes U, V both meet Z , then $U \cap Z$ and $V \cap Z$ are nonempty open subsets of the irreducible variety Z . Their intersection is nonempty, contradicting disjointness. Each nonempty class meets at least one component, so the component count bounds the classes even though a class need not itself be an orbit of any group acting on $\mathfrak{E}(X)$. No prior finiteness theorem or geometric quotient is used in this counting step.

More precisely, if ρ_X is the rank of evaluation at the unit on E_X^+ , then $r_X = \dim E_X^+ - \rho_X$. Connectedness forces $\rho_X = 1$; in general only $\rho_X \geq 1$ is used. All assertions hold in arbitrary characteristic. For $X = 0$, the exchange locus has one point, $r_X = 0$, and there is one Frobenius subalgebra; the first bound still reads $1 \leq 1$, whereas the exponent $\dim \text{End}(X) - 1$ must not be used.

Remark 8.10 (Why the base is two). The direct multiplication-closure equation

$$b \circ m_X \circ (b \otimes b) = m_X \circ (b \otimes b)$$

has degree three in the coordinates of b . Applying the degree bound to that presentation gives a base-three estimate. Frobenius self-duality and the exchange correspondence provide the quadratic presentation (65), yielding the base-two estimate.

Remark 8.11. The general bound (51) also bounds separable Frobenius subalgebras up to inner conjugacy, by forgetting the compatible Frobenius structure and using Lemma 8.1. The exchange bound eliminates the source multiplication and embedding variables from the degree calculation. The joint parameter space is used only to prove local openness; its dimension does not enter the exponent. No hypothesis on the number of subobject isomorphism classes is needed for the Frobenius bound.

9. APPLICATION TO HOPF ALGEBRAS AND FINITE GROUPS

We apply the exchange bound to the regular left comodule of a finite-dimensional semisimple Hopf algebra, and then to function algebras on finite homogeneous spaces. Finite dimensionality makes the relevant endomorphism spaces finite-dimensional; no unrestricted infinite-dimensional extension is claimed.

9.1. Semisimple Hopf algebras. Let $H = (H, m_H, e_H, \delta_H, \epsilon_H, S)$ be a finite-dimensional semisimple Hopf algebra over $\mathbb{C}$. Here $\text{Corep}(H)$ denotes the category of finite-dimensional left H -comodules. A left coideal subalgebra is a unital subalgebra $K \subseteq H$ satisfying

$$\delta_H(K) \subseteq H \otimes K.$$

Theorem 9.1 ([21, Theorem 11.11, Proposition 6.5, and the proof of Corollary 11.24]). *The following statements hold.*

- (i) *The regular comodule H carries a natural connected Frobenius algebra structure $(H, m_H, e_H, \delta_H^F, \epsilon_H^F)$ in $\text{Corep}(H)$. Here the superscript distinguishes its Frobenius coalgebra maps from the Hopf maps δ_H and ϵ_H .*
- (ii) *A unital subalgebra $K \subseteq H$ is a left coideal subalgebra if and only if it is a unital algebra subobject of H in $\text{Corep}(H)$.*
- (iii) *These unital algebra subobjects are precisely the Frobenius subalgebras of H in $\text{Corep}(H)$.*
- (iv) *The category $\text{Corep}(H)$ is an integral, hence pseudo-unitary, fusion category and is therefore positive.*

Following [21, Definition 6.3], positivity means that the category admits a pivotal structure ϕ for which the pivotal dimension $\dim_\phi(Y)$ is positive for every nonzero object Y .

Lemma 9.2 (Connected Frobenius algebras are separable). *Every connected Frobenius algebra in a positive tensor category is separable.*

Proof. Let $A = (A, m_A, e_A, \delta_A, \epsilon_A)$ be such an algebra. Choose a positive pivotal structure ϕ . Through the Frobenius self-duality, identify A^{**} with A ; the pivotal isomorphism $\phi_A : A \rightarrow A^{**}$ then becomes an endomorphism $\alpha \in \text{End}_{\mathcal{C}}(A)$. Put

$$T := m_A \circ (\alpha \otimes \text{id}_A) \circ \delta_A.$$

Naturality and monoidality of ϕ make α an algebra automorphism. Connectedness gives $T \circ e_A = c e_A$ for some $c \in \mathbb{C}$. By [21, Lemma 2.14] and the Frobenius identities,

$$T = m_A \circ ((T \circ e_A) \otimes \text{id}_A) = c \text{id}_A, \quad T \circ m_A = m_A \circ (\alpha \otimes T).$$

Inserting the unit in the second identity yields $T = c\alpha$. The pivotal trace identity [21, Lemmas 2.23–2.24] gives

$$\dim_\phi(A) = \epsilon_A \circ T \circ e_A = c(\epsilon_A \circ e_A).$$

Positivity implies $\dim_\phi(A) > 0$, hence $c \neq 0$. Consequently $\alpha = \text{id}_A$ and $m_A \circ \delta_A = c \text{id}_A$. Since δ_A is an A – A -bimodule morphism, $c^{-1}\delta_A$ is a bimodule section of m_A .

Internal detail 53. The endomorphism α above is the pivotal curl. The same argument proves the stronger statement that a connected Frobenius algebra in a linear pivotal category is separable whenever its pivotal dimension is nonzero. Write

$$\text{ev}_A := \epsilon_A \circ m_A, \quad \text{coev}_A := \delta_A \circ e_A.$$

These morphisms make A self-dual. The Frobenius identities imply $m_A^* = \delta_A$ and $\delta_A^* = m_A$, while the unit and counit identities give $e_A^* = \epsilon_A$ and $\epsilon_A^* = e_A$; compare [21, Lemmas 2.5, 2.9, and 2.11].

Thus, after the Frobenius identification $A^{**} = A$, the pivotal morphism is an endomorphism α satisfying

$$\alpha \circ m_A = m_A \circ (\alpha \otimes \alpha), \quad \alpha \circ e_A = e_A.$$

Set $T = m_A \circ (\alpha \otimes \text{id}_A) \circ \delta_A$ and $x = T \circ e_A$. Connectedness says $\text{Hom}_c(\mathbf{1}, A) = \mathbb{C}e_A$, so $x = ce_A$. The right-module form of the Frobenius identity is

$$\delta_A = (\text{id}_A \otimes m_A) \circ ((\delta_A \circ e_A) \otimes \text{id}_A).$$

Substitution, followed by associativity, gives the first slide [21, Lemma 2.14]:

$$T = m_A \circ (x \otimes \text{id}_A) = c \text{id}_A.$$

For the reverse slide, use

$$\delta_A \circ m_A = (m_A \otimes \text{id}_A) \circ (\text{id}_A \otimes \delta_A)$$

together with the multiplicativity of α to obtain

$$T \circ m_A = m_A \circ (\alpha \otimes T).$$

Precomposing with $\text{id}_A \otimes e_A$ yields

$$T = m_A \circ (\alpha \otimes (T \circ e_A)) = c\alpha.$$

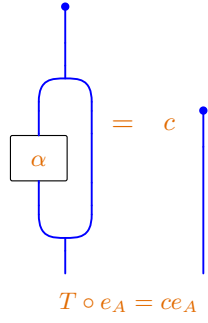
Finally, the pivotal closure of T is, equivalently, the trace identity of [21, Lemmas 2.23–2.24]:

$$\dim_\phi(A) = \epsilon_A \circ T \circ e_A = c(\epsilon_A \circ e_A).$$

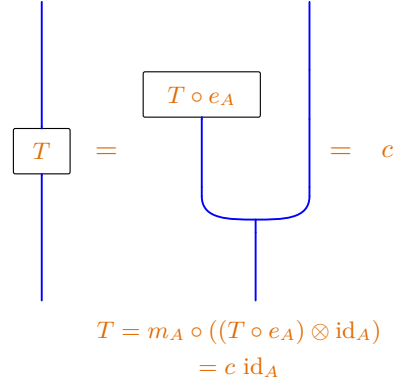
If $\dim_\phi(A) \neq 0$, then $c \neq 0$. Comparing $T = c \text{id}_A$ with $T = c\alpha$ gives $\alpha = \text{id}_A$; only at this point may the pivotal curl be straightened. Hence $m_A \circ \delta_A = c \text{id}_A$, and $c^{-1}\delta_A$ is the required bimodule section.

Here is the same proof in four Ghosh–Palcoux-style panels. Diagrams are read from top to bottom; the α -coupon is retained until the closure has proved that its scalar coefficient is nonzero.

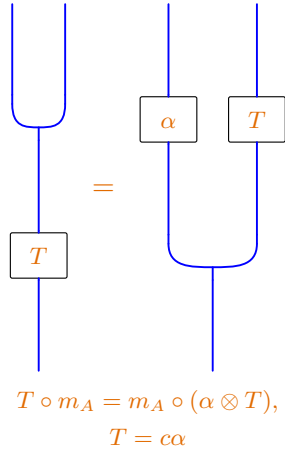
1. Connectedness



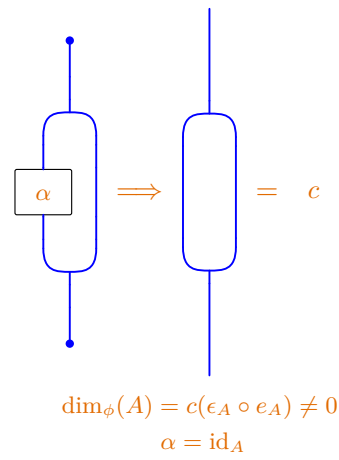
2. First Frobenius slide



3. Reverse Frobenius slide



4. Closure and splitting



□

The ambient endomorphism dimension is particularly simple.

Lemma 9.3. *For the regular left comodule,*

$$\text{End}_{\text{Corep}(H)}(H) \cong H^*$$

as vector spaces. In particular, $\dim_{\mathbb{C}} \text{End}_{\text{Corep}(H)}(H) = \dim_{\mathbb{C}} H$.

Proof. Every $\varphi \in H^*$ defines a comodule endomorphism $(\text{id}_H \otimes \varphi) \circ \delta_H$. Conversely, if T is a comodule endomorphism, then

$$T = (\text{id}_H \otimes (\epsilon_H \circ T)) \circ \delta_H.$$

These assignments are inverse. □

Internal detail 54. For $\varphi \in H^*$, the map $T_\varphi = (\text{id}_H \otimes \varphi) \circ \delta_H$ is a left-comodule morphism because coassociativity gives $\delta_H \circ T_\varphi = (\text{id}_H \otimes T_\varphi) \circ \delta_H$. Moreover $\epsilon_H \circ T_\varphi = \varphi$ by the counit identity. Conversely, if T is a comodule map, then $\delta_H \circ T = (\text{id}_H \otimes T) \circ \delta_H$; applying $\text{id}_H \otimes \epsilon_H$ yields $T = (\text{id}_H \otimes (\epsilon_H \circ T)) \circ \delta_H$. Thus the two constructions are mutually inverse linear maps.

Corollary 9.4 (Semisimple Hopf algebras). *Let H be a semisimple Hopf algebra over $\mathbb{C}$ with $\dim_{\mathbb{C}} H = N$. Then H has at most 2^{N-1} left coideal subalgebras.*

Proof. By Theorem 9.1(i),(iv), H is a connected Frobenius algebra in the positive tensor category $\text{Corep}(H)$. By Theorem 9.1(ii),(iii), every left coideal subalgebra K is a Frobenius subalgebra. Its monomorphism $i : K \rightarrow H$ injects $\text{Hom}_{\text{Corep}(H)}(\mathbf{1}, K)$ into $\text{Hom}_{\text{Corep}(H)}(\mathbf{1}, H) = \mathbb{C}e_H$, while the former space contains the nonzero morphism e_K ; hence K is connected. Lemma 9.2 therefore makes every such K separable. Theorem 8.9 and Lemma 9.3 now give the stated bound. $\square$

Internal detail 55. The same bound holds for right coideal subalgebras, where “right” means $\delta_H(K) \subseteq K \otimes H$. Indeed, if K is a left coideal subalgebra, then the antipode is an anti-algebra map and

$$\delta_H \circ S = (S \otimes S) \circ \delta_H^{\text{op}}$$

shows that $S(K)$ is a right coideal subalgebra. The antipode of a finite-dimensional Hopf algebra is bijective, so $K \mapsto S(K)$ is a bijection between the left and right coideal subalgebras. Thus the same 2^{N-1} estimate holds on the right.

This is an effective strengthening of the qualitative finiteness theorem of Etingof–Walton [17, Theorem 3.6].

Internal detail 56. Etingof–Walton’s Theorem 3.6 states that a semisimple Hopf algebra over a characteristic-zero field has finitely many left coideal subalgebras; their proof does not supply a numerical bound. Corollary 9.4 gives the explicit uniform value $2^{\dim_{\mathbb{C}} H - 1}$ for each handedness, so “effective” here means an actual dimension-only cardinality estimate, not an algorithm for listing the subalgebras. No classification of semisimple Hopf algebras enters the estimate.

9.2. Intermediate subgroups as a check. The group case gives a transparent specialization. Let G be finite and $H \leq G$. The commutative Frobenius algebra $X = \text{Fun}(G/H)$ in $\text{Rep}(G)$ is connected, and its G -stable unital subalgebras correspond to intermediate subgroups $H \leq K \leq G$. By Frobenius reciprocity and the Mackey restriction formula [37, §§7.2–7.3],

$$(71) \quad \dim_{\mathbb{C}} \text{End}_{\text{Rep}(G)}(\text{Fun}(G/H)) = |H \backslash G/H|.$$

Corollary 9.5 (Double-coset bound). *For every finite group G and subgroup $H \leq G$,*

$$(72) \quad |\{K : H \leq K \leq G\}| \leq 2^{|H \backslash G/H| - 1}.$$

Proof. A G -stable subalgebra of $\text{Fun}(G/H)$ is the algebra of functions constant on the blocks of a G -invariant partition of G/H ; such partitions are precisely the quotient maps $G/H \rightarrow G/K$ with $H \leq K \leq G$. Equip each function algebra on a homogeneous space S with the averaged counit $\epsilon_S(f) = |S|^{-1} \sum_{s \in S} f(s)$ and comultiplication $\delta_S(p_s) = |S|p_s \otimes p_s$, where p_s is the characteristic function of s . For pullback j along $G/H \rightarrow G/K$, the Frobenius mate j^* is fibrewise averaging, so $j^*j = \text{id}$; the symmetric pairings also give $j^{**} = j$. Thus $\text{Fun}(G/K)$ is a separable Frobenius subalgebra of $\text{Fun}(G/H)$. The endomorphism dimension in (71) is the number of G -orbits on $(G/H) \times (G/H)$, namely the number of double cosets.

Internal detail 57. For the averaged Frobenius structures and $j : \text{Fun}(G/K) \rightarrow \text{Fun}(G/H)$, the mate is explicitly

$$(j^*f)(gK) = \frac{1}{[K : H]} \sum_{xH \subset gK} f(xH).$$

Indeed, for every $a \in \text{Fun}(G/K)$, $\epsilon_{G/H}(f j(a)) = \epsilon_{G/K}((j^*f)a)$, since $|G/H| = [K : H]|G/K|$. Hence $j^*j = \text{id}$. The pairings are symmetric, so their biduality maps are identities and $j^{**} = j$. Moreover, $m_S \delta_S = |S| \text{id}$; the scalar is nonzero over $\mathbb{C}$, so $|S|^{-1} \delta_S$ is a bimodule section. With

counting-sum counts on both spaces one would instead have $j^*j = [K : H] \text{id}$, which explains why the normalization was specified.

Every unital subalgebra of $\text{Fun}(S)$ for a finite set S is $\text{Fun}(S/\sim)$ for a unique partition: two points are equivalent when all functions in the subalgebra take the same value on them; finite-dimensional semisimplicity supplies the characteristic functions of the resulting blocks. For $S = G/H$, G -stability is exactly invariance of the partition. The block containing H then has stabilizer K with $H \leq K \leq G$, and all blocks are its G -translates, giving the quotient $G/H \rightarrow G/K$. Finally,

$$\text{End}_{\text{Rep}(G)}(\mathbb{C}[G/H]) \cong \text{Fun}((G/H) \times (G/H))^G,$$

as vector spaces, so their $\mathbb{C}$ -dimension is the number of diagonal G -orbits. Sending $(gH, g'H)$ to the double coset $Hg^{-1}g'H$ identifies these orbits with $H \backslash G/H$.

Apply (69). □

The same estimate also follows directly: every intermediate subgroup is a union of H -double cosets containing H , leaving at most $2^{|H \backslash G/H| - 1}$ possibilities.

Internal detail 58. Indeed, if $H \leq K$, then $h_1kh_2 \in K$ for all $k \in K$ and $h_1, h_2 \in H$, so K is a union of H -double cosets. The double coset H itself must occur. There are therefore at most two choices—*included or not*—for each of the remaining $|H \backslash G/H| - 1$ double cosets. Not every such union is a subgroup, which is why this is only an upper bound.

The categorical derivation is a useful consistency check: the dimension appearing in the exponent becomes this familiar permutation-theoretic invariant.

Theorem 9.6 (Sharp finite-group interval bound; Theorem 1.1 of [36]). *Let $H \leq G$ be finite groups, put $n = [G : H] > 1$, and let p be the least prime divisor of n . Set*

$$c(p) := \left(\prod_{j=1}^{\infty} (1 - p^{-j})^{-1} \right) \left(\sum_{z \in \mathbb{Z}} p^{-z^2} \right).$$

Then

$$|\{K : H \leq K \leq G\}| < c(p)n^{\frac{1}{4}\log_p n}.$$

In particular,

$$|\{K : H \leq K \leq G\}| < c(2)n^{\frac{1}{4}\log_2 n} < 7.371968802 n^{\frac{1}{4}\log_2 n}.$$

The coefficient $\frac{1}{4}$ in the exponent and, with this coefficient fixed, the constant $c(p)$ are optimal among inclusions whose index has least prime divisor p : for $H = 1$ and $G = (\mathbb{Z}/p\mathbb{Z})^{2a}$,

$$\frac{|\{K : 1 \leq K \leq G\}|}{|G|^{\frac{1}{4}\log_p |G|}} \longrightarrow c(p) \quad (a \longrightarrow \infty).$$

This index-only estimate is complementary to Corollary 9.5: it is asymptotically much smaller than the worst-case bound 2^{n-1} , whereas the double-coset bound can be sharper for a particular action when $|H \backslash G/H|$ is small. For $H = 1$ it specializes to the sharp absolute estimate of Fusari–Spiga [18]; it is its relative analogue and improves Guerra–Mastrogiacomo–Spiga [22] by removing the factor $n^{1.8919}$. The proof in [36] counts relative generating tuples by their subgroup index and then estimates a discrete Gaussian sum. It is elementary and avoids the classification of finite simple groups, suggesting a possible route to categorical refinements. The cited work also provides a Lean formalization of its principal statements, registered in Palomar.

Internal detail 59. No normality or core-freeness assumption on H is used. The specialization $H = 1$ recovers the absolute subgroup-counting problem. Fusari–Spiga obtain the optimal absolute

exponent using the classification of finite simple groups, while the new relative proof is elementary. The earlier relative bound of Guerra–Mastrogiamomo–Spiga was

$$7.3722 n^{\frac{1}{4} \log_2 n + 1.8919};$$

thus the new result removes its polynomial loss.

10. APPLICATIONS TO UNITARY TENSOR CATEGORIES AND OPERATOR-ALGEBRA INCLUSIONS

We first specialize the general rigidity results to unitary Frobenius algebras, and then apply them to finite-index operator-algebra inclusions.

10.1. Unitary Frobenius algebras. Let $\mathcal{C}$ be a unitary tensor category, with possibly infinitely many simple isomorphism classes; see [15, Remark 9.4.7] for the fusion case. A *normalized unitary Frobenius algebra* is a connected Frobenius algebra $(B, m_B, e_B, \delta_B, \epsilon_B)$ such that

$$\begin{aligned} \delta_B &= m_B^\dagger, & \epsilon_B &= e_B^\dagger, & e_B^\dagger \circ e_B &= \text{id}_1, \\ m_B \circ \delta_B &= \lambda \text{id}_B \quad (\lambda > 0). \end{aligned}$$

The isometric-unit identity is the normalization; the last identity is specialness, with strictly positive scalar. These are the normalized connected Q -system conventions of [8, Definitions 2.12 and 2.16–2.17], expressed in Frobenius-algebra terminology; see also [32, 6, 27, 28].

Internal detail 60. Remark 9.4.7 of [15] states the definition for a unitary fusion category. We use its standard locally finite extension, allowing infinitely many simple isomorphism classes because an infinite-depth subfactor need not generate a fusion category. In the correspondence model used below, [9, Remark 3.13] gives precisely the resulting unitary tensor category when the tensor unit is simple. For categorical dimension and the conjugate equations see [33, §§2–3], and for the Q -system–extension correspondence see [6, §§3.1–3.3].

The normalization $e_B^\dagger e_B = \text{id}_1$ is essential for fixed-object finiteness up to unitary equivalence. On $B = \mathbb{C}$, for every $t > 0$ the scalar structure $(m, e, \delta, \epsilon) = (t, t^{-1}, t, t^{-1})$ is connected, dagger Frobenius, and special. Its unit has norm t^{-1} , so different values of t cannot be related by a unitary algebra isomorphism. The isometric-unit condition removes this freedom. More generally, if $e_B^\dagger e_B = c \text{id}_1$ with $c > 0$, replace $(m_B, e_B, \delta_B, \epsilon_B)$ by $(\sqrt{c} m_B, c^{-1/2} e_B, \sqrt{c} \delta_B, c^{-1/2} \epsilon_B)$. The unit becomes isometric and the Frobenius and dagger identities remain valid. This is the normalization required in [8, Definition 2.7 and Theorem 3.9].

A morphism $u : B \rightarrow B'$ is unitary if $u^\dagger \circ u = \text{id}_B$ and $u \circ u^\dagger = \text{id}_{B'}$.

Specialness gives the bimodule section and separability morphism

$$(73) \quad s_B := \lambda^{-1} \delta_B, \quad p_B := s_B \circ e_B, \quad m_B \circ s_B = \text{id}_B.$$

The Frobenius identities make s_B a B – B -bimodule morphism. Thus every normalized unitary Frobenius algebra is separable in the sense of §2.3.

Internal detail 61. Indeed, $\delta_B = m_B^\dagger$ and the Frobenius equations are

$$\delta_B \circ m_B = (m_B \otimes \text{id}_B) \circ (\text{id}_B \otimes \delta_B) = (\text{id}_B \otimes m_B) \circ (\delta_B \otimes \text{id}_B),$$

the left- and right-linearity identities for $\delta_B : B \rightarrow B \otimes B$. Scalar multiplication preserves both identities, and specialness gives

$$m_B \circ s_B = \lambda^{-1} m_B \circ \delta_B = \text{id}_B.$$

Hence s_B is a bimodule section of multiplication. Lemma 2.1 then gives the separability morphism $p_B = s_B \circ e_B$ and its centrality and normalization identities.

By [8, Theorem 3.9], every algebra isomorphism between normalized connected unitary Frobenius algebras is unitary. The algebraic rigidity results therefore give the following.

Corollary 10.1 (Fixed-object unitary finiteness). *Every object B of $\mathcal{C}$ supports only finitely many unitary-equivalence classes of normalized unitary Frobenius algebra structures.*

Proof. Every such algebra is separable by (73). Apply Theorem 5.1, followed by [8, Theorem 3.9]. $\square$

Corollary 10.2 (Unitary Frobenius subalgebras). *A connected normalized unitary Frobenius algebra X in $\mathcal{C}$ has only finitely many normalized unitary Frobenius subalgebras up to equivalence over X . Moreover, their number is at most*

$$2^{r_X} \leq 2^{\dim_{\mathcal{C}} \text{End}_{\mathcal{C}}(X)-1}.$$

Proof. The ambient algebra X is connected, and every normalized unitary Frobenius subalgebra is separable by (73). Apply Corollary 8.3 and Theorem 8.9. Any algebra isomorphism realizing equivalence over X is itself unitary by [8, Theorem 3.9], so the equivalence still commutes with the embeddings into X . $\square$

Internal detail 62. The local-rigidity principle behind these statements appears to have been folklore in the unitary tensor-category community, but, to the best of our knowledge, the above formulation and proof have not previously appeared in print. The submission version confines the historical attribution to the acknowledgments.

10.2. Finite-index C^* -algebra inclusions. Throughout this subsection, finite index means *finite Watatani index*: the faithful conditional expectation (a norm-one projection) under consideration admits a finite quasi-basis. Let $A \subseteq C$ be a unital inclusion with such an expectation $E : C \rightarrow A$, and assume that $Z(A)$ is finite-dimensional. Equip C_A with the inner product $\langle x, y \rangle_A = E(x^*y)$, let C_1 be the basic construction, and put $z_E := \text{Ind}(E) \in Z(C)_+^\times$. We distinguish this index from the Pimsner–Popa index; see [9, Example 3.10 and the paragraph following Lemma 3.11] and [39].

An intermediate unital C^* -algebra $A \subseteq B \subseteq C$ is *E -compatible* if there is a conditional expectation $F_B : C \rightarrow B$ such that $E = (E|_B)F_B$ [21, Definition 10.4]. Both endpoints are included. Two compatible intermediates B, D are *unitarily conjugate* if $D = uBu^*$ for some unitary $u \in A' \cap C$. This compares compatible intermediates without assuming that every such conjugation preserves E -compatibility.

Let $\mathcal{C}$ be the dualizable part of $C^*\text{Alg}(A, A)$, with adjointable bimodule maps, finite direct sums, and split projections. Since $\text{End}({}_A A_A) = Z(A)$ is finite-dimensional, the conjugate equations give Hom-finiteness; splitting minimal projections gives semisimplicity and finite length. This is the unitary multitensor setting of [9, Lemma 3.11 and Remark 3.13].

Internal detail 63. Here is a direct finite-center argument. Choose a faithful state τ on the finite-dimensional algebra $\text{End}(\mathbf{1}) = Z(A)$. There is a constant $c > 0$ such that $\|z\| \leq c\tau(z)$ for every $z \geq 0$ in this algebra. For a dualizable object W , choose a conjugate solution $R : \mathbf{1} \rightarrow \overline{W} \otimes W$ and $\overline{R} : \mathbf{1} \rightarrow W \otimes \overline{W}$. If $T \geq 0$ in $\text{End}(W)$, the conjugate equation gives

$$T^{1/2} = (\overline{R}^\dagger \otimes \text{id}_W)(\text{id}_W \otimes \text{id}_{\overline{W}} \otimes T^{1/2})(\text{id}_W \otimes R).$$

Consequently, with $\Phi(T) := \tau(R^\dagger(\text{id}_{\overline{W}} \otimes T)R)$,

$$\|T\| \leq \|\overline{R}\|^2 \|R^\dagger(\text{id}_{\overline{W}} \otimes T)R\| \leq c\|\overline{R}\|^2 \Phi(T).$$

If $T_1, \dots, T_k$ are mutually orthogonal positive contractions of norm one, then $\sum_i T_i \leq \text{id}_W$, so $k \leq c\|\overline{R}\|^2 \Phi(\text{id}_W)$. An infinite-dimensional C^* -algebra has arbitrarily large finite families of nonzero orthogonal positive elements; normalizing them contradicts this bound. Thus $\text{End}(W)$ is finite-dimensional. Applying the same argument to direct sums gives Hom-finiteness. Finally, minimal projections in these finite-dimensional endomorphism algebras split each object into finitely many simple summands.

By [9, Lemma 3.11], the correspondence $X = {}_A C_A$, with its usual multiplication m and unit e , is a dagger Frobenius algebra. Its structure maps satisfy

$$(74) \quad e^\dagger = E, \quad mm^\dagger = L_{z_E},$$

where L_x denotes left multiplication by x . The central positive invertibility of z_E makes $m^\dagger L_{z_E}^{-1}$ a bimodule section of m , so X is separable.

Internal detail 64. For a finite right frame (u_i) of C_A , the formula in [9, Lemma 3.11] is

$$m^\dagger(c) = \sum_i u_i \otimes u_i^* c, \quad z_E = \sum_i u_i u_i^*.$$

Thus $mm^\dagger(c) = z_E c$. The Frobenius identities make $m^\dagger$ bimodular, and centrality of z_E makes $L_{z_E}^{-1}$ bimodular. Their composite is therefore a bimodule section of m . These are the ordinary multiplication and unit; the optional central rescaling discussed after the cited lemma is not used.

The inclusion correspondence $Y := {}_A C_C$ has dual $Y^\vee := {}_C C_A$, and $X \cong Y \otimes_C Y^\vee$ [9, Example 3.25]. Adjunction and evaluation at 1 give

$$(75) \quad \text{Hom}_{A-A}(A, X) \cong \text{Hom}_{A-A}(A, Y \otimes_C Y^\vee) \cong \text{End}_{A-C}(Y) \cong A' \cap C.$$

Directly, $x \in A' \cap C$ corresponds to the map $a \mapsto ax$; under this identification $\Gamma(X)$ has the ordinary multiplication of $A' \cap C$. Thus irreducibility is exactly connectedness.

Moreover,

$$(76) \quad \text{End}_C(X) = A' \cap C_1.$$

Indeed, let (u_i) be a finite right frame of C_A and write $e_A(c) = E(c)$. For every $T \in \mathcal{L}_A(C_A)$,

$$T = \sum_i \theta_{T u_i, u_i}, \quad \theta_{x,y}(c) = x E(y^* c), \quad \theta_{x,y} = L_x e_A L_y^*.$$

Hence $\mathcal{L}_A(C_A) = \mathcal{K}_A(C_A) = C_1$. These operators are already right A -linear; they are bimodule maps exactly when they commute with all L_a , $a \in A$, proving (76).

Together with $Z(A) \subseteq A' \cap C \subseteq A' \cap C_1$, this shows that, under finite Watatani index, finite-dimensionality of $Z(A)$, of $A' \cap C$, and of $A' \cap C_1$ are equivalent conditions. Finite index alone implies none of them, as the index-two example below shows.

Internal detail 65. The frame identity $c = \sum_i u_i E(u_i^* c)$ gives $T(c) = \sum_i (T u_i) E(u_i^* c)$, proving the displayed expansion of T . Also $e_A = \theta_{1,1}$ and $L_x = \sum_i \theta_{x u_i, u_i}$. Thus the C^* -algebra generated by the left action of C and the Jones projection e_A is precisely $\mathcal{K}_A(C_A)$, which equals $\mathcal{L}_A(C_A)$ by the frame expansion. Finally, $T L_a = L_a T$ means $T(a c) = a T(c)$. The inclusion $C \hookrightarrow C_1$, $x \mapsto L_x$, is faithful because $L_x(1) = x$.

For the preceding Hom-space identification, a bimodule map $f : A \rightarrow C$ satisfies $f(a) = a f(1) = f(1) a$. Conversely, for $x \in A' \cap C$, the map $f_x(a) = a x$ is adjointable, with $f_x^\dagger(c) = E(x^* c)$. An A - C endomorphism of $Y = {}_A C_C$ is likewise left multiplication by its value at 1. Both descriptions identify the adjunction chain with the same relative commutant, and $m(f_x \otimes f_y)(1) = x y$ gives its ordinary multiplication.

If B is E -compatible, its inclusion $j : {}_A B_A \rightarrow X$ is adjointable, with $j^\dagger = F_B$ [21, Proposition 10.5]. Thus B_A is a direct summand of C_A , hence finitely generated projective. The restricted expectation makes ${}_A B_A$ a separable dagger Frobenius algebra. Compatibility and bimodularity identify the Frobenius transpose of j with F_B , so $j^* j = \text{id}$ and $j^{**} = j$. Consequently every E -compatible intermediate gives a separable Frobenius subalgebra of X .

Internal detail 66. The adjointability calculation does not require irreducibility. For $b \in B$ and $c \in C$, compatibility and B -bimodularity give

$$\langle j(b), c \rangle_C = E(b^* c) = (E|_B)(b^* F_B(c)) = \langle b, F_B(c) \rangle_B.$$

Thus $j^\dagger = F_B$. For the bilinear Frobenius pairings, similarly,

$$E(cj(b)) = (E|_B)(F_B(c)b), \quad E(j(b)c) = (E|_B)(bF_B(c)).$$

Their nondegeneracy shows respectively that the Frobenius transpose of j is F_B , and that the transpose of F_B is j . Therefore $j^*j = F_Bj = \text{id}$ and $j^{**} = j$. The projection jF_B makes B_A a direct summand of the finitely generated projective module C_A . Applying [9, Lemma 3.11] to $E|_B$ supplies its dagger Frobenius structure; its positive invertible central index gives its separability section exactly as in (74).

Theorem 10.3 (General C^* -algebra bound). *Let $A \subseteq C$ be a unital inclusion with a faithful conditional expectation E of finite Watatani index, and suppose that $Z(A)$ is finite-dimensional. The number of E -compatible intermediate unital C^* -algebras, up to conjugation by unitaries in $A' \cap C$, is at most*

$$2^{\dim_{\mathbb{C}}(A' \cap C_1) - 1}.$$

If $Z(A) = \mathbb{C}1$, this bound is at most

$$2^{\lfloor \|\text{Ind}(E)\|^2 \rfloor - 1} < 2^{\|\text{Ind}(E)\|^2}.$$

Proof. Theorem 8.9 and (76) give $2^{\dim_{\mathbb{C}}(A' \cap C_1) - 1}$ for conjugacy by invertible elements of $A' \cap C$. Invertible and unitary conjugacy agree for these subalgebras. Indeed, if $uBu^{-1} = D$, self-adjointness gives $hBh^{-1} = B$, where $h = u^*u$. The relative commutant is finite-dimensional, so h has finite spectrum. Polynomial interpolation on the finitely many spectral ratios of h expresses $\text{Ad}_{h^{-1/2}}$ as a polynomial in Ad_h . Hence $h^{-1/2}$ normalizes B , and the unitary $uh^{-1/2}$ still sends B onto D .

Internal detail 67. To spell out the polar-decomposition step, taking adjoints in $uBu^{-1} = D$ gives $uBu^{-1} = (u^*)^{-1}Bu^*$, hence $hBh^{-1} = B$ for $h = u^*u$. Write $h = \sum_i \lambda_i e_i$ with $\lambda_i > 0$. On $e_i C e_j$, Ad_h acts by λ_i/λ_j . Choose polynomials $p_{\pm}$ satisfying $p_{\pm}(\lambda_i/\lambda_j) = (\lambda_i/\lambda_j)^{\pm 1/2}$. Then $\text{Ad}_{h^{\pm 1/2}} = p_{\pm}(\text{Ad}_h)$. Both maps preserve the linear space B , and they are inverse, so they carry B onto itself. The element $v = uh^{-1/2}$ is unitary, lies in $A' \cap C$, and satisfies $vBv^* = D$. No invariance of E under this conjugation is asserted or used.

If $Z(A) = \mathbb{C}1$, the Frobenius cup $R = m^\dagger e$ is a selfdual conjugate solution for X , and

$$(77) \quad R^\dagger R = E(z_E), \quad d(X) \leq \|E(z_E)\| \leq \|\text{Ind}(E)\|.$$

Writing $X = \bigoplus_i S_i^{\oplus n_i}$ and using $d(S_i) \geq 1$ gives

$$\dim_{\mathbb{C}}(A' \cap C_1) = \sum_i n_i^2 \leq \left(\sum_i n_i \right)^2 \leq d(X)^2 \leq \|\text{Ind}(E)\|^2.$$

Since $\dim_{\mathbb{C}}(A' \cap C_1)$ is an integer, this proves the second bound.

Internal detail 68. The evaluation $e^\dagger m$ and coevaluation $m^\dagger e$ satisfy the two conjugate equations by the Frobenius and unit identities. Their norms therefore bound the intrinsic dimension, even when they are not standard solutions. Since

$$(m^\dagger e)^\dagger (m^\dagger e) = e^\dagger L_{z_E} e = E(z_E),$$

we obtain $d(X) \leq \|E(z_E)\|$ when $Z(A) = \mathbb{C}1$. This is an upper bound, not an identification of a nonstandard expectation with the minimal-index expectation. In the irreducible case $z_E = \|z_E\|1_C$, so the upper bound is exactly $\|\text{Ind}(E)\|$.

□

Corollary 10.4 (Irreducible C^* -algebra bound). *Let $A \subseteq C$ be a unital irreducible inclusion with a faithful conditional expectation E of finite Watatani index. Its Watatani index is scalar, and the number of E -compatible intermediate unital C^* -algebras is at most*

$$2^{\dim_{\mathbb{C}}(A' \cap C_1) - 1} \leq 2^{\|\text{Ind}(E)\| - 1} < 2^{\|\text{Ind}(E)\|}.$$

Neither algebra is required to be simple.

Proof. Irreducibility implies $Z(A) = Z(C) = \mathbb{C}1$ and makes X connected by (75). Inner conjugation is trivial. The dimension comparison in Proposition 10.7 and (77) give $\dim_{\mathbb{C}}(A' \cap C_1) \leq d(X) \leq \|\text{Ind}(E)\|$. $\square$

The qualitative finiteness in this corollary was proved in [21, Corollary 10.6]. The restriction to E -compatible intermediates is essential: the same paper gives an irreducible finite-index inclusion with uncountably many unrestricted intermediate C^* -algebras [21, Example 10.7]. The finite-center hypothesis in the general theorem also has content. For the diagonal inclusion $C(K) \subseteq C(K) \oplus C(K)$ with the averaging expectation, the Watatani index is 2, whereas $A' \cap C = C(K) \oplus C(K)$ and $A' \cap C_1 = C_1 \cong M_2(C(K))$. Thus neither relative commutant is finite-dimensional when K is infinite. Every clopen $U \subseteq K$ gives an E -compatible intermediate

$$B_U = \{(f, g) : f = g \text{ on } K \setminus U\}.$$

If K is the Cantor set, these give infinitely many distinct classes, since the ambient algebra is commutative. If K has n points, they give 2^n classes at the same index 2.

Internal detail 69. For the diagonal inclusion, the two elements $\sqrt{2}(1, 0)$ and $\sqrt{2}(0, 1)$ form a quasi-basis for the averaging expectation; their squared products sum to $2 \cdot 1_C$. This frame identifies C_A with A^2 and the left action of A with scalar matrices. Hence $C_1 \cong M_2(A)$ and $A' \cap C_1 = C_1$; also $A' \cap C = C$ because C is commutative. For a clopen U , let F_U act as the identity over U and as averaging over $K \setminus U$. This is a continuous unital positive B_U -bimodule projection, and $EF_U = E$. Distinct clopen subsets give distinct B_U . All inner conjugations are trivial because C is commutative. This proves both the infinite-center counterexample and the absence of a uniform index-only bound independent of the center size, even when the center is finite-dimensional.

Remark 10.5 (Von Neumann algebra inclusions). The preceding C^* -algebra bounds apply to the E -compatible intermediate von Neumann algebras of a unital inclusion $N \subseteq M$ equipped with a faithful normal conditional expectation $E : M \rightarrow N$ of finite Watatani index, with $Z(N)$ finite-dimensional. Indeed, these intermediates belong to the collection of intermediate C^* -algebras already counted. The unitary inner-conjugacy convention is unchanged, and no factoriality assumption is imposed.

If M admits a faithful normal tracial state τ and E is τ -preserving, every intermediate von Neumann algebra P is E -compatible: its τ -preserving conditional expectation $E_P : M \rightarrow P$ satisfies $(E|_P) \circ E_P = E$, by [1, Theorem 9.1.2 and Lemma 9.1.4(v)].

Finally, an irreducible inclusion of von Neumann algebras is already an inclusion of factors, since $Z(N), Z(M) \subseteq N' \cap M = \mathbb{C}1$.

Internal detail 70. A von Neumann subalgebra is norm closed, so it is an intermediate C^* -algebra; the unitary conjugacy relation on this smaller collection is the restriction of the relation already used above. If $F : M \rightarrow P$ is E -compatible, it is automatically normal. Indeed, for a bounded increasing net $0 \leq x_i \uparrow x$, put $y = \sup_i F(x_i) \in P$. Then $0 \leq y \leq F(x)$, and normality of E gives

$$E(F(x) - y) = E(x) - \sup_i E(x_i) = 0.$$

Faithfulness of E gives $y = F(x)$.

In the tracial case, Theorem 9.1.2 of [1] identifies the prescribed expectation E with E_N by uniqueness. Lemma 9.1.4(v), applied with $B = P$ and $D = N$, then gives $E_N \circ E_P = E_N$, exactly the required compatibility.

10.3. Finite-index subfactors. Let $N \subseteq M$ be a finite-index inclusion of II_1 factors, let $I := [M : N]$ be its Jones index, and let $E_N : M \rightarrow N$ be the trace-preserving conditional expectation. Its Watatani index is $I1_M$. Every intermediate von Neumann algebra $N \subseteq P \subseteq M$ is E_N -compatible, because its trace-preserving expectation satisfies $E_N = E_N|_P \circ E_P$. Such an intermediate need not be a factor when $N \subseteq M$ is reducible.

Let $\mathcal{L}(N \subseteq M)$ denote all these intermediate von Neumann algebras, including N and M , and let $M_1 := \langle M, e_N \rangle$ be the basic construction. Let $\mathcal{C}$ now denote the unitary tensor category of dualizable N - N correspondences. The correspondence realization above gives

$$(78) \quad H = {}_N M_M, \quad \overline{H} = {}_M M_N, \quad X = H \otimes_M \overline{H} \cong {}_N M_N,$$

and the identities

$$(79) \quad \text{Hom}_{\mathcal{C}}(\mathbf{1}, X) \cong \text{End}_{N-M}(H) \cong N' \cap M, \quad \text{End}_{\mathcal{C}}(X) \cong N' \cap M_1.$$

In particular, irreducibility is exactly connectedness. The cup estimate (77) gives $d(X) \leq I$; this is all that is needed below, with no standardness assumption on this chosen Frobenius duality.

Internal detail 71. The notation ${}_N M_N$ uses finitely generated projective right Hilbert modules, with the N -valued inner product from E_N . This is the finite-index correspondence model in [9, Remark 2.20, Example 3.10, Lemma 3.11, and Example 3.25]. Its finite right frame identifies the basic construction with the full adjointable endomorphism algebra, giving $\text{End}_{N-N}({}_N M_N) = N' \cap M_1$. For the properly infinite factor formulation, the corresponding reconstruction and intermediate-extension correspondence are [6, Theorem 3.11 and Proposition 4.11]; the dimension there is related to minimal index as explained in [6, §2.2]. Let τ be the normalized trace of M . For a finite right frame (u_i) , the canonical basic-construction trace Tr satisfies $\text{Tr}(xe_N y) = \tau(yx)$ and $\text{Tr}(1) = [M : N]$. Hence

$$[M : N] = \sum_i \tau(u_i^* u_i) = \tau\left(\sum_i u_i u_i^*\right) = \tau(z_{E_N}).$$

Since z_{E_N} is central in the factor M , this proves $z_{E_N} = [M : N]1_M$, including in the nonirreducible case.

Theorem 10.6 (General intermediate-algebra bound). *For a finite-index inclusion $N \subseteq M$ of II_1 factors, the number of intermediate von Neumann algebras, up to conjugation by unitaries in $N' \cap M$, is at most*

$$2^{\dim_{\mathbb{C}}(N' \cap M_1) - 1} \leq 2^{\lfloor [M:N]^2 \rfloor - 1} < 2^{[M:N]^2}.$$

The same bounds hold for intermediate subfactors up to this conjugacy.

Proof. Apply Theorem 10.3 to the inclusion with its trace-preserving expectation, and restrict to intermediate von Neumann algebras. In this case conjugation by $\mathcal{U}(N' \cap M)$ preserves the trace and the set of compatible intermediates. $\square$

The general theorem strengthens the qualitative statement to an explicit estimate. Compare the related partial finiteness result for two distinguished families in [30]. Under irreducibility, conjugation becomes trivial, recovering the qualitative finiteness theorem of [40].

10.4. Endomorphism dimension versus categorical dimension. The following categorical statement gives the required bound.

Proposition 10.7 (Dimension comparison). *Let $\mathcal{C}$ be a positive semisimple tensor category over $\mathbb{C}$, in the sense of [21, Definition 6.3], with positive pivotal dimension d , and let A be a connected finite-length separable algebra object in $\mathcal{C}$. Write*

$$(80) \quad A \cong \mathbf{1} \oplus \bigoplus_{i=1}^t S_i^{\oplus n_i},$$

where the $S_i \not\cong \mathbf{1}$ are pairwise nonisomorphic simple objects. Then

$$(81) \quad n_i \leq d(S_i) \quad (1 \leq i \leq t),$$

and consequently

$$(82) \quad \dim_{\mathbb{C}} \text{End}_{\mathcal{C}}(A) \leq d(A).$$

More precisely,

$$(83) \quad d(A) - \dim_{\mathbb{C}} \text{End}_{\mathcal{C}}(A) = \sum_{i=1}^t n_i(d(S_i) - n_i).$$

Proof. Let $\mathcal{C}_A$ be the category of right A -modules. Choose a separability morphism $p_A : \mathbf{1} \rightarrow A \otimes A$. If $q : M \rightarrow N$ is a module map and $t : N \rightarrow M$ is a splitting in $\mathcal{C}$, let $r_Y : Y \otimes A \rightarrow Y$ denote the action on a right module Y and set

$$\sigma_Y := (r_Y \otimes \text{id}_A) \circ (\text{id}_Y \otimes p_A), \quad \tilde{t} := r_M \circ (t \otimes \text{id}_A) \circ \sigma_N.$$

Centrality of p_A makes σ_Y and $\tilde{t}$ right A -linear, while normalization gives $r_Y \circ \sigma_Y = \text{id}_Y$ and $q \circ \tilde{t} = \text{id}_N$.

Internal detail 72. The four assertions can be checked as follows. Writing $R_p = (\text{id}_A \otimes m_A) \circ (p_A \otimes \text{id}_A)$ and using centrality of p_A , right module associativity gives

$$\sigma_Y \circ r_Y = (r_Y \otimes \text{id}_A) \circ (\text{id}_Y \otimes R_p) = (\text{id}_Y \otimes m_A) \circ (\sigma_Y \otimes \text{id}_A),$$

which is right A -linearity of σ_Y . Normalization gives

$$r_Y \circ \sigma_Y = r_Y \circ (r_Y \otimes \text{id}_A) \circ (\text{id}_Y \otimes p_A) = r_Y \circ (\text{id}_Y \otimes (m_A \circ p_A)) = r_Y \circ (\text{id}_Y \otimes e_A) = \text{id}_Y.$$

Using the displayed right-linearity identity for σ_N and right module associativity gives

$$\begin{aligned} \tilde{t} \circ r_N &= r_M \circ (t \otimes \text{id}_A) \circ (\text{id}_N \otimes m_A) \circ (\sigma_N \otimes \text{id}_A) \\ &= r_M \circ (r_M \otimes \text{id}_A) \circ (t \otimes \text{id}_A \otimes \text{id}_A) \circ (\sigma_N \otimes \text{id}_A) \\ &= r_M \circ (\tilde{t} \otimes \text{id}_A), \end{aligned}$$

so $\tilde{t}$ is right A -linear. Because q is right A -linear,

$$q \circ \tilde{t} = r_N \circ (q \otimes \text{id}_A) \circ \sigma_N = r_N \circ \sigma_N = \text{id}_N.$$

Thus every split epimorphism of underlying objects splits in the module category; in a semisimple ambient category this proves semisimplicity of $\mathcal{C}_A$.

Thus splittings can be averaged to module splittings, and $\mathcal{C}_A$ is semisimple. The regular right module A_A is simple, because

$$\text{End}_{\mathcal{C}_A}(A_A) \cong \text{Hom}_{\mathcal{C}}(\mathbf{1}, A) = \mathbb{C}$$

by connectedness.

Internal detail 73. The isomorphism sends a right A -linear map $T : A \rightarrow A$ to $T \circ e_A$. Its inverse sends $a : \mathbf{1} \rightarrow A$ to left multiplication by a , namely $m_A \circ (a \otimes \text{id}_A)$; this is a right A -module

map. Right A -linearity of T gives $T = m_A \circ ((T \circ e_A) \otimes \text{id}_A)$, so the maps are inverse. In a semisimple category an object is simple exactly when its endomorphism algebra is $\mathbb{C}$ under the Schur convention of Definition 2.2; hence connectedness makes A_A simple.

For each i , the free–forgetful adjunction gives

Internal detail 74. The free right module on V is $V \otimes A$, and the adjunction is the explicit bijection

$$\text{Hom}_{\mathcal{C}_A}(V \otimes A, M) \longleftrightarrow \text{Hom}_{\mathcal{C}}(V, M), \quad F \mapsto F \circ (\text{id}_V \otimes e_A),$$

whose inverse sends $g : V \rightarrow M$ to $r_M \circ (g \otimes \text{id}_A)$. Taking $V = S_i$ and $M = A$ gives the equality below directly; no dual-object identification is needed.

$$\dim_{\mathbb{C}} \text{Hom}_{\mathcal{C}_A}(S_i \otimes A, A) = \dim_{\mathbb{C}} \text{Hom}_{\mathcal{C}}(S_i, A) = n_i.$$

Hence, in the semisimple module category,

$$S_i \otimes A \cong A^{\oplus n_i} \oplus Y_i$$

for some right A -module Y_i . Applying the positive pivotal dimension to the underlying objects gives

$$d(S_i)d(A) = n_id(A) + d(Y_i),$$

so $n_i \leq d(S_i)$.

Internal detail 75. The forgetful functor $\mathcal{C}_A \rightarrow \mathcal{C}$ sends a module isomorphism to an isomorphism of its underlying objects. Pivotal dimension is invariant under isomorphism, additive on direct sums, and multiplicative on tensor products [15, Proposition 4.7.12]. Positivity gives $d(A) > 0$ and $d(Y_i) \geq 0$. Thus the underlying-object isomorphism gives the displayed equality, and division by $d(A)$ yields $d(S_i) = n_i + d(Y_i)/d(A) \geq n_i$.

Finally,

$$d(A) = 1 + \sum_i n_i d(S_i), \quad \dim_{\mathbb{C}} \text{End}_{\mathcal{C}}(A) = 1 + \sum_i n_i^2,$$

and subtraction proves (82) and (83). $\square$

From this point through the end of this section, assume that $N \subseteq M$ is irreducible. Then the inclusion correspondence H is simple. A balanced conjugate solution for a simple correspondence is standard; the dimension–index identity therefore gives

$$(84) \quad d(X) = d(H)^2 = [M : N] = I.$$

Here the Jones and minimal indices agree; see [6, §2.2]. Applying Proposition 10.7 to the connected unitary Frobenius algebra $X = {}_N M_N$ gives

$$(85) \quad \dim_{\mathbb{C}}(N' \cap M_1) = \dim_{\mathbb{C}} \text{End}_{\mathcal{C}}(X) \leq d(X) = [M : N].$$

Internal detail 76. The equality just used is scoped to irreducible inclusions. For $H = {}_N M_M$, adjunction identifies $\text{End}_{N-M}(H)$ with $N' \cap M = \mathbb{C}$, so H is simple. Balance then makes its conjugate solution standard: any two such solutions differ by an invertible element of this one-dimensional endomorphism algebra, and balancing fixes its absolute value. The trace-preserving inclusion solution has dimension product $[M : N]$; hence $d(H)^2 = [M : N]$. Tensor multiplicativity gives $d(X) = d(H)d(\overline{H}) = [M : N]$. The earlier nonirreducible theorem required only the upper bound $d(X) \leq I$, and did not use this equality.

The proof also makes the difference transparent. If

$$(86) \quad X \cong {}_N M_N \oplus \bigoplus_{\alpha} V_{\alpha}^{\oplus n_{\alpha}},$$

then

$$(87) \quad [M : N] - \dim_{\mathbb{C}}(N' \cap M_1) = \sum_{\alpha} n_{\alpha}(d(V_{\alpha}) - n_{\alpha}).$$

Indeed, bimodule Frobenius reciprocity gives

$$n_{\alpha} = \dim_{\mathbb{C}} \operatorname{Hom}_{\mathcal{C}}(V_{\alpha}, X) = \dim_{\mathbb{C}} \operatorname{Hom}_{N-M}(V_{\alpha} \otimes_N H, H).$$

Irreducibility gives $\operatorname{End}_{N-M}(H) \cong N' \cap M = \mathbb{C}$, so H is simple. Thus

$$V_{\alpha} \otimes_N H \cong H^{\oplus n_{\alpha}} \oplus W_{\alpha}$$

and

$$(88) \quad d(V_{\alpha}) - n_{\alpha} = \frac{d(W_{\alpha})}{d(H)} \geq 0.$$

In principal-graph language, W_{α} records the other odd vertices adjacent to V_{α} .

Internal detail 77. The adjunction $H \dashv \overline{H}$ gives the explicit Frobenius-reciprocity bijection

$$\operatorname{Hom}_{N-N}(V_{\alpha}, H \otimes_M \overline{H}) \cong \operatorname{Hom}_{N-M}(V_{\alpha} \otimes_N H, H).$$

Because H is simple, the common dimension is precisely the multiplicity of H in $V_{\alpha} \otimes_N H$, so the complementary bimodule W_{α} exists. Additivity and multiplicativity of statistical dimension give

$$d(V_{\alpha})d(H) = n_{\alpha}d(H) + d(W_{\alpha}).$$

Division by $d(H) > 0$ proves (88). On the principal graph, H is the distinguished odd vertex and the simple summands of W_{α} are the remaining adjacent odd vertices.

Thus (87) records branching beyond the distinguished depth-one vertex. In particular, equality holds in (85) for depth-two inclusions.

10.5. A direct bound for the intermediate lattice. Under the standing irreducibility assumption, $\mathcal{L}(N \subset M)$ is the lattice of intermediate subfactors, including N and M .

Theorem 10.8 (Irreducible intermediate-subfactor bound). *Let $N \subseteq M$ be a finite-index irreducible II_1 subfactor. Then*

$$(89) \quad |\mathcal{L}(N \subset M)| \leq 2^{r_X} \leq 2^{\dim_{\mathbb{C}}(N' \cap M_1) - 1} \leq 2^{\lfloor [M:N] \rfloor - 1} < 2^{[M:N]}.$$

Proof. Irreducibility makes X connected and the conjugation action trivial. It also forces every intermediate von Neumann algebra to be a factor, since $Z(P) \subseteq N' \cap M = \mathbb{C}1$. Theorem 8.9 gives the first two inequalities. Proposition 10.7 gives $\dim_{\mathbb{C}}(N' \cap M_1) \leq d(X) \leq I$, proving the rest.

Internal detail 78. Every compatible intermediate gives the projection jF_P on X ; its image recovers P , so different intermediates give different exchange idempotents. In the irreducible subfactor setting this is also Bisch's biprojection correspondence [5, Corollary 3.3], followed by the connected unitary exchange correspondence [20, Theorem 4.8]. Set $D = \dim_{\mathbb{C}}(N' \cap M_1)$. The exchange theorem gives $|\mathcal{L}(N \subset M)| \leq 2^{r_X} \leq 2^{D-1}$. The dimension comparison gives $D \leq I$. Since D is an integer, $D \leq \lfloor I \rfloor$, proving every inequality in (89). □

The relative-commutant form is considerably sharper than its index-only corollary whenever X has few simple summands. For example, a proper 2-supertransitive inclusion has $\dim_{\mathbb{C}}(N' \cap M_1) = 2$, and

Internal detail 79. By definition, 2-supertransitivity means that the first two relative commutants agree with those of the Temperley–Lieb standard invariant. At the 2-box level this algebra is spanned by the identity and the first Jones projection. For a proper inclusion these are linearly independent, so the algebra has complex dimension two. The estimate therefore reads $2^{2-1} = 2$;

since the endpoints N and M already contribute two lattice elements, no further intermediate subfactor can occur.

Theorem 10.8 gives $|\mathcal{L}(N \subset M)| \leq 2$; hence there is no nontrivial intermediate subfactor. At the opposite extreme, for a depth-two inclusion the difference (87) vanishes, so replacing the relative-commutant dimension by the index loses no information.

10.6. Comparison with angle rigidity. Throughout this comparison, $N \subset M$ is a finite-index irreducible II_1 subfactor. Set

$$I := [M : N], \quad D := \dim_{\mathbb{C}}(N' \cap M_1) = \dim_{\mathbb{R}}(N' \cap M_1)_{\text{sa}}.$$

Bakshi–Das–Liu–Ren proved that the number of minimal intermediate subfactors is at most the kissing number τ_D [4, Theorem 4.1]. Their elementary estimate $\tau_D < 3^D$, iterated over successive subintervals, gives the whole-lattice bound

$$(90) \quad |\mathcal{L}(N \subset M)| \leq 9^I \quad [4, \text{Theorem 4.6}].$$

Internal detail 80. The published numbering is Theorems 4.1 and 4.6. If E is a finite-dimensional complex C^* -algebra, the decomposition

$$x = \frac{x + x^*}{2} + i \frac{x - x^*}{2i}$$

is unique, so $E = E_{\text{sa}} \oplus iE_{\text{sa}}$ as real vector spaces and $\dim_{\mathbb{C}} E = \dim_{\mathbb{R}} E_{\text{sa}}$.

For the recursion, let $\Lambda(x)$ be the supremum of the sizes of irreducible intermediate lattices of index at most x , counting both endpoints. For $x \geq 4$, let $m(x)$ be the larger of 2 and the supremum of the number of proper minimal intermediates in these lattices. Each strict subfactor step has index at least 2. Thus every upper interval above a proper minimal intermediate has index at most $x/2$. If there are $q \geq 2$ such intermediates, every element except the bottom lies above one, and counting the endpoints separately gives

$$|\mathcal{L}| \leq 2 + q(\Lambda(x/2) - 1) \leq q\Lambda(x/2) \leq m(x)\Lambda(x/2).$$

If $q = 1$, the bound $1 + \Lambda(x/2) \leq m(x)\Lambda(x/2)$ suffices; if $q = 0$, the lattice has at most two elements. Therefore

$$\Lambda(x) \leq m(x)\Lambda(x/2) \quad (x \geq 4), \quad \Lambda(x) = 2 \quad (2 \leq x < 4).$$

The same convention for m is compatible with the kissing-number bound: $D \leq x$, the kissing numbers are nondecreasing with dimension, and $\tau_{[x]} \geq 2$, so $m(x) \leq \tau_{[x]}$. This recovers the recursion underlying the published 9^I estimate.

The bound $\tau_D \leq (\sqrt{6})^D$ in [35], whose Euclidean-volume proof is formalized in Lean 4/Mathlib and registered in Palomar, gives $|\mathcal{L}(N \subset M)| \leq 6^I$ by the same recursion.

Internal detail 81. Put $a = \sqrt{6}$. For $I \geq 2$, choose $J \geq 0$ so that $2 \leq I/2^J < 4$. The bound $m(x) \leq a^x$ and the finite recursion give

$$\Lambda(I) \leq 2 \prod_{j=0}^{J-1} a^{I/2^j} = 2a^{2I(1-2^{-J})} \leq a^{2I} = 6^I,$$

because the omitted exponent is $2I/2^J \geq 4$, so $2 \leq a^{2I/2^J}$. The index-one lattice has one element; there is no proper subfactor inclusion of index strictly between 1 and 2. Thus the terminal factor has been included, rather than silently discarded. The base $\sqrt{6}$ is optimal among coefficient-one bounds $\tau_n \leq c^n$ valid in every positive dimension because $\tau_2 = 6$. The registered snapshot in [35] covers eight principal declarations, including the bound and optimality of the universal base, proved through the Euclidean-volume construction. The current version of the

cited kissing-number manuscript also contains a shorter spherical-cap proof that is not part of that certificate. Neither asymptotic optimality nor human peer review is claimed.

The quadratic exchange bound instead counts the whole lattice directly:

$$(91) \quad |\mathcal{L}(N \subset M)| \leq 2^{D-1} \leq 2^{\lfloor I \rfloor - 1} < 6^I.$$

Thus it improves these elementary index-only bounds and is sensitive to the second relative commutant. The two methods nevertheless capture different information: angular separation controls minimal intermediates, while local rigidity isolates all exchange idempotents at once.

For asymptotic comparison, the Kabatiansky–Levenshtein bound [29], combined with the larger-angle comparison for spherical codes [10, §2, (2.5)–(2.6) and footnote 1], gives $\tau_D \leq 2^{(0.401+o(1))D}$. The angle recursion therefore yields $2^{(0.802+o(1))I}$, asymptotically sharper than the index-only specialization of (89). The local bound remains direct and can be much smaller when $D \ll I$. It may be possible to improve it further by exploiting the sparse or multihomogeneous structure of the exchange equations. When $N' \cap M_1$ is abelian, the quasi-polynomial whole-lattice estimate of [4] is asymptotically stronger than these index-only exponential estimates. The angle and local rigidity methods are therefore complementary.

Internal detail 82. The coefficient 0.401 uses the larger-angle improvement; it is not a rounding down of the direct exponent at angle $\pi/3$. To see this, put

$$F(\theta) = \frac{1 + \sin \theta}{2 \sin \theta} \log_2 \frac{1 + \sin \theta}{2 \sin \theta} - \frac{1 - \sin \theta}{2 \sin \theta} \log_2 \frac{1 - \sin \theta}{2 \sin \theta}.$$

The direct spherical-code estimate is $\limsup_{n \rightarrow \infty} n^{-1} \log_2 A(n, \theta) \leq F(\theta)$, where $A(n, \theta)$ is the largest size of a code on S^{n-1} with minimum angle θ . A cap of angular radius $\theta/2$ has normalized area $2^{(\log_2 \sin(\theta/2) + o(1))n}$. The larger-angle comparison of cap packing densities therefore implies, for $\pi/3 < \phi < \pi/2$,

$$\limsup_{n \rightarrow \infty} \frac{\log_2 \tau_n}{n} \leq 1 + \log_2 \sin(\phi/2) + F(\phi);$$

see [10, §2, (2.3), (2.5)–(2.6) and footnote 1]. For example, take $\phi = \arcsin(8/9) > \pi/3$. The right side becomes

$$\frac{5}{4} + \frac{1}{2} \log_2 \frac{9 - \sqrt{17}}{18} + \frac{17}{16} \log_2 \frac{17}{16} = 0.4009481312 \dots < 0.401.$$

In contrast, direct substitution of $\theta = \pi/3$ into F gives $0.4014135460 \dots$. This distinction justifies the smaller rounded coefficient used in the main text.

For the recursive asymptotic estimate, fix $\varepsilon > 0$ and choose $C_\varepsilon \geq 1$ so that

$$\log_2 \tau_D \leq (0.401 + \varepsilon)D + C_\varepsilon \quad (D \geq 1).$$

Such a constant covers the finitely many small dimensions as well as the asymptotic range. The same upper bound, with D replaced by x , holds for $\log_2 m(x)$. There are $J = O(\log I)$ stages before $2 \leq I/2^J < 4$, so

$$\log_2 \Lambda(I) \leq 1 + \sum_{j=0}^{J-1} ((0.401 + \varepsilon)I/2^j + C_\varepsilon) \leq 2(0.401 + \varepsilon)I + O_\varepsilon(\log I).$$

Taking $I \rightarrow \infty$ and then $\varepsilon \rightarrow 0$ gives $\Lambda(I) \leq 2^{(0.802+o(1))I}$. This does not contradict the exchange estimate: the angle bound is asymptotic and index-only, whereas the direct bound has exponent $D - 1$ and can be much smaller when $D \ll I$. Further improvements of the latter would require using the sparse supports or multihomogeneous blocks of the exchange equations.

11. EXAMPLES AND APPLICATIONS

The examples below separate the local mechanism from the additional hypotheses used in the applications.

Example 11.1 (Vector spaces). In $\text{Vec}_{\mathbb{k}}$, separability is equivalent to semisimplicity [14, Chapter II, §1], and the Wedderburn–Artin theorem [31, Chapter 1, §§2–3] classifies finite-dimensional semisimple $\mathbb{k}$ -algebras as finite direct sums of full matrix algebras over $\mathbb{k}$. Thus, for fixed $d = \dim_{\mathbb{k}} A$, there are only finitely many isomorphism classes.

Internal detail 83. For a finite-dimensional algebra over the algebraically closed field $\mathbb{k}$, separability is equivalent to semisimplicity; see [14, Chapter II, §1]. The Wedderburn–Artin theorem then gives a unique isomorphism type $\bigoplus_{j=1}^q M_{n_j}(\mathbb{k})$ up to reordering of the positive integers n_j ; see [31, Chapter 1, §§2–3]. For fixed $d = \dim_{\mathbb{k}} A$, these integers satisfy $\sum_j n_j^2 = d$, and there are only finitely many finite tuples of positive integers satisfying this equation. This proves the stated finiteness directly.

Example 11.2 (Why inner conjugacy is necessary). For $A = \mathbb{k}^2$ and $X = M_2(\mathbb{k})$, the embeddings $A \rightarrow X$ form an infinite algebraic family of rank-one idempotents, but a single $\text{GL}_2(\mathbb{k})$ -conjugacy class.

Internal detail 84. A unital map $\mathbb{k}^2 \rightarrow M_2(\mathbb{k})$ is determined by the image p of $(1, 0)$; it is an embedding exactly when p is a nontrivial idempotent, necessarily of rank one, and the other primitive idempotent maps to $1 - p$. Rank-one idempotents vary with the choice of an image line and a complementary kernel line, so they form a positive-dimensional family. Any two are conjugate: choose a linear isomorphism carrying the image and kernel decomposition of the first to that of the second. Conjugation by its matrix carries one idempotent, hence one embedding, to the other.

Hence inner conjugacy is the correct general conclusion for a nonconnected ambient algebra.

Application 11.3 (Rational representations of linearly reductive groups). Let G be a linearly reductive algebraic group over the algebraically closed field $\mathbb{k}$, and let X be a finite-dimensional rational G -algebra. If $X^G = \mathbb{k}1$, then X has only finitely many G -stable separable unital subalgebras. More generally, for a fixed rational G -module B , there are only finitely many equivariant separable algebra structures on B up to equivariant algebra isomorphism. For a G -stable algebra, linear reductivity allows an ordinary separability section to be chosen G -equivariantly. The subalgebra assertion therefore follows from Corollary 7.2, whereas the fixed-module assertion follows from Theorem 5.1, both applied in $\text{Rep}(G)$.

Internal detail 85. The external input is the defining complete-reducibility property of a linearly reductive group: every finite-dimensional rational representation is a direct sum of irreducibles. In characteristic zero every reductive algebraic group is linearly reductive; see [25, §15]. Morphism spaces are ordinary finite-dimensional spaces of equivariant linear maps, and subobjects are invariant subspaces. Moreover $\text{Hom}_{\text{Rep}(G)}(\mathbf{1}, X) = X^G$, so the hypothesis $X^G = \mathbb{k}1$ is exactly connectedness of the algebra object in $\text{Rep}(G)$. For a finite-dimensional rational G -algebra A , let $W = \text{Hom}_{A-A}(A, A \otimes A)$, with the outer actions on $A \otimes A$. This is a finite-dimensional rational G -module, and $m_A \circ - : W \rightarrow \text{End}_{A-A}(A)$ is G -equivariant. An ordinary separability section says that id_A is in its image. Exactness of G -invariants, which is linear reductivity, gives a G -invariant preimage of id_A . Thus ordinary and equivariant separability agree here. Equivalence over X is equality of the embedded G -stable subalgebras, so the categorical corollary gives the stated assertion. The fixed-module assertion instead applies Theorem 5.1 directly to the object B in $\text{Rep}(G)$.

The same corollary applies to every semisimple tensor category. In particular, the ambient category need not be finite; that is, it need not be a fusion category.

12. SCOPE, SHARPNESS, AND OPEN DIRECTIONS

12.1. Where each assumption enters. It is useful to separate the hypotheses needed for local rigidity from those used only in the global applications. Hom-finiteness makes the relevant morphism spaces finite-dimensional affine spaces, while separability contracts Hochschild cohomology in degrees one and two. These are the inputs in the two local orbit calculations; neither semisimplicity nor unitarity is used there. Connectedness of the ambient algebra has a different role: it makes $\Gamma(X)$ one-dimensional, so inner conjugation is trivial. It is not needed for fixed-object rigidity.

To pass from one fixed source object to all arbitrary separable algebra subobjects of X by the argument of §7, one needs finitely many underlying subobject isomorphism classes. Semisimplicity and finite length are sufficient for this step. Compatible Frobenius subalgebras instead have split embeddings and are encoded by exchange idempotents in $\text{End}_{\mathcal{C}}(X)$. The resulting bound on inner-conjugacy classes in Theorem 8.9 requires neither semisimplicity, finite length, nor a finite-subobject hypothesis; connectedness turns that bound into an actual count. Positivity enters the categorical-dimension estimates and ensures separability of connected Frobenius algebras. Unitarity supplies the $\dagger$ -compatible structures used in the operator-algebraic applications.

Finally, degree estimates play no part in rigidity itself. They turn openness of classes and isolation of points into numerical bounds, whose sharpness depends on the equations chosen to present the parameter space.

12.2. The limits of local rigidity. The following examples explain why separability is essential and what openness of the separable orbits does not imply. First, an open isomorphism orbit need not be closed: a family of isomorphic separable algebras can have a nonseparable limit. On the fixed vector space $\mathbb{k}1 \oplus \mathbb{k}x$, let A_t be the algebra defined by $x^2 = tx$. If $t \neq 0$, then $t^{-1}x$ and $1 - t^{-1}x$ are orthogonal idempotents, so $A_t \cong \mathbb{k} \times \mathbb{k}$ and is separable. At $t = 0$ one obtains $A_0 = \mathbb{k}[x]/(x^2)$, the dual-number algebra; its generator x is nonzero and square-zero, so A_0 is nonsemisimple and nonseparable. Here “degeneration” simply means that the multiplication table depends polynomially on t and is specialized at $t = 0$.

Internal detail 86. In the fixed basis $1, x$, the only varying structure constant is t ; hence $t \mapsto A_t$ is a regular map from $\mathbb{A}^1$ to the variety of algebra structures. When $t \neq 0$, putting $p = t^{-1}x$ gives $p^2 = p$, and p and $1 - p$ are orthogonal idempotents with sum 1. At $t = 0$, the ideal $\mathbb{k}x$ is nonzero and square-zero, and hence lies in the Jacobson radical. Thus the special fiber is nonsemisimple.

Second, separability of the source is essential for homomorphism rigidity. The algebra maps

$$\mathbb{k}[\zeta]/(\zeta^2) \longrightarrow \mathbb{k}[u]/(u^2), \quad \zeta \longmapsto au, \quad a \in \mathbb{k},$$

form an affine line of distinct maps. Since the codomain is commutative, these maps lie in distinct inner-conjugacy classes.

Internal detail 87. Each assignment is unital and multiplicative because $(au)^2 = a^2u^2 = 0$, so it respects the defining relation $\zeta^2 = 0$. Distinct scalars give distinct linear maps because $u \neq 0$ in the codomain. That algebra is commutative, so its inner-conjugation action is trivial. Hence this is genuinely an affine line of inequivalent homomorphisms, showing exactly where separability of the source is needed.

12.3. The nonsemisimple ambient case. Finite length alone does not imply that an object has only finitely many subobject isomorphism classes, even in a Hom-finite tensor category. Consider finite-dimensional rational representations of $G = \mathbb{G}_a^2$ over $\mathbb{k}$. On $V = \mathbb{k}e_0 \oplus \mathbb{k}e_1 \oplus \mathbb{k}e_2$, define

$$\rho(s, t)e_0 = e_0, \quad \rho(s, t)e_1 = e_1 + se_0, \quad \rho(s, t)e_2 = e_2 + te_0.$$

These formulas define a rational representation in every characteristic. For $[a : b] \in \mathbb{P}^1(\mathbb{k})$, the subspace

$$W_{[a:b]} = \mathbb{k}e_0 \oplus \mathbb{k}(ae_1 + be_2)$$

is G -stable. On its second displayed basis vector the action adds $(as + bt)e_0$. Its fixed subspace is exactly $\mathbb{k}e_0$, so an isomorphism between two such representations must preserve this line. Equivariance then forces their pairs (a, b) to be proportional. Thus the $W_{[a:b]}$ are pairwise nonisomorphic as $[a : b]$ varies, although all are subobjects of the single finite-length object V .

Internal detail 88. In the ordered basis (e_0, e_1, e_2) , the representation matrix is

$$\rho(s, t) = \begin{pmatrix} 1 & s & t \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \rho(s, t)\rho(s', t') = \rho(s + s', t + t').$$

This verifies the group law without using exponentials or a characteristic-zero assumption. For $w = ae_1 + be_2$, every intertwiner $F : W_{[a:b]} \rightarrow W_{[a':b']}$ that is an isomorphism has the form

$$F(e_0) = \lambda e_0, \quad F(w) = \nu e_0 + \mu(a'e_1 + b'e_2), \quad \lambda\mu \neq 0.$$

Comparing $F\rho(s, t)w$ and $\rho(s, t)F(w)$ gives $\lambda(as + bt) = \mu(a's + b't)$ for all $s, t \in \mathbb{k}$, hence $\lambda(a, b) = \mu(a', b')$. Conversely, proportional pairs give the same subspace. The projective-line parameter therefore records isomorphism classes, not merely different embeddings of one object.

This example concerns underlying objects; it does not assert that V has infinitely many separable algebra subobjects. It identifies precisely why the proof for arbitrary separable algebra subobjects cannot replace the finite-subobject-isomorphism-class hypothesis by finite length alone. The local theorems for each fixed object and each fixed separable source remain valid. Compatible Frobenius subalgebras admit a different argument: their split embeddings are encoded in the single exchange locus, and Theorem 8.9 bounds their inner-conjugacy classes without a finite list of underlying objects or a semisimplicity assumption.

12.4. Sources of loss and possible refinements. The upper bounds are universal but can be far from sharp in specific families. There are three main sources of loss. First, Bézout bounds the cumulative degree of the entire parameter variety, including nonseparable components irrelevant to the classification. Second, it treats the equations as though they were dense and independent, whereas associativity and the exchange relations have substantial sparsity and algebraic dependencies. Third, it does not use stabilizer dimensions, fusion-rule symmetries, positivity, or integrality.

Internal detail 89. The base two is sharp if the ambient category is allowed to vary without a simple-unit assumption. Take $\mathcal{C}_n = \text{Vec}_{\mathbb{C}}^n$ with componentwise tensor product and $X_n = (\mathbb{C}^2, \dots, \mathbb{C}^2)$. In each component use coordinatewise multiplication, $\epsilon(a, b) = a + b$, and $\delta(e_j) = e_j \otimes e_j$ for the coordinate idempotents e_1, e_2 . The only unital subalgebras of $\mathbb{C}^2$ are $\mathbb{C}(1, 1)$ and $\mathbb{C}^2$. The former has counit $z \mapsto 2z$ and comultiplication $1 \mapsto \frac{1}{2}(1 \otimes 1)$; for its diagonal embedding i , one has $i^*(a, b) = (a + b)/2$, hence $i^*i = \text{id}$ and $i^{**} = i$. Both subalgebras are separable, and Frobenius uniqueness excludes any additional compatible structures. Independent choices in the n components give exactly 2^n Frobenius subalgebras. Inner conjugation is trivial because $\Gamma(X_n) = (\mathbb{C}^2)^n$ is commutative.

The Frobenius dual on each matrix algebra $M_2(\mathbb{C})$ is transpose. A symmetric matrix annihilates $(1, 1)^t$ precisely when it has the form

$$a \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \quad a \in \mathbb{C}.$$

Thus $r_{X_n} = n$ and $N_F^{\text{inn}}(X_n) = 2^n = 2^{r_{X_n}}$. Any uniform improvement over all ambient categories therefore requires additional hypotheses.

Several refinements are therefore possible in principle. One can eliminate the linear equations before counting, use multihomogeneous or sparse Bézout bounds, decompose the parameter space by ranks and symmetries, or replace the universal estimate by the actual cumulative degree of the relevant reduced locus. For a concrete Frobenius algebra X , one can compute directly in the r_X intrinsic coordinates of (63), where the linear conditions have already been eliminated. When X is connected, the isolated separable points can be studied by a Gröbner basis, a rational univariate representation, or a certified numerical decomposition. For general X , the count instead uses the irreducible components meeting the open inner-conjugacy classes.

Internal detail 90. For Gröbner bases, elimination, and exact solution of zero-dimensional systems, see [11, Chapters 2–4]; for homotopy continuation and certified numerical decomposition, see [38]. These are possible refinements only: none is used in the proofs of the universal bounds in this paper.

There is also an asymptotic version of this problem. For a sequence (X_n) with $r_n := r_{X_n} \rightarrow \infty$, the present estimate gives

$$\limsup_{n \rightarrow \infty} \frac{\log_2 \max\{1, N_F^{\text{inn}}(X_n)\}}{r_n} \leq 1.$$

For connected X_n , this is the bound for $N_F(X_n)$ itself. The local-rigidity step makes the relevant inner-conjugacy classes open and, in the connected case, makes their points isolated; it supplies neither a separation scale nor an asymptotic counting rate. Any improvement of the exponent must therefore use uniform global information about the exchange ideals, or additional categorical structure. Likewise, a meaningful asymptotic comparison with the angle method requires optimizing both arguments in the same parameter.

The subfactor application illustrates why the intrinsic parameter matters. The dimension of the second relative commutant may be far smaller than the Jones index, and the exact difference formula (87) measures the loss incurred by replacing the former with the latter. In concrete standard invariants, the exchange equations already live in a space visible at depth two.

Remark 8.10 explains the earlier structural improvement from cubic multiplication-closure equations to quadratic exchange relations. The loss discussed here occurs after that reduction: the product-of-degrees estimate still ignores the special form of the quadratic exchange system. A further improvement of the base must exploit that structure.

12.5. Further questions. The preceding discussion singles out the following questions.

1. For which nonsemisimple tensor categories, and for which objects X in them, does X have only finitely many subobject isomorphism classes, so that the general argument also controls arbitrary separable algebra subobjects without compatible Frobenius structures?
2. In positive tensor categories, or more specifically integral fusion categories, can $N_F^{\text{inn}}(X) \leq 2^{r_X}$ be replaced asymptotically by a uniform bound c^{r_X} with $c < 2$, or even a subexponential bound? For irreducible subfactors, how do the best asymptotic exchange and angle bounds compare when both are measured in the Jones index?

3. Can arbitrary separable algebra subobjects of X , without a chosen compatible Frobenius structure, be encoded by an intrinsic quadratic system whose dimension is controlled by $\dim_{\mathbb{k}} \text{End}_{\mathcal{C}}(X)$?

Internal detail 91. The following more specialized questions are retained for possible later use.

- I1. When can a unitary Frobenius algebra contain a Frobenius subalgebra that is not unitary? In the finite-dimensional Hopf-algebra setting, can a C^* -Hopf algebra possess a left coideal subalgebra that is not $*$ -closed? This question is already posed in the introduction of [21].
- I2. Can planar-algebraic or principal-graph information improve the bound $2^{\dim_{\mathbb{C}}(N' \cap M_1) - 1}$ while still counting the whole intermediate lattice in a single step?
- I3. In the nonirreducible operator-algebra setting, can the effective bounds on inner-conjugacy classes proved here be sharpened using more of the standard invariant, and how do these refinements compare with angle-based methods?

12.6. Local Ocneanu rigidity and the formal-angle method. The local Ocneanu-rigidity argument gives a direct intrinsic-dimension bound on inner-conjugacy classes and, in the connected case, on the whole Frobenius subalgebra poset. The formal-angle method captures additional structure. The framework of [20, 21] produces exchange relations, a linear-monoidal version of Landau’s theorem, coherent pairs and coherent sublattices, rigidity invariance, and formal-angle invariants. It also isolates natural Hopf-algebraic questions, including whether left coideal subalgebras of finite-dimensional C^* -Hopf algebras must be $*$ -closed and whether their lattice is coherent.

By contrast, the present argument makes algebraic and differential geometric ideas, such as tangent spaces, algebraic group orbits, and affine degree, explicit inside tensor category theory. The local Ocneanu-rigidity and formal-angle methods should therefore be viewed as complementary rather than competing.

Internal detail 92. The deformation-theoretic core may be summarized as follows. Separability contracts Hochschild cohomology in degrees one and two: the second degree controls algebra structures, while the first controls algebra maps modulo inner conjugacy. Algebraic geometry turns the tangent identities into open orbits, and affine degree converts the resulting finite-dimensional parameter spaces into explicit bounds. For Frobenius subalgebras, the exchange relations compress the source and embedding data into a quadratic locus of self-dual idempotents.

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